THE ARTIN–DAVENPORT CONJECTURE ON CUBIC FORMS

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Abstract. We show that any integral cubic form in ten or more variables has a nontrivial integer zero. Like earlier authors on this topic, we use geometric methods to produce a zero in some cases, and the circle method in all other cases. New ingredients are required on both fronts: in the first case, bounds on the dimension of singular loci and incidence varieties, and in the second case, a stratified geometric sieve, together with stratification results for points counts over finite fields and a cube-root Hensel lifting analysis.

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1. Introduction

In his presidential address to the London Mathematical Society in November 1959, it was conjectured by Davenport [Dav60] that any integral cubic form in $n \geqslant 10$ variables always possesses a nontrivial integer zero. The exact same conjecture was attributed to Artin by Lang [Lan60], in his address to the American Mathematical Society in February 1960. For $n \leqslant 9$, there exists cubic forms which fail to have nontrivial integer zeros. For example, if $K/\mathbb{Q}$ is a cubic number field with $\mathbb{Z}$-basis $\omega_1, \omega_2, \omega_3$ for the ring of integers $\mathfrak{o}_K$, and if p is a rational prime that is inert in $\mathfrak{o}_K$, then the nonary cubic form

$$\mathsf{N}_{K/\mathbb{Q}}(x_1, x_2, x_3) + p\mathsf{N}_{K/\mathbb{Q}}(x_4, x_5, x_6) + p^2\mathsf{N}_{K/\mathbb{Q}}(x_7, x_8, x_9)$$

has only the trivial zero over $\mathbb{Q}_p$, where $\mathsf{N}_{K/\mathbb{Q}}(x, y, z) = N_{K/\mathbb{Q}}(x\omega_1 + y\omega_2 + z\omega_3)$ is the associated norm form. Thanks to work of Demyanov [Dem50] (when $p \neq 3$) and Lewis [Lew52] (in general), it has been known since the 1950s that cubic forms in $n \geqslant 10$ variables have a nontrivial p-adic zero for every prime p.

In 1959, Davenport [Dav59] adapted the circle method to prove that $n \geqslant 32$ suffices for any integral cubic form in n variables to possess a nontrivial integer zero. This was improved to $n \geqslant 16$ by Davenport [Dav63] in 1963, before being extended to $n \geqslant 14$ by Heath-Brown [HB07] in 2007, using an averaged van der Corput variant of the circle method. The following is our main result.

Theorem 1.1. *The Artin–Davenport conjecture is true.*

Suppose that $X \subset \mathbb{P}^{n-1}$ is a cubic hypersurface defined by a cubic form $F \in \mathbb{Z}[x_1, \dots, x_n]$. The arithmetic of X has been the subject of intense focus over the years, with many results that work towards the Artin–Davenport conjecture, or that apply in smaller dimensions. Compelling evidence is provided by Heath-Brown [HB83], who establishes the conjecture for the special class of smooth cubic hypersurfaces X. It follows from Hooley [Hoo88, Hoo91] that we have the same conclusion if X has at worst isolated singularities.

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Assuming that X is geometrically integral, not equal to a cone, and defined over $\mathbb{Q}$, the existence of a single rational point on X is enough to demonstrate the $\mathbb{Q}$-unirationality of X, and so the Zariski density of $X(\mathbb{Q})$ in X. Variants of this result have been known for many years, and can be found in the work of Colliot-Thélène and Salberger [CTS89, Proposition 1.3] and Kollár [Kol02].

For a large class of possibly singular cubic hypersurfaces $X \subset \mathbb{P}^{n-1}$, Colliot-Thélène [Bro10, Appendix] has calculated the Brauer group $\mathrm{Br}(Y)$ associated to a smooth model Y of X. (This calculation is based on earlier unpublished work of Salberger.) As a consequence of this and the conjecture of Colliot-Thélène [CT03] that the Brauer–Manin obstruction should control the Hasse principle for rationally connected varieties, one is led to the following conjecture.

Conjecture 1.2. *Let $X \subset \mathbb{P}^{n-1}$ be a cubic hypersurface defined over $\mathbb{Q}$ which is not a cone, with $n \geqslant 5$ and singular locus which is empty or of codimension at least 4 in X. Then the Hasse principle holds for the locus of smooth points on X.*

This conjecture is formulated in [Bro10, Section 1]. As a crucial auxiliary result, we shall prove the following result in Lemma 3.8, which may prove to be of independent interest.

Theorem 1.3. *Let $n \geqslant 2$ and let $X \subset \mathbb{P}^{n-1}$ be a cubic hypersurface defined over $\mathbb{Q}$, with singular locus of codimension at most $\lfloor (n-1)/3 \rfloor$ in X. Then $X(\mathbb{Q}) \neq \emptyset$.*

We note here that $\lfloor (n-1)/3 \rfloor = 3$ for $10 \leqslant n \leqslant 12$ and $\lfloor (n-1)/3 \rfloor = 4$ for $13 \leqslant n \leqslant 15$.

Overview of the paper. Our paper may be read linearly as usual, but the following non-linear outline of the main ideas may be useful. Associated to a cubic form $F \in \mathbb{Z}[x_1, \dots, x_n]$ is the $n \times n$ Hessian matrix $H(\boldsymbol{x})$, which is formed from the second partial derivatives of F. Throughout this paper, let $|\boldsymbol{x}| := \max\{|x_1|, \dots, |x_n|\}$ for $\boldsymbol{x} \in \mathbb{R}^n$. We will say that the cubic form F is *good* if for any $H \geqslant 1$ and any $\varepsilon > 0$ we have the upper bound

$$\#\{\boldsymbol{x} \in \mathbb{Z}^n : |\boldsymbol{x}| \leqslant H, \operatorname{rank} H(\boldsymbol{x}) = r\} \ll H^{r+\varepsilon},$$

for each integer $0 \leqslant r \leqslant n$. A crucial step in Davenport's [Dav63] treatment of general cubic forms is a proof of the fact that forms that fail to be good automatically possess nontrivial integer zeros. This allows us to focus our attention on good cubic forms in all that follows.

The circle method is introduced in §12. We shall adopt the Kloosterman circle method approach that forms the backbone of the works by Heath-Brown [HB83] and Hooley [Hoo88], and which rely on an application of Poisson summation to bring into play the complete exponential sums

$$S_q(\boldsymbol{v}) := \sum_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \sum_{\boldsymbol{x} \in (\mathbb{Z}/q\mathbb{Z})^n} e_q(aF(\boldsymbol{x}) + \boldsymbol{v} \cdot \boldsymbol{x}), \tag{1.1}$$

for integers $q \geqslant 1$ and vectors $\boldsymbol{v} \in \mathbb{Z}^n$. When q is prime the estimation of these sums in [HB83, Hoo88] uses the full force of Deligne's resolution of the Weil conjectures [Del80]. This is one of several places where tight control over the singular locus of the cubic hypersurface X enters the argument. Our main idea can be summarised as using the stratification theory of exponential sums over finite fields as a proxy for the breakdown in control over $S_q(\boldsymbol{v})$ whenever X has a large-dimensional singular locus.

Works of Fouvry, Katz and Laumon [KL85, FK01] produce a stratification that is nearly sufficient for our purposes. Their construction is based on a general induction procedure. In our projective setting, a different construction, based on the theory of singular support developed by Beilinson [Bei16], Hu–Yang [HY18] and Raskin–Smith [RS22], gives sharper control on the strata Z_j, enabling us to use classical projective duality to show that if $X(\mathbb{Q})$ were empty, then the dimension of the Zariski closure of $Z_j(\mathbb{Q})$ in Z_j would be strictly less than the standard bound for $\dim Z_j$ itself. A similar idea with smooth $\mathbb{Q}$-points on dual varieties was previously used in the circle method for smooth cubic hypersurfaces in 9 variables by Hooley [Hoo88], before he found a more robust argument avoiding any assumption on the dual variety [Hoo94]. The perspective of Raskin–Smith [RS22], connecting the singular support to Gysin maps of singular

varieties, is convenient for us. However, one could alternatively adopt a more Fourier-theoretic approach based on Laumon [Lau87], Sawin [Saw19] and Zhou [Zho24].

It is well known that homogeneity of F gives $S_q(u\boldsymbol{v}) = S_q(\boldsymbol{v})$ for all $u \in (\mathbb{Z}/q\mathbb{Z})^*$, whence the Chinese remainder theorem gives

$$S_{q_1q_2}(\boldsymbol{v}) = S_{q_1}(\boldsymbol{v})S_{q_2}(\boldsymbol{v}), \tag{1.2}$$

whenever $\gcd(q_1, q_2) = 1$. The circle method will lead us to estimate sums of the shape

$$\sum_{\boldsymbol{v}}\sum_{q} S_q(\boldsymbol{v})I_q(\boldsymbol{v}) = \sum_{\boldsymbol{v}}\sum_{\text{cubefull } r} S_r(\boldsymbol{v}) \sum_{\substack{\text{cube-free } d \\ \gcd(d,r)=1}} S_d(\boldsymbol{v})I_{dr}(\boldsymbol{v}),$$

where we temporarily write $I_q(\boldsymbol{v})$ for a certain oscillatory integral. See (12.9) for the specific ranges involved. Our overall strategy is then to combine precise stratifications for $|S_d(\boldsymbol{v})|$, on the one hand, with weaker but more robust estimates for $|S_r(\boldsymbol{v})|$ and $|I_{dr}(\boldsymbol{v})|$.

The stratifications are given in Lemma 5.3 for prime squares, and in Appendix A for primes. A hybrid level-set decomposition for $|S_d(\boldsymbol{v})|$, based on the global stratum of $\boldsymbol{v}$ and the local strata-based factorisation type of d, is given in §10. Moreover, in §10, the level-set estimates are uniform over adelic boxes, which turns out to be crucial for our approach.

The estimates for cubefull factors $|S_r(\boldsymbol{v})|$ tend to depend not exactly on $\boldsymbol{v} \bmod r$, but only roughly on $\boldsymbol{v} \bmod r^{1/3}$, an insight that we formalise in §7 away from a finite set of primes p. The remaining primes are handled separately in §8 by a localisation trick. In §6 we also give stronger estimates that depend roughly on $\boldsymbol{v} \bmod r^{1/2}$. However, for most of the paper, it turns out to be useful for us to use weaker estimates that depend on less information.

Moreover, the integral estimates in §12 tend to control a large window (box neighbourhood) of vectors $\boldsymbol{v}$ at once, not just a single real point $\boldsymbol{v}$. An adelic box $\mathcal{A}$ of vectors $\boldsymbol{v}$, on which $|S_r(\boldsymbol{v})|$ and $|I_{dr}(\boldsymbol{v})|$ are controlled, is then fertile ground for stratification methods to take place: bounds via point counts, sieves, and traces. Most of our cubefull and integral estimates may be fruitfully restricted to various natural subsets $\mathcal{S} \subset \mathcal{A}$, giving bounds that are sensitive to the density of $\mathcal{S}$ in $\mathcal{A}$. A notable exception is the Poisson summation based squarefull bound (Lemma 6.7), which is reasonably strong in short boxes but often weak in small subvarieties thereof. Our work will require combining different estimates in different ranges.

In Lemma 3.8 we prove Theorem 1.3 and obtain further information about the singular loci of our cubic hypersurface X and its hyperplane sections, assuming that $X(\mathbb{Q}) = \emptyset$. This leads to useful exponential sum estimates via work of Hooley, Katz, et al., as we recall in §5.

In §10 we prove the required cube-free estimates in ten variables, using a stratified geometric sieve, Proposition 2.9. In §11, this is combined with the corresponding ten variable cubefull estimates of §7. The proof of Theorem 1.1 is then completed in §12 by incorporating the integral factor, as well as a main term calculation based on work of Bernert [Ber25].

Final remarks. The role of stratifications can be illustrated already for prime moduli. In this case our goal is to show that

$$\mathcal{T}(B) := \sum_{B^{3/2}<p\leqslant 2B^{3/2}} \sum_{\mathbf{0}\neq\boldsymbol{v}\in[-B^{1/2},B^{1/2}]^n} |p^{-n}S_p(\boldsymbol{v})| = o(1).$$

If we had square-root cancellation $S_p(\boldsymbol{v}) \ll p^{(n+1)/2}$ everywhere, then we would immediately get

$$\mathcal{T}(B) \ll B^{(3+n)/2}(B^{3/2})^{(1-n)/2} = B^{(9-n)/4} = o(1)$$

for $n \geqslant 10$. However, square-root cancellation does not hold pointwise. Stratification theory controls the extent to which failures of square-root cancellation can hold. Moreover, when counting $\mathbb{Q}$-points on strata using dimension growth bounds, we must avoid certain bad configurations. The dimension of any stratum with $\mathbb{Q}$-points is controlled using singular support in Appendix A. Rational linear subspaces of strata are controlled using a second moment in §9.

Many questions remain open, even for cubic forms in $n \geqslant 10$ variables over $\mathbb{Q}$. We have been able to show that for any such cubic form, either geometric methods produce a $\mathbb{Q}$-point, or else

the circle method produces a localised asymptotic formula. It would be interesting to determine the weakest conditions under which an asymptotic formula can be proven; does goodness suffice? A related question is that of obtaining lower bounds for point counts and a question of Wooley, for which we refer to recent work of Kumaraswamy and Rome [KR24].

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Supplementary Lean code. The interested reader may consult

`https://github.com/wangyangvictor/cubic-lean-mid`

for a Codex Lean auto-formalisation of Theorem 1.1 and its proof, modulo five inputs from the literature in "elementary" form. This builds on the vast existing Lean infrastructure, including the strong foundations of Mathlib in areas such as commutative algebra.

AI statement. The structure of the proof and most of the new ideas in this paper are due to the authors and were developed from June 2024 until September 2026. However, a key ingredient in our work is a suitable upper bound for the dimension of the incidence variety

$$I = \{(\boldsymbol{x}, \boldsymbol{y}) \in \mathbb{A}^n \times \mathbb{A}^n \colon H(\boldsymbol{x})\boldsymbol{y} = 0\},$$

with the inequalities $n \leqslant \dim I \leqslant 2n$ being essentially trivial. When the cubic hypersurface X is smooth it is straightforward to prove that $\dim I = n$, a fact that underpins work of Heath-Brown [HB83, Lemma 1]. In general, assuming only that $X(\mathbb{Q}) = \emptyset$, we were able to prove the upper bound $\dim I \leqslant \lfloor (3n-2)/2 \rfloor$. This was improved to $\dim I \leqslant \lfloor (3n-3)/2 \rfloor$ in a private communication from Heath-Brown, which we believe would be enough to handle cubic forms in at least eleven variables, once combined with the other ingredients in our paper. After prompting ChatGPT 6.0, we were led to the improved bound $\dim I \leqslant \lfloor (4n-3)/3 \rfloor$ in Lemma 3.8, under the same assumption $X(\mathbb{Q}) = \emptyset$. ChatGPT 6.0 suggested an approach via the Galois descent procedure in Lemma 3.6, which is inspired by work of Bate–Martin–Röhrle–Tange [BMRT13]. Our work involves combining many different estimates that are effective in different ranges of parameters. ChatGPT 5.5, 5.6, 6.0 and 6.1 have been helpful with this process, as well as in proofreading our work, and conducting the Lean auto-formalisation of this work. We are very grateful to Mark Sellke at OpenAI for giving us access to the Pro models in May 2026.

2. Technical lemmas

2.1. **Counting points over finite fields.** The following result provides point counts for cubic hypersurfaces over finite fields, under various assumptions.

Lemma 2.1. *Let $\mathbb{F}_q$ be a finite field of order q and let $X \subset \mathbb{P}^n$ be a geometrically integral cubic hypersurface defined over $\mathbb{F}_q$. If $\sigma = \dim \mathrm{Sing}(X) \in \{-1, \dots, n-2\}$, where $\mathrm{Sing}(X)$ is the singular locus of X, then*

$$\#X(\mathbb{F}_q) - \frac{q^n - 1}{q - 1} \ll_n q^{(n+\sigma)/2}.$$

Moreover,

$$\#X(\mathbb{F}_q) - \frac{q^n - 1}{q - 1} \ll_n \begin{cases} q^{n-2}, & \text{if } n \geqslant 3 \text{ and } X \text{ is not a cone,} \\ q^{n-3/2}, & \text{otherwise.} \end{cases}$$

Proof. The first result can be extracted from Theorem 1 of [HK91]. The second is the main result of [Bro13] and the last follows from the Lang–Weil estimate [LW54]. □

Remark 2.2. Although Lemma 2.1 will suffice for the present paper, it might be possible to obtain sharper results for $n \geqslant 4$ by using [Wan23, Theorem 5.5].

2.2. **Basic geometric facts.** Let us record some useful facts about geometric integrality, cones, and singular loci. The first part of [Bro10, Lemma 1] states the following.

Proposition 2.3. *Let $N \geqslant 3$. If $X \subset \mathbb{P}^N_{\mathbb{Q}}$ is a cubic hypersurface with $X(\mathbb{Q}) = \emptyset$, then X is geometrically integral and is not a cone.*

Remark 2.4. It will be useful to know how to compute the maximal vertex of a conical cubic hypersurface. If $C(\boldsymbol{y})$ is a cubic form in m variables over a perfect field K, then its maximal vertex over $\overline{K}$ coincides with the base change of its maximal vertex over K. This follows from Galois descent, since the maximal vertex over a field L/K corresponds to the L-vector space

$$M(L) = \{\boldsymbol{z} \in L^m : C(\boldsymbol{y} + t\boldsymbol{z}) - C(\boldsymbol{y}) = 0 \in L[\boldsymbol{y}, t]\},$$

which satisfies $M(\overline{K})^{\mathrm{Gal}(\overline{K}/K)} = M(K)$. This is particularly clean if K has characteristic $\neq 2, 3$, in which case by Taylor expansion the condition in $M(L)$ is equivalent to $H_C(\boldsymbol{z}) = \boldsymbol{0}$.

Thanks in part to Lemma 2.1, singular loci of hyperplane sections of a cubic hypersurface will be relevant. The following result is best interpreted scheme-theoretically.

Theorem 2.5 (Zak)**.** *Let $Y \subset \mathbb{P}^N$ be a geometrically integral hypersurface of degree $\geqslant 2$ over a field. Let $H \subset \mathbb{P}^N$ be a hyperplane. Then* $\dim \mathrm{Sing}(Y \cap H) \leqslant \dim \mathrm{Sing}(Y) + 1$. *In particular, if* $\dim Y - \dim \mathrm{Sing}(Y) \geqslant 4$, *then $Y \cap H$ is geometrically integral.*

Proof. This follows from Zak's bound recorded in Theorem 2 of Katz's appendix to [HK91]. □

For further results in this direction, we refer the reader to Zak [Zak93, Theorem 1.7].

2.3. **A geometric sieve in adelic boxes.** We need to count points uniformly in adelic boxes. It will be convenient to have the following version of Noether normalisation to hand.

Lemma 2.6. *Let $Y \subseteq \mathbb{A}^N_{\mathbb{Z}}$ be a closed subscheme whose generic fibre $Y_{\mathbb{Q}} \subseteq \mathbb{A}^N_{\mathbb{Q}}$ has dimension $0 \leqslant d \leqslant N-1$. Let $\pi\colon \mathbb{A}^N \to \mathbb{A}^d$ be the projection onto the first d coordinates. There exist a matrix $A \in \mathrm{Mat}_{N\times N}(\mathbb{Z}) \cap \mathrm{GL}_N(\mathbb{Q})$, an integer $C \geqslant 1$, and polynomials $F_j \in \mathbb{Z}[x_1, \dots, x_d, x_{d+j}]$ for $1 \leqslant j \leqslant N-d$, such that the following hold:*

(1) The composite map $Y_{\mathbb{Q}} \xrightarrow{A} \mathbb{A}^N_{\mathbb{Q}} \xrightarrow{\pi} \mathbb{A}^d_{\mathbb{Q}}$ is finite.
(2) For each $1 \leqslant j \leqslant N-d$, we have $F_j = Cx_{d+j}^C + O_{x_1,\dots,x_d}(x_{d+j}^{C-1})$, and the polynomial $F_j(A\boldsymbol{x}) \in \mathbb{Z}[\boldsymbol{x}] = \mathbb{Z}[x_1, \dots, x_N]$ vanishes on Y.

Proof. Since the field $\mathbb{Q}$ is infinite, it is a well-known consequence of Noether normalisation that there exists a dense open subvariety $U \subset \mathrm{GL}_N$ over $\mathbb{Q}$ such that (1) holds for all $A \in U(\mathbb{Q})$, as remarked in [Eis95, p. 284]. Since $\mathrm{Mat}_{N\times N}(\mathbb{Z}) \cap \mathrm{GL}_N(\mathbb{Q})$ is Zariski dense in $\mathrm{GL}_N(\overline{\mathbb{Q}})$, we can satisfy (1) with a matrix $A \in \mathrm{Mat}_{N\times N}(\mathbb{Z}) \cap \mathrm{GL}_N(\mathbb{Q})$. It follows from (1) that each x_{d+j} satisfies a monic non-constant polynomial relation in $\mathbb{Q}[x_1, \dots, x_d][x_{d+j}]$ on the variety $A(Y_{\mathbb{Q}})$. After multiplying each relation by a suitable monomial in x_{d+j}, we are led to (2), since every element of the ideal $I(Y_{\mathbb{Q}}) = I(Y) \otimes \mathbb{Q}$ is a nonzero rational scalar times a element of $I(Y)$. □

Lemma 2.6 immediately gives the following variant of the Schwartz–Zippel bound.

Lemma 2.7. *Let $Y \subseteq \mathbb{A}^N_{\mathbb{Z}}$ be a closed subscheme whose generic fibre has dimension $0 \leqslant d \leqslant N$. Let $m \geqslant 1$ be an integer. Let $\boldsymbol{b} \in \mathbb{Z}^N$, $L \geqslant 1$ and $\boldsymbol{v}_0 \in \mathbb{R}^N$. Then*

$$\#\{\boldsymbol{v} \in Y(\mathbb{Z}) : |\boldsymbol{v} - \boldsymbol{v}_0| \leqslant L,\ \boldsymbol{v} \equiv \boldsymbol{b} \bmod m\} \ll_Y (1 + L/m)^d.$$

Proof. This is trivial if $d = N$. If $d \leqslant N-1$, use Lemma 2.6. □

The following version of the Ekedahl sieve for large square-free moduli also becomes available.

Lemma 2.8. *Let $Y \subseteq \mathbb{A}^N_{\mathbb{Z}}$ be a closed subscheme whose generic fibre has dimension at most d, where $0 \leqslant d \leqslant N$. Let $m \geqslant 1$ be an integer. Let $\boldsymbol{b} \in \mathbb{Z}^N$, $L \geqslant 1$ and $\boldsymbol{v}_0 \in \mathbb{R}^N$. If $S \gg 1 + L/m$, then*

$$\sum_{\substack{q \sim S \\ \gcd(q,m)=1}} \sum_{\substack{\boldsymbol{v} \in \mathbb{Z}^N \setminus Y(\mathbb{Z}) \\ |\boldsymbol{v}-\boldsymbol{v}_0| \leqslant L \\ \boldsymbol{v} \equiv \boldsymbol{b} \bmod m \\ \boldsymbol{v} \bmod q \in Y(\mathbb{Z}/q\mathbb{Z})}} \mu^2(q) \ll_{Y,\varepsilon} (LS + |\boldsymbol{v}_0|)^{\varepsilon}(1 + L/m)^{d+1}.$$

Proof. Since the ideal $I(Y)$ is finitely generated, the divisor bound on a suitable polynomial in $I(Y)$ for each vector $\boldsymbol{v} \in \mathbb{Z}^N \setminus Y(\mathbb{Z})$ implies that

$$\sum_{\substack{q \sim S \\ \gcd(q,m)=1}} \sum_{\substack{\boldsymbol{v} \in \mathbb{Z}^N \setminus Y(\mathbb{Z}) \\ |\boldsymbol{v}-\boldsymbol{v}_0| \leqslant L \\ \boldsymbol{v} \equiv \boldsymbol{b} \bmod m \\ \boldsymbol{v} \bmod q \in Y(\mathbb{Z}/q\mathbb{Z})}} \mu^2(q) \ll_{Y,\varepsilon} \sum_{\substack{\text{admissible } \boldsymbol{v} \in \mathbb{Z}^N \setminus Y(\mathbb{Z}) \\ |\boldsymbol{v}-\boldsymbol{v}_0| \leqslant L \\ \boldsymbol{v} \equiv \boldsymbol{b} \bmod m}} (L + |\boldsymbol{v}_0|)^{\varepsilon},$$

where we call a vector $\boldsymbol{v} \in \mathbb{Z}^N$ admissible if there exists a square-free integer $q \sim S$ with $\gcd(q, m) = 1$ such that $\boldsymbol{v} \bmod q \in Y(\mathbb{Z}/q\mathbb{Z})$. Lemma 2.8 will follow if we can prove that

$$f(Y) := \sum_{\substack{\text{admissible } \boldsymbol{v} \in \mathbb{Z}^N \\ |\boldsymbol{v}-\boldsymbol{v}_0| \leqslant L \\ \boldsymbol{v} \equiv \boldsymbol{b} \bmod m}} 1 \ll_{Y,\varepsilon} (LS + |\boldsymbol{v}_0|)^{\varepsilon}(1 + L/m)^{d+1}. \tag{2.1}$$

We include vectors $\boldsymbol{v} \in Y(\mathbb{Z})$ in (2.1) so that $f(Y)$ is monotone, in the sense that if $Y \subseteq Y'$ then $f(Y) \leqslant f(Y')$. The inequality (2.1) is trivial if $d = N$, so suppose that $d \leqslant N - 1$. We may assume $\dim Y_{\mathbb{Q}} = d$, by adding a linear space of generic dimension d to Y if necessary. After changing variables as in Lemma 2.6, and further enlarging Y if necessary, we may assume in (2.1) that Y is the simultaneous zero locus of integer polynomials of the form

$$F_j(x_1, \dots, x_d, x_{d+j}) = Cx_{d+j}^C + O_{x_1,\dots,x_d}(x_{d+j}^{C-1}),$$

for $1 \leqslant j \leqslant N - d$, where $C \geqslant 1$. The contribution of $\boldsymbol{v} \in Y(\mathbb{Z})$ to the left-hand side of (2.1) is negligible by Lemma 2.7. If $\boldsymbol{v} \notin Y(\mathbb{Z})$, then let j_* be the minimal index j for which $F_j(v_1, \dots, v_d, v_{d+j}) \neq 0$. Once $v_1, \dots, v_d, v_{d+j_*}$ are chosen, there are $O_{Y,\varepsilon}((L + |\boldsymbol{v}_0|)^{\varepsilon})$ choices for $q \mid F_{j_*}(v_1, \dots, v_d, v_{d+j_*})$, by the divisor bound. For each $j \neq j_*$, the congruence $F_j(v_1, \dots, v_d, y_{d+j}) \equiv 0 \bmod q$ has at most $\gcd(C, q)C^{\omega(q)}$ solutions $y_{d+j} \bmod q$, by the Chinese remainder theorem. Each residue $y_{d+j} \bmod q$ has $O(1 + L/(qm)) = O(1)$ integral lifts v_{d+j} satisfying $v_{d+j} \equiv y_{d+j} \bmod q$ and $v_{d+j} \equiv b_{d+j} \bmod m$, since $\gcd(q, m) = 1$ and $q \sim S \gg L/m$. Thus the contribution to (2.1) from the index j_* is

$$\ll_{Y,\varepsilon} \sum_{v_1,\dots,v_d,v_{d+j_*}} (L + |\boldsymbol{v}_0|)^{\varepsilon} S^{\varepsilon} \ll_{\varepsilon} (LS + |\boldsymbol{v}_0|)^{2\varepsilon}(1 + L/m)^{d+1}.$$

Summing over $1 \leqslant j_* \leqslant N - d$ completes the proof of (2.1). □

2.4. **A geometric sieve for stratifications.** We now give a stratified geometric sieve, concerning an arbitrary finite collection of closed subschemes.

Proposition 2.9. *Let $Y_0, Y_1, \dots, Y_s \subseteq \mathbb{A}^N_{\mathbb{Z}}$ be closed subschemes whose generic fibres have dimensions at most $d_0, d_1, \dots, d_s$, respectively. Suppose that*

$$N \geqslant d_0 \geqslant d_1 \geqslant \cdots \geqslant d_s \geqslant 0.$$

Let $\mathcal{U} \subseteq Y_0(\mathbb{Z})$ and $0 \leqslant \alpha \leqslant d_0$. Suppose that uniformly over $\boldsymbol{v}_0 \in \mathbb{R}^N$, $L \geqslant 1$, $\boldsymbol{a} \in \mathbb{Z}^N$ and integers $q \geqslant 1$,

$$\#\{\boldsymbol{v} \in \mathcal{U} : |\boldsymbol{v} - \boldsymbol{v}_0| \leqslant L,\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod q\} \ll_{\mathcal{U},\varepsilon} (L + |\boldsymbol{v}_0|)^{\varepsilon}(1 + L/q)^{\alpha}. \tag{2.2}$$

Let $m \geqslant 1$ be an integer. Let $\boldsymbol{b} \in \mathbb{Z}^N$, $L \geqslant 1$ and $\boldsymbol{v}_0 \in \mathbb{R}^N$. Let

$$\mathcal{C} := \{\boldsymbol{v} \in \mathcal{U} \setminus \textstyle\bigcup_{i=1}^s Y_i(\mathbb{Z}) : |\boldsymbol{v} - \boldsymbol{v}_0| \leqslant L,\ \boldsymbol{v} \equiv \boldsymbol{b} \bmod m\}.$$

For $R_1, \dots, R_s \geqslant 1/2$, let

$$\mathcal{N}(\boldsymbol{R}) := \sum_{\substack{q_i \sim R_i \\ \gcd(q_i, q_j)=1 \ (i \neq j) \\ \gcd(q_i, m)=1}} \sum_{\boldsymbol{v} \in \mathcal{C}} \prod_{i=1}^{s} \mu^2(q_i) \mathbf{1}_{\boldsymbol{v} \bmod q_i \in Y_i(\mathbb{Z}/q_i\mathbb{Z})}.$$

Then

$$\mathcal{N}(\boldsymbol{R}) \ll_{\mathcal{U}, Y_0, \dots, Y_s, \alpha, \varepsilon} (LR_1 \cdots R_s + |\boldsymbol{v}_0|)^{\varepsilon} \left(1 + \frac{(L/m)^{\alpha}}{\prod_{i=1}^{s} R_i^{\alpha - d_i - 1}} + \sum_{j=1}^{s} \frac{(L/m)^{d_j + 1}}{\prod_{i=j+1}^{s} R_i^{d_j - d_i}} \right), \quad (2.3)$$

where empty products are interpreted as 1.

Proof. The contribution to $\mathcal{N}(\boldsymbol{R})$ from any individual vector $\boldsymbol{v} \in \mathcal{C}$ is $\ll (L + |\boldsymbol{v}_0|)^{\varepsilon}$, by the divisor bound. Thus (2.3) is trivial if $L \leqslant m$. We may therefore assume $L > m$.

By the Lang–Weil bound, we have $\#Y_i(\mathbb{F}_p) \ll p^{d_i}$ for all p. If $R_1 \cdots R_s \leqslant L/m$, then there are $\ll \prod_i q_i^{d_i + \varepsilon}$ available residue classes $\boldsymbol{v}$ mod $q_1 \cdots q_s$. Applying (2.2) with $q = mq_1 \cdots q_s \ll L$ gives

$$\frac{\mathcal{N}(\boldsymbol{R})}{(LR_1 \cdots R_s + |\boldsymbol{v}_0|)^{\varepsilon}} \ll \sum_{q_i \sim R_i} (L/q)^{\alpha} \prod_i q_i^{d_i} \ll \frac{(L/m)^{\alpha}}{\prod_{i=1}^{s} R_i^{\alpha - d_i - 1}},$$

which is satisfactory. If $R_1 \cdots R_s > L/m$, choose $1 \leqslant j \leqslant s$ such that

$$R_{j+1} \cdots R_s \ll L/m \ll R_j R_{j+1} \cdots R_s.$$

This is possible because $R_1, \dots, R_s \gg 1$ and $1 \ll L/m \ll R_1 \cdots R_s$. Given $\boldsymbol{v}$, the number of available moduli q_i is $\ll (L + |\boldsymbol{v}_0|)^{\varepsilon}$ for each $i < j$, by the divisor bound. On the other hand, there are $\ll \prod_{i>j} q_i^{d_i + \varepsilon}$ available residue classes $\boldsymbol{v}$ mod $\prod_{i>j} q_i$. Lemma 2.8, with $M := m \prod_{i>j} q_i \gg L/R_j$ in place of m, with $Y := Y_j$ and with $S := R_j \gg 1 + L/M$, gives

$$\frac{\mathcal{N}(\boldsymbol{R})}{(LR_1 \cdots R_s + |\boldsymbol{v}_0|)^{\varepsilon}} \ll \sum_{q_i \sim R_i \ (i>j)} (1 + L/M)^{d_j + 1} \prod_{i>j} q_i^{d_i} \ll \sum_{q_i \sim R_i \ (i>j)} (L/M)^{d_j + 1} \prod_{i>j} q_i^{d_i}$$

since $L/M \gg 1$. This is $\ll (L/m)^{d_j+1} \prod_{i>j} R_i^{d_i - d_j}$, which is satisfactory. □

2.5. **Oscillatory integrals.** We proceed by looking at cubic weighted exponential integrals. Let $w : \mathbb{R}^n \to \mathbb{R}_{\geqslant 0}$ be a smooth weight function of compact support. Let $P \geqslant 1$ and consider the exponential integral

$$I(\theta; \boldsymbol{\beta}) = \int w(\boldsymbol{x}/P) e(\theta F(\boldsymbol{x}) - \boldsymbol{\beta}.\boldsymbol{x}) \mathrm{d}\boldsymbol{x}. \quad (2.4)$$

for arbitrary $\theta \in \mathbb{R}$ and $\boldsymbol{\beta} \in \mathbb{R}^n$. We extract the following result from [BHB09].

Lemma 2.10. *Let $w(\boldsymbol{x}) = \omega((\boldsymbol{x}_0 - \boldsymbol{x})/\ell)$, where $\omega \in C_c^{\infty}(\mathbb{R}^n)$ is supported on $[-1, 1]^n$, and $\ell > 0$ and $\boldsymbol{x}_0 \in \mathbb{R}^n$ are fixed. Let $\varepsilon > 0$ and let $N \geqslant 1$. Assume $1 \leqslant q \leqslant P^2$ and $|\theta| \leqslant q^{-1} P^{-1}$. Then we have*

$$I(\theta; q^{-1}\boldsymbol{v}) \ll_{F, \varepsilon, N} |\boldsymbol{v}|^{-N},$$

if $|\boldsymbol{v}| \gg P^3$. Alternatively, when $|\boldsymbol{v}| \ll P^3$, we have

$$I(\theta; q^{-1}\boldsymbol{v}) \ll_{F, \varepsilon, N} P^{-N} + \mathrm{Vol}\left\{ \boldsymbol{x} \in \mathbb{R}^n : |\boldsymbol{x}/P - \boldsymbol{x}_0| \leqslant 2\ell, \ |q\theta \nabla F(\boldsymbol{x}) - \boldsymbol{v}| \leqslant P^{\varepsilon} V_0 \right\},$$

where $V_0 = qP^{-1} \max\{1, \sqrt{|\theta| P^3}\}$. Finally, we have

$$I(\theta; q^{-1}\boldsymbol{v}) \ll_{F, \varepsilon, N} P^{-N},$$

if $|\boldsymbol{v}| \geqslant P^{2\varepsilon} q P^{-1} \max\{1, |\theta| P^3\}$. The implied constants above may depend on ω, ℓ and $\boldsymbol{x}_0$ in addition to F, ε, N.

Proof. We may assume that P is sufficiently large, the bounded P case following from the trivial bound and integration by parts. The first two parts directly follow from taking $g = F$ and $H = O(1)$ in [BHB09, Lemma 9 and its proof]. (Although [BHB09, Lemma 9] is written without the condition $|\boldsymbol{x}/P - \boldsymbol{x}_0| \leqslant 2\ell$, it is clear from the proof that we may include it, since $w(\boldsymbol{x}/P)$ is supported on the strictly smaller box $|\boldsymbol{x}/P - \boldsymbol{x}_0| \leqslant \ell$.) For the final part, it suffices to show that the volume is over the empty set in the second estimate for $I(\theta; q^{-1}\boldsymbol{v})$, if $|\boldsymbol{v}| \geqslant P^{2\varepsilon} q P^{-1} \max\{1, |\theta| P^3\}$. Note that $q\theta \nabla F(\boldsymbol{x}) \ll q|\theta|P^2$ if $|\boldsymbol{x}/P - \boldsymbol{x}_0| \leqslant 2\ell$. If $|\theta|P^3 < 1$, then the claim is obvious. If $|\theta|P^3 \geqslant 1$, then the claim is also clear, since then $|\boldsymbol{v}| \geqslant P^{2\varepsilon} q P^{-1} \cdot |\theta|P^3 = q|\theta|P^{2+2\varepsilon} \gg q|\theta|P^2$, which would imply the contradictory upper bound $|\boldsymbol{v}| \ll P^{\varepsilon} V_0 = P^{\varepsilon} q P^{-1} \sqrt{|\theta|P^3}$. □

3. Geometry of cubic forms

In this section we collect some facts about the geometry of cubic forms over $\mathbb{Q}$. In later sections, we will be able to spread out much information from $\mathbb{Q}$ to finite fields $\mathbb{F}_p$, for all but finitely many primes p. Let us now introduce the main objects of interest.

Let $F \in \mathbb{Q}[x_1, \dots, x_n]$ be a nonzero cubic form in $n \geqslant 2$ variables. We make no further assumptions about F, except where specified otherwise. This level of generality will be useful for a slicing argument in §4. Let

$$H(\boldsymbol{x}) \in \mathrm{Mat}_{n\times n}(\mathbb{Q}[x_1, \dots, x_n])$$

be the Hessian matrix of F, as in §1, formed from the second partial derivatives of F. This gives a vector-valued bilinear form $H(\boldsymbol{x})\boldsymbol{y} = H(\boldsymbol{y})\boldsymbol{x}$ on $\mathbb{Q}^n \times \mathbb{Q}^n$.

Let $X \subset \mathbb{P}^{n-1}$ be the closed subscheme $F = 0$ over $\overline{\mathbb{Q}}$. In this section, we work with schemes in order to avoid edge cases concerning the definition of $\mathrm{Sing}\, X$, and we work over $\overline{\mathbb{Q}}$ in order to avoid cumbersome base-change notation. Thus X is a $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-stable scheme over $\overline{\mathbb{Q}}$, and $X(\mathbb{Q})$ consists of the $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-invariant $\overline{\mathbb{Q}}$-points of X. Let

$$I := \{(\boldsymbol{x}, \boldsymbol{y}) \in \mathbb{A}^n \times \mathbb{A}^n \colon H(\boldsymbol{x})\boldsymbol{y} = 0\} \tag{3.1}$$

be the Hessian incidence correspondence; its scheme structure is of no matter to us. On the other hand, given an integer $0 \leqslant r \leqslant n$, define

$$V_r := \{\boldsymbol{x} \in \mathbb{A}^n \colon \operatorname{rank} H(\boldsymbol{x}) \leqslant r\}, \tag{3.2}$$

with the *reduced* scheme structure. The tangent space $T_{\boldsymbol{x}}V_r$ at a point $\boldsymbol{x} \in V_r$ is thus defined in terms of the *radical* of the determinantal ideal associated to V_r. Explicitly, $T_{\boldsymbol{x}}V_r$ may be identified with the set of $\overline{\mathbb{Q}}[\varepsilon]/(\varepsilon^2)$-valued points of V_r whose reduction modulo ε is $\boldsymbol{x}$.

We define $r(F)$ to be the maximum of the rank of $H(\boldsymbol{x})$ as $\boldsymbol{x}$ ranges over $X(\overline{\mathbb{Q}})$. Building on work of Zak [Zak04], we will show in Lemma 3.4 that $r(F) \geqslant 2\lceil n/3 \rceil$ if $X(\mathbb{Q}) = \emptyset$. Indeed, some results in this section will only be proven under the assumption $X(\mathbb{Q}) = \emptyset$, in which case this assumption will always be clearly stated.

The next lemma is a geometric reformulation of a key identity [BDE12, §3, equation (3.3)] from Davenport's bad-form analysis [Dav63, proof of Lemma 3]. This formulation has been observed before by Rydin Myerson [RM19, §1.4]. For completeness, we supply a proof.

Lemma 3.1. *Let $0 \leqslant r \leqslant n$. Put $V_{-1} := \emptyset$. If $\boldsymbol{x} \in V_r \setminus V_{r-1}$, $\boldsymbol{v} \in T_{\boldsymbol{x}}V_r$, and $\boldsymbol{y}, \boldsymbol{z} \in \ker H(\boldsymbol{x})$, then*

$$\boldsymbol{y}^t H(\boldsymbol{v})\boldsymbol{z} = 0.$$

Proof. If $\boldsymbol{y} = \boldsymbol{0}$ or $\boldsymbol{z} = \boldsymbol{0}$ the result is trivial, so assume $\boldsymbol{y}, \boldsymbol{z} \neq \boldsymbol{0}$. In particular, $r < n$, since $\boldsymbol{y}, \boldsymbol{z} \in \ker H(\boldsymbol{x})$. If $r = 0$, then V_0 is a linear space, so $\boldsymbol{v} \in V_0$, whence $H(\boldsymbol{v}) = 0$ and again the result is trivial.

Suppose $0 < r < n$. Let $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$ be the standard basis vectors of $\overline{\mathbb{Q}}^n$. After a linear change of variables in the cubic form $F(\boldsymbol{x})$, which induces corresponding changes of variables on $\boldsymbol{v}, \boldsymbol{y}, \boldsymbol{z}$, we may assume that $\boldsymbol{y} \in \overline{\mathbb{Q}}^* \boldsymbol{e}_i$ and $\boldsymbol{z} \in \overline{\mathbb{Q}}^* \boldsymbol{e}_j$ for some $1 \leqslant i, j \leqslant n$, possibly with $i = j$. After re-scaling $\boldsymbol{y}, \boldsymbol{z}$, we may assume $\boldsymbol{y} = \boldsymbol{e}_i$ and $\boldsymbol{z} = \boldsymbol{e}_j$. Choose row and column sets I, J with

$|I| = |J| = r$ such that the sub-matrix $M := H(\boldsymbol{x})_{I,J}$ of $H(\boldsymbol{x})$, drawn from rows I and columns J, is invertible. Define the matrix

$$G(\boldsymbol{w}) := \begin{pmatrix} H(\boldsymbol{w})_{I,J} & H(\boldsymbol{w})_{I,\{j\}} \\ H(\boldsymbol{w})_{\{i\},J} & H(\boldsymbol{w})_{\{i\},\{j\}} \end{pmatrix},$$

for any $\boldsymbol{w} \in \overline{\mathbb{Q}}^n$. If $i \in I$ or $j \in J$, then $G(\boldsymbol{w})$ has a repeated row or column, respectively, so $\det G(\boldsymbol{w}) = 0$. Otherwise, if $i \notin I$ and $j \notin J$, then $G(\boldsymbol{w})$ is an $(r+1) \times (r+1)$ sub-matrix of $H(\boldsymbol{w})$, so $\det G(\boldsymbol{w}) = 0$ for all $\boldsymbol{w} \in V_r$. In either case, $\det G$ vanishes on V_r.

At $\boldsymbol{w} = \boldsymbol{x}$, the last row and column of $G(\boldsymbol{w})$ vanish. Indeed, $H(\boldsymbol{x})_{I,\{j\}}$ and $H(\boldsymbol{x})_{\{i\},J}$ are sub-matrices of $H(\boldsymbol{x})\boldsymbol{e}_j = H(\boldsymbol{x})\boldsymbol{z} = \boldsymbol{0}$ and $\boldsymbol{e}_i^t H(\boldsymbol{x}) = \boldsymbol{y}^t H(\boldsymbol{x}) = \boldsymbol{0}^t$, and $H(\boldsymbol{x})_{\{i\},\{j\}} = \boldsymbol{y}^t H(\boldsymbol{x})\boldsymbol{z} = 0$. Consequently, using the linearity of H,

$$\det G(\boldsymbol{x} + \varepsilon\boldsymbol{v}) = \det \begin{pmatrix} M + \varepsilon H(\boldsymbol{v})_{I,J} & \varepsilon H(\boldsymbol{v})_{I,\{j\}} \\ \varepsilon H(\boldsymbol{v})_{\{i\},J} & \varepsilon \boldsymbol{y}^t H(\boldsymbol{v})\boldsymbol{z} \end{pmatrix} \equiv \varepsilon \det(M)\, \boldsymbol{y}^t H(\boldsymbol{v})\boldsymbol{z} \bmod \varepsilon^2$$

by Laplace expansion along the last row of $G(\boldsymbol{x} + \varepsilon\boldsymbol{v})$. Since $\det G|_{V_r} = 0$ and $\boldsymbol{v} \in T_{\boldsymbol{x}} V_r$, the coefficient of ε in $\det G(\boldsymbol{x} + \varepsilon\boldsymbol{v})$ is zero. Lemma 3.1 follows, since $\det(M) \neq 0$. □

Remark 3.2. Let us briefly sketch a more conceptual, but less elementary, proof of Lemma 3.1. By general machinery or explicit calculation, the vector spaces $\ker H(\boldsymbol{x})$ form a vector bundle over $\boldsymbol{x} \in V_r \setminus V_{r-1}$. Extend $\boldsymbol{z}$ to a local section $\boldsymbol{\zeta}(\boldsymbol{w})$ for $\boldsymbol{w}$ near $\boldsymbol{x}$, with $H(\boldsymbol{w})\boldsymbol{\zeta}(\boldsymbol{w}) = \boldsymbol{0}$ and $\boldsymbol{\zeta}(\boldsymbol{x}) = \boldsymbol{z}$. Kähler differentiation at $\boldsymbol{x}$ in the direction $\boldsymbol{v}$, using the product rule, gives $H(\boldsymbol{v})\boldsymbol{z} + H(\boldsymbol{x})d\boldsymbol{\zeta}_{\boldsymbol{x}}(\boldsymbol{v}) = \boldsymbol{0}$. Multiplying by $\boldsymbol{y}^t$ gives $\boldsymbol{y}^t H(\boldsymbol{v})\boldsymbol{z} = 0$, since $\boldsymbol{y}^t H(\boldsymbol{x}) = 0$.

The following definition formalises a notion that we already met in the introduction.

Definition 3.3. We say that the cubic form F is *good* if for any $H > 1$ and any $\varepsilon > 0$ we have the upper bound

$$\#\{\boldsymbol{z} \in \mathbb{Z}^n : |\boldsymbol{z}| \leqslant H,\ \operatorname{rank}(H(\boldsymbol{z})) = r\} \ll_{F,\varepsilon} H^{r+\varepsilon},$$

for each $0 \leqslant r \leqslant n$, where the implied constant only depends on F and ε.

It follows from work of Davenport [Dav63] that if $X(\mathbb{Q}) = \emptyset$, then F is good. This implication is valid for any $n \geqslant 2$. A similar idea to Davenport's leads to the following result.

Lemma 3.4. *Assume $X(\mathbb{Q}) = \emptyset$. If $n \geqslant 2$ and F is irreducible, then $r(F) \geqslant 2\lceil n/3 \rceil$.*

Proof. First, it follows from Zak [Zak04, Proposition 4.2 and Corollary 4.4] (with $N' = n-1$) that

$$\det H \equiv 0 \bmod F^{n-r(F)}. \tag{3.3}$$

For completeness, let us unravel Zak's proof of (3.3). If $r(F) = n$, then (3.3) is trivial, so assume $r(F) < n$. By definition of $r(F)$, every $(r(F)+1) \times (r(F)+1)$ minor of H is divisible by F. The divisibility (3.3) then follows from [Zak04, Proposition 4.3], an elementary result on matrices, applied to the matrix H over the integral domain $A = \mathbb{Q}[\boldsymbol{x}]$, with principal prime ideal $\mathfrak{p} = (F)$.

Since $\det H \neq 0$ by goodness (Definition 3.3), comparing degrees in (3.3) gives $n \geqslant 3(n - r(F))$ and hence $r(F) \geqslant 2n/3$. If $n \equiv 0 \bmod 3$ then $r(F) \geqslant 2n/3 = 2\lceil n/3 \rceil$, and if $n \equiv 2 \bmod 3$ then $r(F) \geqslant \lceil 2n/3 \rceil = (2n+2)/3 = 2\lceil n/3 \rceil$. It remains to prove Lemma 3.4 when $n \equiv 1 \bmod 3$. In this case, $r(F) \geqslant \lceil 2n/3 \rceil = (2n+1)/3$, whereas $2\lceil n/3 \rceil = (2n+4)/3$. Assume for contradiction that $r(F) = (2n+1)/3$. Then $n - r(F) = (n-1)/3$, so (3.3) gives

$$\det H = F^{(n-1)/3} L$$

for some nonzero linear form $L \in \mathbb{Q}[\boldsymbol{x}]$, since $\deg(\det H) = n$. Let

$$M := \{\boldsymbol{x} \in \mathbb{A}^n : L(\boldsymbol{x}) = 0\} \cong \mathbb{A}^{n-1}.$$

Then $\dim M = n - 1 \geqslant 1$, so $M(\mathbb{Q}) \setminus \{\boldsymbol{0}\}$ is Zariski dense in M. If M is contained in V_s but not in V_{s-1}, then it follows that $M(\mathbb{Q}) \setminus \{\boldsymbol{0}\}$ is contained in V_s but not in V_{s-1}. Here $s < n$, since $M \subseteq \{\boldsymbol{x} \in \mathbb{A}^n : \det H(\boldsymbol{x}) = 0\}$. Choose a vector $\boldsymbol{x} \in M(\mathbb{Q}) \setminus \{\boldsymbol{0}\}$ with $\boldsymbol{x} \in V_s - V_{s-1}$. Since $s < n$, there exists a vector $\boldsymbol{y} \in \mathbb{Q}^n \setminus \{\boldsymbol{0}\}$ with $H(\boldsymbol{x})\boldsymbol{y} = \boldsymbol{0}$. Then $H(\boldsymbol{y})\boldsymbol{x} = H(\boldsymbol{x})\boldsymbol{y} = \boldsymbol{0}$, whence $\det H(\boldsymbol{y}) = 0$. But $F(\boldsymbol{y}) \neq 0$, since $X(\mathbb{Q}) = \emptyset$. Hence $L(\boldsymbol{y}) = 0$, so $\boldsymbol{y} \in M$.

Since M is a linear space contained in V_s, we have $M = T_{\boldsymbol{x}}M \subseteq T_{\boldsymbol{x}}V_s$. Lemma 3.1, with $(s, \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{y}, \boldsymbol{y})$ in place of $(r, \boldsymbol{x}, \boldsymbol{v}, \boldsymbol{y}, \boldsymbol{z})$, gives $0 = \boldsymbol{y}^t H(\boldsymbol{y})\boldsymbol{y} = 6F(\boldsymbol{y})$. This contradicts $F(\boldsymbol{y}) \neq 0$. It follows that $r(F) > (2n+1)/3$, which implies Lemma 3.4. □

Remark 3.5. As stated, [Zak04, Proposition 4.3] is false for general integral domains A and prime ideals $\mathfrak{p}$. Its proof relies on the identity $v_{\mathfrak{p}}((\det M)^{n-1}) = (n-1)v_{\mathfrak{p}}(\det M)$, where $v_{\mathfrak{p}}(a) := \max\{k \in \mathbb{Z}_{\geqslant 0} \cup \{\infty\} : a \in \mathfrak{p}^k\}$ for $a \in A$. This is true if $\mathfrak{p}$ is principal, for instance, but not in general.

If $X(\mathbb{Q}) = \emptyset$, we claim that $T_{\boldsymbol{x}}V_r \cap \ker H(\boldsymbol{x}) = 0$ for all $\boldsymbol{x} \in (V_r \setminus V_{r-1})(\mathbb{Q})$. Since $\boldsymbol{x}$ is rational, the intersection $T_{\boldsymbol{x}}V_r \cap \ker H(\boldsymbol{x})$ is a linear space defined over $\mathbb{Q}$. If it were nonzero, it would contain a nonzero rational vector $\boldsymbol{y}$, and Lemma 3.1 would give $6F(\boldsymbol{y}) = \boldsymbol{y}^t H(\boldsymbol{y})\boldsymbol{y} = 0$, contradicting $X(\mathbb{Q}) = \emptyset$. In this way, Davenport obtained a strong restriction on $T_{\boldsymbol{x}}V_r$ at rational points $\boldsymbol{x}$. However, it turns out that one can also derive useful geometric information at irrational points $\boldsymbol{x}$. Lemma 3.1 constrains the linear equivalence class of F over $\overline{\mathbb{Q}}$, giving a normal form in which many coefficients vanish. On the other hand, if too many coefficients vanish, then one might expect X to have a $\mathbb{Q}$-point for relatively simple reasons, generalising the familiar fact that if F is reducible over $\overline{\mathbb{Q}}$ then X must have a $\mathbb{Q}$-point if $n \geqslant 4$ (as in Proposition 2.3). The combinatorics of which coefficients can vanish, while keeping $X(\mathbb{Q}) = \emptyset$, may be studied through the lens of weighted linear optimisation. This is the idea behind Lemma 3.6, which we now state.

Lemma 3.6. *Assume that $X(\mathbb{Q}) = \emptyset$. Let $A \in \mathrm{GL}_n(\overline{\mathbb{Q}})$, and assign a real weight w_i to each variable x_i. The weight of a cubic monomial $x_ix_jx_k$ is defined to be $w_i + w_j + w_k$. If every monomial occurring with a nonzero coefficient in $F(A\boldsymbol{x})$ has nonnegative weight, then*

$$w_1 + \cdots + w_n \geqslant 0. \tag{3.4}$$

Proof. First suppose $A \in \mathrm{GL}_n(\mathbb{Q})$. Then $A\boldsymbol{e}_i \in \mathbb{Q}^n \setminus \{\boldsymbol{0}\}$ for each standard basis vector $\boldsymbol{e}_i \in \mathbb{Q}^n$. Thus $F(A\boldsymbol{e}_i) \neq 0$, since $X(\mathbb{Q}) = \emptyset$. But $F(A\boldsymbol{e}_i)$ is precisely the coefficient of x_i^3 in $F(A\boldsymbol{x})$, which has weight $3w_i \geqslant 0$ for all i, by assumption. Summing over i leads to (3.4) in this case. It remains to reduce the general case to the case $A \in \mathrm{GL}_n(\mathbb{Q})$, by a careful Galois descent procedure.

Let I be a subset of the set of all monomials $x_ix_jx_k$, where $1 \leqslant i, j, k \leqslant n$. (After reordering i, j, k we could assume $i \leqslant j \leqslant k$, but that would cause confusion below.) Let $W \in \mathbb{R}$. Given I, let S_I be the set of matrices $A \in \mathrm{GL}_n(\overline{\mathbb{Q}})$ for which every monomial appearing in $F(A\boldsymbol{x})$ lies in the set I. Given I and W, let $T_{I,W}$ be the set of vectors $\boldsymbol{v} \in \mathbb{R}^n$ for which the following hold:

(1) Every monomial $x_ix_jx_k \in I$ satisfies $v_i + v_j + v_k \geqslant 0$.
(2) We have $\sum_{1 \leqslant i \leqslant n} v_i = W$.

Letting I range over all possible subsets of monomials, define the (not necessarily disjoint) union

$$\mathcal{S}_W := \bigcup_I (S_I \times T_{I,W}).$$

Every pair $(A, \boldsymbol{w})$ in the statement of Lemma 3.6 lies in $\mathcal{S}_W$ for some choice of W. We then wish to show that $W \geqslant 0$. Note however that in the definition of S_I, the monomial support of $F(A\boldsymbol{x})$ is only required to be a subset of I, not necessarily equal to I.

Two general symmetries will be useful. First, if $(A, \boldsymbol{v}) \in S_I \times T_{I,W}$, then

$$(AP, P^{-1}\boldsymbol{v}) \in S_{I \circ P} \times T_{I \circ P, W}$$

for every permutation matrix P, where $I \circ P := \{m \circ P : m \in I\}$. Indeed, if the $(i, \pi(i))$ entry of P is 1, where π is a permutation, then $(P\boldsymbol{x})_i = x_{\pi(i)}$ and $(P^{-1}\boldsymbol{v})_{\pi(i)} = v_i$, and a monomial $m = x_ix_jx_k$ in $F(A\boldsymbol{x})$ is transformed to $m \circ P = x_{\pi(i)}x_{\pi(j)}x_{\pi(k)}$ in $F(AP\boldsymbol{x})$, so that

$$(P^{-1}\boldsymbol{v})_{\pi(i)} + (P^{-1}\boldsymbol{v})_{\pi(j)} + (P^{-1}\boldsymbol{v})_{\pi(k)} = v_i + v_j + v_k \geqslant 0.$$

Second, let $\mathcal{B}(\boldsymbol{v})$ be the set of matrices $U = (u_{ij}) \in \mathrm{GL}_n(\overline{\mathbb{Q}})$ such that $u_{ij} \neq 0 \Rightarrow v_i \leqslant v_j$. For example, if $v_1 \leqslant v_2 \leqslant \cdots \leqslant v_n$, then any invertible upper triangular matrix lies in $\mathcal{B}(\boldsymbol{v})$, and if

$v_1 < v_2 < \cdots < v_n$ then every matrix in $\mathcal{B}(\boldsymbol{v})$ is upper triangular. We claim that if $U \in \mathcal{B}(\boldsymbol{v})$ and J denotes the monomial support of $F(AU\boldsymbol{x})$, then

$$(AU, \boldsymbol{v}) \in S_J \times T_{J,W}.$$

Indeed, if $x_i x_j x_k \in I$, then every monomial appearing in the transformed polynomial

$$(x_i x_j x_k) \circ U = (U\boldsymbol{x})_i (U\boldsymbol{x})_j (U\boldsymbol{x})_k$$

may be written in the form $x_{i'} x_{j'} x_{k'}$, where $v_i \leqslant v_{i'}$, $v_j \leqslant v_{j'}$ and $v_k \leqslant v_{k'}$, by the definition of $\mathcal{B}(\boldsymbol{v})$. It follows that $v_{i'} + v_{j'} + v_{k'} \geqslant v_i + v_j + v_k \geqslant 0$, as desired.

With these two general symmetries established, suppose now that $(A, \boldsymbol{v}), (A', \boldsymbol{v}') \in S_I \times T_{I,W}$. Then there exist $U \in \mathcal{B}(\boldsymbol{v})$, $U' \in \mathcal{B}(\boldsymbol{v}')$ and a permutation matrix P such that $AUP = A'U'$. Indeed, first order the coordinates for $\boldsymbol{v}$ and $\boldsymbol{v}'$ separately so that their entries are non-decreasing. In these orderings, both $\mathcal{B}(\boldsymbol{v})$ and $\mathcal{B}(\boldsymbol{v}')$ contain the group of invertible upper triangular matrices. The usual *Bruhat decomposition* therefore gives

$$A^{-1}A' = UPV,$$

with $U \in \mathcal{B}(\boldsymbol{v})$, $V \in \mathcal{B}(\boldsymbol{v}')$ and P a permutation matrix. This remains a decomposition of the same form after undoing the coordinate permutations. Taking $U' = V^{-1}$ gives $AUP = A'U'$.

Let $A'' := AUP = A'U'$. By the symmetries in the penultimate paragraph, $(A'', P^{-1}\boldsymbol{v}) = (AUP, P^{-1}\boldsymbol{v}) \in \mathcal{S}_W$ and $(A'', \boldsymbol{v}') = (A'U', \boldsymbol{v}') \in \mathcal{S}_W$. If I'' denotes the monomial support of $F(A''\boldsymbol{x})$, then

$$(A'', \tfrac{1}{2}(P^{-1}\boldsymbol{v} + \boldsymbol{v}')) \in S_{I''} \times T_{I'',W} \subseteq \mathcal{S}_W,$$

since $T_{I'',W}$ is convex. Moreover, by the parallelogram law and the identity $\|P^{-1}\boldsymbol{v}\|_2 = \|\boldsymbol{v}\|_2$,

$$\|\tfrac{1}{2}(P^{-1}\boldsymbol{v} + \boldsymbol{v}')\|_2^2 = \tfrac{1}{2}(\|P^{-1}\boldsymbol{v}\|_2^2 + \|\boldsymbol{v}'\|_2^2) - \|\tfrac{1}{2}(P^{-1}\boldsymbol{v} - \boldsymbol{v}')\|_2^2 \leqslant \tfrac{1}{2}(\|\boldsymbol{v}\|_2^2 + \|\boldsymbol{v}'\|_2^2),$$

with equality if and only if $P^{-1}\boldsymbol{v} = \boldsymbol{v}'$.

Each set $T_{I,W}$ is closed in $\mathbb{R}^n$, and thus if $T_{I,W} \neq \emptyset$ then $T_{I,W}$ contains a vector of minimal Euclidean length. Since there are only finitely many possible sets I, it follows that there exists $(A, \boldsymbol{v}) \in \mathcal{S}_W$ for which $\|\boldsymbol{v}\|_2$ is minimal. Suppose $(A, \boldsymbol{v}) \in S_I \times T_{I,W}$. Then for any $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, we have $(\sigma(A), \boldsymbol{v}) \in S_I \times T_{I,W}$. Taking $(\sigma(A), \boldsymbol{v})$ for $(A', \boldsymbol{v}')$ in the previous paragraph, we get $P^{-1}\boldsymbol{v} = \boldsymbol{v}$, since $\|\frac{1}{2}(P^{-1}\boldsymbol{v} + \boldsymbol{v})\|_2 \geqslant \|\boldsymbol{v}\|_2$ by minimality of $\|\boldsymbol{v}\|_2$. Since $AUP = \sigma(A)U'$, where $U, U' \in \mathcal{B}(\boldsymbol{v})$, it follows from the definition of $\mathcal{B}(\boldsymbol{v})$ that the filtration

$$D_{\leqslant r} := \mathrm{span}_{\overline{\mathbb{Q}}}\{A\boldsymbol{e}_i : v_i \leqslant r\} \subseteq \overline{\mathbb{Q}}^n$$

is $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-stable, where r ranges over all real numbers. By Galois descent, there exists a filtration of $\mathbb{Q}$-vector spaces $E_{\leqslant r} \subseteq \mathbb{Q}^n$ such that $E_{\leqslant r} \otimes \overline{\mathbb{Q}} = D_{\leqslant r}$. Let $B \in \mathrm{GL}_n(\mathbb{Q})$ be a matrix whose columns $B\boldsymbol{e}_i$ split this filtration, so that

$$\mathrm{span}_{\mathbb{Q}}\{B\boldsymbol{e}_i : v_i \leqslant r\} = E_{\leqslant r}.$$

By the definition of $\mathcal{B}(\boldsymbol{v})$, we may thus write $B = AM$ for some $M \in \mathcal{B}(\boldsymbol{v})$. Therefore,

$$(B, \boldsymbol{v}) = (AM, \boldsymbol{v}) \in \mathcal{S}_W,$$

by the symmetry properties observed earlier. On replacing A with B, it follows that we may assume $A \in \mathrm{GL}_n(\mathbb{Q})$, whence $W \geqslant 0$, completing the proof of Lemma 3.6. □

Recall the affine incidence correspondence I from (3.1). It will be convenient to have the following concentration result for $\dim I$. We follow the standard convention that an irreducible set must be nonempty, and that the affine cone of the empty set is $\{\mathbf{0}\}$.

Lemma 3.7. *There exist an integer $0 \leqslant r \leqslant n$ and an irreducible or empty closed subset $Z \subseteq X$, whose affine cone $\widehat{Z}$ lies in V_r but not in V_{r-1}, such that $\dim I = \dim \widehat{Z} + n - r$.*

Proof. Since $\{\mathbf{0}\} \times \mathbb{A}^n \subseteq I$, we have $\dim I \geqslant n$. If $\dim I = n$, then we may take $Z = \emptyset$ and $r = 0$, in the light of our convention $\widehat{\emptyset} = \{\mathbf{0}\}$. Thus, we henceforth assume that $\dim I > n$.

Let $\boldsymbol{v} := \boldsymbol{y} + \boldsymbol{x}$ and $\boldsymbol{w} := \boldsymbol{y} - \boldsymbol{x}$. Clearly $\nabla F(\boldsymbol{v}) - \nabla F(\boldsymbol{w}) = 2H(\boldsymbol{y})\boldsymbol{x}$. Thus

$$I \cong J := \{(\boldsymbol{v}, \boldsymbol{w}) \in \mathbb{A}^{2n} : \nabla F(\boldsymbol{v}) = \nabla F(\boldsymbol{w})\},$$

by changing variables from $(\boldsymbol{x}, \boldsymbol{y})$ to $(\boldsymbol{v}, \boldsymbol{w})$, since $H(\boldsymbol{y})\boldsymbol{x} = H(\boldsymbol{x})\boldsymbol{y}$. Moreover,

$$F(\boldsymbol{v}) - F(\boldsymbol{w}) \equiv 2F(\boldsymbol{x}) \bmod H(\boldsymbol{y})\boldsymbol{x},$$

since $2\nabla F(\boldsymbol{y}) \cdot \boldsymbol{x} = \boldsymbol{y}^t H(\boldsymbol{y})\boldsymbol{x} \equiv 0 \bmod H(\boldsymbol{y})\boldsymbol{x}$ and $\boldsymbol{y} \cdot \nabla F(\boldsymbol{x}) - \boldsymbol{y} \cdot \nabla F(-\boldsymbol{x}) = 0$. Therefore,

$$I|_{\widehat{X} \times \mathbb{A}^n} \cong J_0 := \{(\boldsymbol{v}, \boldsymbol{w}) \in J : F(\boldsymbol{v}) = F(\boldsymbol{w})\}.$$

We claim that $\dim J_0 = \dim J$. For $\boldsymbol{z} \in \overline{\mathbb{Q}}^n$, put

$$Y_{\boldsymbol{z}} := \{\boldsymbol{x} \in \mathbb{A}^n : \nabla F(\boldsymbol{x}) = \boldsymbol{z}\}.$$

We claim that F is constant on each irreducible component C of $Y_{\boldsymbol{z}}$. Work over $\mathbb{C}$, and let $\boldsymbol{x}(t)$ be a smooth path in the smooth locus of C. Since $\nabla F(\boldsymbol{x}(t)) = \boldsymbol{z}$, Euler's identity gives

$$3F(\boldsymbol{x}(t)) = \boldsymbol{z} \cdot \boldsymbol{x}(t).$$

Differentiating and using the chain rule, we obtain

$$3\frac{\mathrm{d}}{\mathrm{d}t}F(\boldsymbol{x}(t)) = \boldsymbol{z} \cdot \boldsymbol{x}'(t) = \nabla F(\boldsymbol{x}(t)) \cdot \boldsymbol{x}'(t) = \frac{\mathrm{d}}{\mathrm{d}t}F(\boldsymbol{x}(t)).$$

Thus F is constant along every such path, and hence on a small connected analytic neighbourhood of a smooth point of C. Since this neighbourhood is Zariski dense in C, the claim follows. Let $\pi : J \to \mathbb{A}^n$ send $(\boldsymbol{v}, \boldsymbol{w})$ to their common gradient. For each nonempty $Y_{\boldsymbol{z}}$, choose an irreducible component C with $\dim C = \dim Y_{\boldsymbol{z}}$. Since F is constant on C, every pair in $C \times C$ belongs to $(J_0)_{\boldsymbol{z}}$. Hence

$$2\dim Y_{\boldsymbol{z}} = \dim(C \times C) \leqslant \dim(J_0)_{\boldsymbol{z}} \leqslant \dim J_{\boldsymbol{z}} = 2\dim Y_{\boldsymbol{z}}.$$

Thus the two fibres, $(J_0)_{\boldsymbol{z}}$ and $J_{\boldsymbol{z}}$, have the same dimension. Put $S_d := \{\boldsymbol{z} \in \mathbb{A}^n : \dim Y_{\boldsymbol{z}} = d\}$, for $0 \leqslant d \leqslant n$. Each S_d is constructible and over every nonempty S_d, both projections have fibres of dimension $2d$. The sets $\pi^{-1}(S_d)$ form a finite constructible partition of J, and their intersections with J_0 partition J_0. Since the dimension of a finite union is the maximum of the dimensions of its pieces, we conclude that

$$\dim J = \max_{S_d \neq \emptyset}(\dim S_d + 2d) = \dim J_0.$$

It now follows that $\dim I = \dim I|_{\widehat{X} \times \mathbb{A}^n}$. Stratifying the projective incidence correspondence on X by rank, and taking affine cones on a top-dimensional stratum, completes the proof of Lemma 3.7. □

Lemma 3.8. *Assume $X(\mathbb{Q}) = \emptyset$. We have*

$$1 + \dim \operatorname{Sing} X \leqslant \max\{0, (2n-3)/3\}, \tag{3.5}$$

$$\dim I|_{\operatorname{Sing}(\widehat{X}) \times \mathbb{A}^n} \leqslant \max\{n, (4n-6)/3\}, \tag{3.6}$$

$$\dim I \leqslant \max\{n, (4n-3)/3\}. \tag{3.7}$$

Moreover, fix an integer $0 \leqslant r \leqslant n$ and an irreducible closed subset $Z \subseteq X$, whose affine cone $\widehat{Z}$ lies in V_r but not in V_{r-1}. If $Z \subseteq \operatorname{Sing} X$, then

$$\dim \widehat{Z} \leqslant \min\{n - r, 2r - 3\}. \tag{3.8}$$

If $Z \not\subseteq \operatorname{Sing} X$, then

$$\dim \widehat{Z} \leqslant \min\{r + (n-3)/3, 2r - 2\}. \tag{3.9}$$

Proof. The bound (3.5) is trivial if $\operatorname{Sing} X = \emptyset$, and otherwise it follows from (3.8), since

$$\min\{n-r, 2r-3\} \leqslant \tfrac{2}{3}(n-r) + \tfrac{1}{3}(2r-3) = \tfrac{1}{3}(2n-3).$$

The bound (3.6) is trivial if $\dim I \leqslant n$, and otherwise it follows from (3.8), since

$$\min\{n-r, 2r-3\} + n - r \leqslant \tfrac{1}{3}(n-r) + \tfrac{2}{3}(2r-3) + n - r = \tfrac{1}{3}(4n-6).$$

The bound (3.7) is trivial if $\dim I \leqslant n$, and otherwise it follows from Lemma 3.7, together with (3.6) and the first bound in (3.9), since $r + (n-3)/3 + n - r = (4n-3)/3$.

It remains to prove (3.8) and (3.9). Write $\Phi(\boldsymbol{y}, \boldsymbol{v}, \boldsymbol{z}) := \boldsymbol{y}^t H(\boldsymbol{v})\boldsymbol{z}$ for the symmetric trilinear form associated to F. Choose a nonzero smooth point $\boldsymbol{x}$ of $\widehat{Z}$ with $\operatorname{rank} H(\boldsymbol{x}) = r$, and with $\nabla F(\boldsymbol{x}) \neq \boldsymbol{0}$ if $Z \not\subseteq \operatorname{Sing} X$. Set

$$T := T_{\boldsymbol{x}}(\widehat{Z}), \qquad L := \ker H(\boldsymbol{x}).$$

Lemma 3.1 gives $\Phi(L, T, L) = 0$. Also, $\Phi(\overline{\mathbb{Q}}^n, \boldsymbol{x}, L) = 0$, because $H(\boldsymbol{x})L = \boldsymbol{0}$. Finally, since $F|_{\widehat{Z}} = 0$, we have $F(\boldsymbol{x} + \varepsilon\boldsymbol{v}) \equiv 0 \bmod \varepsilon^2$ for all $\boldsymbol{v} \in T$, so $\boldsymbol{v}^t \nabla F(\boldsymbol{x}) = 0$. Thus $\Phi(T, \boldsymbol{x}, \boldsymbol{x}) = 0$, in view of the identity $H(\boldsymbol{x})\boldsymbol{x} = 2\nabla F(\boldsymbol{x})$.

Before proceeding, we collect some basic facts about T and L. Since $\widehat{Z}$ is cut out by homogeneous equations, Euler's identity gives $\boldsymbol{x} \in T$. By smoothness of $\widehat{Z}$ at $\boldsymbol{x}$, we have $\dim T = \dim \widehat{Z}$. Since $\operatorname{rank} H(\boldsymbol{x}) = r$, we have $\dim L = n - r$.

Suppose first that $Z \subseteq \operatorname{Sing} X$. Then $\nabla F(\boldsymbol{x} + \varepsilon\boldsymbol{v}) \equiv 0 \bmod \varepsilon^2$ for all $\boldsymbol{v} \in T$, so $H(\boldsymbol{x})\boldsymbol{v} = \boldsymbol{0}$. Thus $T \subseteq L$, so $\dim \widehat{Z} \leqslant n - r$. It remains to prove that $\dim \widehat{Z} \leqslant 2r - 3$. Split the flag $\overline{\mathbb{Q}}\boldsymbol{x} \subseteq T \subseteq L \subseteq \overline{\mathbb{Q}}^n$, writing

$$W_{-4} := \overline{\mathbb{Q}}\boldsymbol{x}, \qquad T = W_{-4} \oplus W_{-1}, \qquad L = T \oplus W_0, \qquad \overline{\mathbb{Q}}^n = L \oplus W_2,$$

so that $\overline{\mathbb{Q}}^n = W_{-4} \oplus W_{-1} \oplus W_0 \oplus W_2$. Let $A \in \mathrm{GL}_n(\overline{\mathbb{Q}})$ be a matrix such that

$$\{A\boldsymbol{e}_1, \ldots, A\boldsymbol{e}_n\} \subseteq W_{-4} \cup W_{-1} \cup W_0 \cup W_2.$$

Let $w_i := \nu$ if $A\boldsymbol{e}_i \in W_\nu$, where $1 \leqslant i \leqslant n$ and $\nu \in \{-4, -1, 0, 2\}$. Let $1 \leqslant i, j, k \leqslant n$ with $w_i \leqslant w_j \leqslant w_k$ and suppose $\Phi(A\boldsymbol{e}_i, A\boldsymbol{e}_j, A\boldsymbol{e}_k) \neq 0$. If $w_i = -4$, then $w_j = w_k = 2$, since $\Phi(\boldsymbol{x}, L, \overline{\mathbb{Q}}^n) = 0$ and $L = W_{-4} \oplus W_{-1} \oplus W_0$. Thus $w_i + w_j + w_k = -4 + 2 + 2 = 0$ in this case. If $w_i = -1$, then $w_k = 2$, since $\Phi(T, L, L) = 0$. Thus $w_i + w_j + w_k \geqslant -1 - 1 + 2 = 0$ in this case. If $w_i \geqslant 0$, then $w_i + w_j + w_k \geqslant 0$ trivially. Lemma 3.6 therefore gives

$$-4 - \dim W_{-1} + 2\dim W_2 \geqslant 0.$$

Since $\dim W_{-1} = \dim \widehat{Z} - 1$ and $\dim W_2 = r$, we get $\dim \widehat{Z} \leqslant 2r - 3$.

Suppose now that $Z \not\subseteq \operatorname{Sing} X$. Since $H(\boldsymbol{x})\boldsymbol{x} = 2\nabla F(\boldsymbol{x}) \neq \boldsymbol{0}$, we have $\boldsymbol{x} \notin L$. Split the flag $(T \cap L) \oplus \overline{\mathbb{Q}}\boldsymbol{x} \subseteq T \subseteq T + L \subseteq \overline{\mathbb{Q}}^n$, writing

$$W_{-2} := (T \cap L) \oplus \overline{\mathbb{Q}}\boldsymbol{x}, \qquad T = W_{-2} \oplus W_1, \qquad L = (T \cap L) \oplus W_0, \qquad \overline{\mathbb{Q}}^n = (T + L) \oplus W_4,$$

so that $T + L = T \oplus W_0$. By trilinearity of Φ, we have $\Phi(W_{-2}, L + \overline{\mathbb{Q}}\boldsymbol{x}, T + L) = 0$ since

(1) $\Phi(T \cap L, L, T) \subseteq \Phi(L, L, T) = 0$,
(2) $\Phi(T \cap L, L, L) \subseteq \Phi(T, L, L) = 0$,
(3) $\Phi(T \cap L, \boldsymbol{x}, T + L) \subseteq \Phi(L, \boldsymbol{x}, \overline{\mathbb{Q}}^n) = 0$,
(4) $\Phi(\boldsymbol{x}, L, T + L) \subseteq \Phi(\boldsymbol{x}, L, \overline{\mathbb{Q}}^n) = 0$,
(5) $\Phi(\boldsymbol{x}, \boldsymbol{x}, T) = 0$, and
(6) $\Phi(\boldsymbol{x}, \boldsymbol{x}, L) \subseteq \Phi(\boldsymbol{x}, \overline{\mathbb{Q}}^n, L) = 0$.

Since $L + \overline{\mathbb{Q}}\boldsymbol{x} = W_{-2} \oplus W_0$ and $T + L = W_{-2} \oplus W_0 \oplus W_1$, it follows that

$$\Phi(W_{-2}, W_{-2} \oplus W_0, W_{-2} \oplus W_0 \oplus W_1) = 0. \tag{3.10}$$

Let $A \in \mathrm{GL}_n(\overline{\mathbb{Q}})$ be a matrix such that

$$\{A\boldsymbol{e}_1, \ldots, A\boldsymbol{e}_n\} \subseteq W_{-2} \cup W_0 \cup W_1 \cup W_4.$$

Let $w_i := \nu$ if $A\boldsymbol{e}_i \in W_\nu$, where $1 \leqslant i \leqslant n$ and $\nu \in \{-2, 0, 1, 4\}$. Let $1 \leqslant i, j, k \leqslant n$ with $w_i \leqslant w_j \leqslant w_k$ and suppose $\Phi(A\boldsymbol{e}_i, A\boldsymbol{e}_j, A\boldsymbol{e}_k) \neq 0$. If $w_i = -2$ and $w_j \leqslant 0$, then $w_k = 4$ by

(3.10). Thus $w_i + w_j + w_k \geqslant -2 - 2 + 4 = 0$ in this case. If $w_i = -2$ and $w_j \geqslant 1$, then $w_i + w_j + w_k \geqslant -2 + 1 + 1 = 0$. If $w_i \geqslant 0$, then $w_i + w_j + w_k \geqslant 0$ trivially. Lemma 3.6 therefore gives

$$-2 \dim W_{-2} + \dim W_1 + 4 \dim W_4 \geqslant 0.$$

Let $a := \dim(T \cap L)$ and $t := \dim T = \dim \widehat{Z}$. Then $\dim W_{-2} = a + 1$, $\dim W_1 = t - a - 1$ and $\dim W_4 = n - \dim(T + L) = n - t - \dim L + a = r - t + a$. Thus

$$0 \leqslant -2(a + 1) + (t - a - 1) + 4(r - t + a) = -3 + a - 3t + 4r.$$

Since $\boldsymbol{x} \in T$ but $\boldsymbol{x} \notin L$, we have $a \leqslant t - 1$, so $0 \leqslant -4 - 2t + 4r$ and $t \leqslant 2r - 2$. On the other hand, $a \leqslant \dim L = n - r$, so $0 \leqslant -3 + n - 3t + 3r$ and $t \leqslant r + (n - 3)/3$. □

Corollary 3.9. *Assume that $n \geqslant 3$ and F is irreducible. Let $1 \leqslant r \leqslant n - 1$ and suppose that $X(\mathbb{Q}) = \emptyset$. Then*

$$\dim V_r \leqslant \min\left\{ \left\lfloor \frac{n}{3} \right\rfloor + r - 1, n - 1 \right\}$$

and $V_0 = V_1 \cap \widehat{X} = \{\mathbf{0}\}$. Moreover, $\dim(V_r \cap \widehat{X}) \leqslant \min\{2r - 2, n - 2\}$ if $r < 2\lceil n/3 \rceil$.

Proof. Since F is good (see Definition 3.3), the polynomial $\det(H(\boldsymbol{x}))$ does not vanish identically. In particular, if $\boldsymbol{x} \in V_r$, then $\det(H(\boldsymbol{x})) = 0$ must hold and hence $\dim V_r \leqslant n - 1$. Furthermore, V_0 is linear space defined over $\mathbb{Q}$, so $V_0 = \{\mathbf{0}\}$ by the $r = 0$ case of Definition 3.3.

Next consider the projection $\pi\colon I \to \mathbb{A}^n$ defined by $(\boldsymbol{x}, \boldsymbol{y}) \mapsto \boldsymbol{x}$. Then clearly

$$V_r = \{\boldsymbol{x} \in \mathbb{A}^n\colon \dim \pi^{-1}(\boldsymbol{x}) \geqslant n - r\}.$$

Thus $\dim V_r \leqslant \dim I - (n - r) \leqslant \lfloor n/3 \rfloor + r - 1$ by (3.1) and (3.7), since $n \leqslant (4n - 3)/3$.

The claims $\dim(V_r \cap \widehat{X}) \leqslant 2r - 2$ and $V_1 \cap \widehat{X} = \{\mathbf{0}\}$ follow from (3.8) and (3.9). The final claim follows directly from Lemma 3.4. □

It will also be useful to have some control on linear spaces in $\operatorname{Sing} X$, going beyond (3.5).

Lemma 3.10. *Assume $X(\mathbb{Q}) = \emptyset$. If ∇F vanishes on a $\overline{\mathbb{Q}}$-vector space $L \subseteq \overline{\mathbb{Q}}^n$, then*

$$\dim L \leqslant n/3.$$

Proof. Let Φ be as in the proof of Lemma 3.8. Then $\Phi(L, L, \overline{\mathbb{Q}}^n) = 0$, since

$$2H(\boldsymbol{x})\boldsymbol{y} = \nabla F(\boldsymbol{x} + \boldsymbol{y}) - \nabla F(\boldsymbol{x} - \boldsymbol{y}) = \mathbf{0} - \mathbf{0} = \mathbf{0}$$

for all $\boldsymbol{x}, \boldsymbol{y} \in L$. Write $\overline{\mathbb{Q}}^n = L \oplus M$. Choose $A \in \mathrm{GL}_n(\overline{\mathbb{Q}})$ with $\{A\boldsymbol{e}_1, \dots, A\boldsymbol{e}_n\} \subseteq L \cup M$. Let $w_i := -2$ if $A\boldsymbol{e}_i \in L$, and $w_i := 1$ if $A\boldsymbol{e}_i \in M$. Let $1 \leqslant i, j, k \leqslant n$ with $w_i \leqslant w_j \leqslant w_k$ and suppose $\Phi(A\boldsymbol{e}_i, A\boldsymbol{e}_j, A\boldsymbol{e}_k) \neq 0$. If $w_i = -2$, then $w_j = w_k = 1$, since $\Phi(L, L, \overline{\mathbb{Q}}^n) = 0$. Thus $w_i + w_j + w_k = 0$ in this case. If $w_i = 1$, then $w_i + w_j + w_k = 3$. Lemma 3.6 therefore gives $-2 \dim L + \dim M \geqslant 0$. since $\dim M = n - \dim L$, it follows that $3 \dim L \leqslant n$. □

4. Families of singular hyperplane sections

Adopt the notation of §3. Assume $n \geqslant 4$ and $X(\mathbb{Q}) = \emptyset$. By Proposition 2.3, the $\overline{\mathbb{Q}}$-variety $X \subset \mathbb{P}^{n-1}$ is integral. Write $H_{\boldsymbol{v}} := \{[\boldsymbol{x}] \in \mathbb{P}^{n-1} : \boldsymbol{v} \cdot \boldsymbol{x} = 0\}$ for $\boldsymbol{v} \neq \mathbf{0}$. For each integer $t \geqslant 0$, put

$$B_t := \{[\boldsymbol{v}] \in \mathbb{P}^{n-1} : \dim \operatorname{Sing}(X \cap H_{\boldsymbol{v}}) \geqslant t\}, \tag{4.1}$$

where $\operatorname{Sing}(X \cap H_{\boldsymbol{v}})$ is interpreted scheme-theoretically as in §3. Consider the closed incidence correspondence

$$S := \{([\boldsymbol{x}], [\boldsymbol{v}]) \in X \times \mathbb{P}^{n-1} : \boldsymbol{v} \cdot \boldsymbol{x} = 0, \quad v_i F_j(\boldsymbol{x}) - v_j F_i(\boldsymbol{x}) = 0 \text{ for all } i, j\}, \tag{4.2}$$

where $F_i = \partial F / \partial x_i$. Its fibre over $[\boldsymbol{v}]$ is $\operatorname{Sing}(X \cap H_{\boldsymbol{v}})$. Thus B_t is closed and defined over $\mathbb{Q}$, by upper semicontinuity of fibre dimension for a proper morphism. The scheme structures on B_t and S are unimportant; for instance, the reduced scheme structure would suffice.

Proposition 4.1. *Suppose $t \geqslant \max\{0, \dim \operatorname{Sing} X\}$, and let A_t be the Zariski closure of $B_t(\mathbb{Q})$ in $\mathbb{P}^{n-1}$. Then A_t is defined over $\mathbb{Q}$. Moreover, $A_t(\mathbb{Q}) = B_t(\mathbb{Q})$ and*

$$1 + \dim A_t \leqslant \max\left\{0, n - \left\lceil \frac{3(t+2)}{2} \right\rceil \right\}. \tag{4.3}$$

Proof. The set A_t is contained in the intersection of its Galois conjugates. Thus A_t is Galois-stable, and hence defined over $\mathbb{Q}$. Moreover, $A_t \subseteq B_t$, since B_t is closed. Thus $A_t(\mathbb{Q}) \subseteq B_t(\mathbb{Q})$. Since $B_t(\mathbb{Q}) \subseteq A_t(\mathbb{Q})$ by definition, it follows that $A_t(\mathbb{Q}) = B_t(\mathbb{Q})$. It remains to prove (4.3).

The set $A_t(\mathbb{Q}) = B_t(\mathbb{Q})$ is Zariski dense in A_t. Thus every nonempty open subset of A_t contains a $\mathbb{Q}$-point. This, together with (3.5), will be the main arithmetic input used below.

Let Z be an irreducible component of $(A_t)_{\mathrm{sm}}$. As in §3, we work geometrically over $\overline{\mathbb{Q}}$. Let $\pi\colon S \to \mathbb{P}^{n-1}$ be the projection $([\boldsymbol{x}], [\boldsymbol{v}]) \mapsto [\boldsymbol{v}]$. For each $\boldsymbol{v} \in Z$, we have $\dim \pi^{-1}(\boldsymbol{v}) = \dim \operatorname{Sing}(X \cap H_{\boldsymbol{v}}) \geqslant t \geqslant 0$. Since Z is irreducible, generic smoothness of the map $\pi|_{\pi^{-1}(Z)}$, for the reduced scheme structures on Z and $\pi^{-1}(Z)$, produces an irreducible open subset U of $\pi^{-1}(Z)$ for which $\pi|_U$ is smooth of relative dimension $\geqslant t$ and the image Z' of $\pi|_U$ is a dense open subset of Z. In particular, $\pi|_U$ is flat, with some constant fibre dimension $e \geqslant t$.

Let $\widehat{Z}'$ denote the affine cone of Z'. Let $\boldsymbol{v} \in \widehat{Z}' \setminus \{\mathbf{0}\}$ and $M_{\boldsymbol{v}} := (T_{\boldsymbol{v}}\widehat{Z}')^{\perp} \subseteq \mathbb{A}^n$. We claim that

$$\dim\{\boldsymbol{x} \in M_{\boldsymbol{v}} : \nabla F(\boldsymbol{x}) \in M_{\boldsymbol{v}}^{\perp}\} \geqslant e + 1. \tag{4.4}$$

For a linear subspace $L \subseteq \overline{\mathbb{Q}}^n$, we have $\frac{\mathrm{d}}{\mathrm{d}t} F(\boldsymbol{x} + t\boldsymbol{v})|_{t=0} = \boldsymbol{v} \cdot \nabla F(\boldsymbol{x})$ for all $\boldsymbol{x}, \boldsymbol{v} \in L$. If $F|_L \neq 0$, then $F|_L$ is singular at $\boldsymbol{x}$ if and only if $\nabla F(\boldsymbol{x})$ is orthogonal to L. Thus if $F|_{M_{\boldsymbol{v}}} \neq 0$, then the left-hand side of (4.4) equals $\dim \operatorname{Sing}\{\boldsymbol{x} \in M_{\boldsymbol{v}} : F(\boldsymbol{x}) = 0\}$.

Suppose first that U is not contained in $\operatorname{Sing}(X) \times Z'$. Then by shrinking U and Z' if necessary, we may assume that $U \subseteq X_{\mathrm{sm}} \times Z'$. For $([\boldsymbol{x}], [\boldsymbol{v}]) \in U$, each $\boldsymbol{w} \in T_{\boldsymbol{v}}\widehat{Z}'$ lifts via $\pi|_U$ to a tangent vector $(\boldsymbol{y}, \boldsymbol{w})$ of (the bi-affine cone of U in) $\widehat{X}_{\mathrm{sm}} \times \widehat{Z}'$ at $(\boldsymbol{x}, \boldsymbol{v})$, since $\pi|_U$ is smooth. By the definition of S, we have $F(\boldsymbol{x}) = \boldsymbol{x} \cdot \boldsymbol{v} = 0$, and $\nabla F(\boldsymbol{x}) = \alpha \boldsymbol{v}$ for some $\alpha \neq 0$. Differentiating the first two equations in the direction $(\boldsymbol{y}, \boldsymbol{w})$, using the chain and product rules, gives

$$0 = \boldsymbol{y} \cdot \nabla F(\boldsymbol{x}) = \alpha \boldsymbol{y} \cdot \boldsymbol{v} = -\alpha \boldsymbol{x} \cdot \boldsymbol{w}.$$

Thus $\boldsymbol{x} \in (T_{\boldsymbol{v}}\widehat{Z}')^{\perp} = M_{\boldsymbol{v}}$. On the other hand, $\nabla F(\boldsymbol{x}) = \alpha \boldsymbol{v} \in T_{\boldsymbol{v}}\widehat{Z}'$, since $\boldsymbol{v} \in \widehat{Z}'$ and $\widehat{Z}'$ is a cone. Thus $\nabla F(\boldsymbol{x})$ is orthogonal to $M_{\boldsymbol{v}}$. The affine cone of the fibre $\pi|_U^{-1}(\boldsymbol{v})$ therefore lies in the locus in (4.4), proving that inequality.

Now suppose $U \subseteq \operatorname{Sing}(X) \times Z'$. Then $\dim U \leqslant \dim \operatorname{Sing} X + \dim Z'$. On the other hand, $\dim U = e + \dim Z' \geqslant t + \dim Z' \geqslant \dim \operatorname{Sing} X + \dim Z'$, since $e \geqslant t$ and $t \geqslant \dim \operatorname{Sing} X$. Thus equality holds. The closure of U in $\operatorname{Sing}(X) \times Z'$ therefore contains some irreducible component $X' \times Z'$ of $\operatorname{Sing}(X) \times Z'$, with $\dim X' = \dim \operatorname{Sing} X$. Since $\boldsymbol{x} \cdot \boldsymbol{v}$ vanishes on U, it vanishes on $X' \times Z'$ by continuity. Differentiating the equation $\boldsymbol{x} \cdot \boldsymbol{v} = 0$ with respect to $\boldsymbol{v} \in \widehat{Z}' \setminus \{\mathbf{0}\}$, keeping $\boldsymbol{x} \in \widehat{X}'$ fixed, shows that $\widehat{X}' \subseteq (T_{\boldsymbol{v}}\widehat{Z}')^{\perp} = M_{\boldsymbol{v}}$. On the other hand, the definition of $\operatorname{Sing} X$ implies that $\nabla F(\boldsymbol{x}) = \mathbf{0}$ vanishes for all $\boldsymbol{x} \in \widehat{X}'$, so that $\nabla F(\boldsymbol{x})$ is trivially orthogonal to $M_{\boldsymbol{v}}$. It follows that $\widehat{X}'$ lies in the locus in (4.4), proving that inequality.

Since Z' is open in A_t, we may choose $\boldsymbol{v}$ to be a $\mathbb{Q}$-point. The vector space $M_{\boldsymbol{v}}$ is then defined over $\mathbb{Q}$. The inequality (4.4) trivially implies $\dim M_{\boldsymbol{v}} \geqslant 1$, since $e \geqslant t \geqslant 0$. Moreover, $F|_{M_{\boldsymbol{v}}} \neq 0$, since $X(\mathbb{Q}) = \emptyset$. Thus (4.4) in fact implies $\dim M_{\boldsymbol{v}} - 1 \geqslant \dim \operatorname{Sing}\{\boldsymbol{x} \in M_{\boldsymbol{v}} : F(\boldsymbol{x}) = 0\} \geqslant e + 1$, whence $\dim M_{\boldsymbol{v}} \geqslant 2$. Plugging the inequality (3.5) for $F|_{M_{\boldsymbol{v}}} \neq 0$ into (4.4) now gives

$$\frac{2(n - 1 - \dim Z) - 3}{3} = \frac{2\dim M_{\boldsymbol{v}} - 3}{3} \geqslant e + 1 \geqslant t + 1,$$

whence $2(1 + \dim Z) \leqslant 2n - 3(t+2)$. This proves (4.3), after taking into account the fact that A_t could be empty, in which case $1 + \dim A_t = 0$. □

Consider the closed Gaussian incidence correspondence

$$\begin{array}{ccccc} & & \Gamma := \overline{\{(x,H) \in X_{\mathrm{sm}} \times \check{\mathbb{P}}^{n-1} : T_x X \subset H\}} \subseteq X \times \check{\mathbb{P}}^{n-1} & & \\ & \swarrow^{\pi_1} & & \searrow^{\pi_2} & \\ X & & & & X^*, \end{array} \tag{4.5}$$

where $X^* \subset \check{\mathbb{P}}^{n-1}$ denotes the dual variety of X, which by definition is the image of Γ under π_2. Equivalently, since we are working in characteristic $\neq 3$,

$$\Gamma = \overline{\{([\boldsymbol{x}], [\nabla F(\boldsymbol{x})]) : [\boldsymbol{x}] \in X_{\mathrm{sm}}\}} \subseteq X \times \mathbb{P}^{n-1},$$

and X^* is the closure of the image of X_{sm} under the Gauss map $[\boldsymbol{x}] \mapsto [\nabla F(\boldsymbol{x})]$.

Lemma 4.2. *We have* $\dim X^* \geqslant 2\lceil n/3 \rceil - 2$.

Proof. Thanks to a result of Segre [Seg51], we know that $\dim X^* = r(F) - 2$. The claim now follows from Lemma 3.4. □

Let $\Gamma_{[\boldsymbol{v}]} \subseteq X$ be the fibre of Γ over $[\boldsymbol{v}] \in \mathbb{P}^{n-1}$. We view $\Gamma_{[\boldsymbol{v}]}$ as a closed subset of X. Let

$$W_t := \{[\boldsymbol{v}] \in \mathbb{P}^{n-1} : \dim \Gamma_{[\boldsymbol{v}]} \geqslant t\}. \tag{4.6}$$

Again W_t is closed and defined over $\mathbb{Q}$, by repeating the argument for B_t from after (4.2). In later sections, $\mathrm{Sing}(X_{\mathrm{sm}} \cap H_{\boldsymbol{v}})$ will be more directly relevant than $\Gamma_{[\boldsymbol{v}]}$. However,

$$\mathrm{Sing}(X_{\mathrm{sm}} \cap H_{\boldsymbol{v}}) \subseteq \Gamma_{[\boldsymbol{v}]}, \tag{4.7}$$

and there are technical advantages to working with the closed set $\Gamma_{[\boldsymbol{v}]}$. Moreover, Euler's identity $\boldsymbol{x} \cdot \nabla F(\boldsymbol{x}) = 3F(\boldsymbol{x})$ implies, via the definition (4.2) of S, that

$$\Gamma \subseteq S, \qquad \Gamma_{[\boldsymbol{v}]} \subseteq S_{[\boldsymbol{v}]} = \mathrm{Sing}(X \cap H_{\boldsymbol{v}}). \tag{4.8}$$

Proposition 4.3. *If* $t \geqslant (n+1)/3$, *then*

$$1 + \dim W_t \leqslant \max\{0, n - t - 2\}. \tag{4.9}$$

If $t \geqslant \max\{(n+1)/3, \dim \mathrm{Sing}\, X\}$, *then*

$$1 + \dim B_t \leqslant \max\{0, n - t - 2\}. \tag{4.10}$$

Proof. Suppose first that $t \geqslant (n+1)/3$. Since X_{sm} is irreducible, the closure Γ is irreducible, with $\dim \Gamma = \dim X_{\mathrm{sm}} = n - 2$. Yet $\dim X^* \geqslant 2n/3 - 2$ by Lemma 4.2. Consequently the general fibre of $\pi_2 \colon \Gamma \to X^*$ has dimension at most $n/3 < t$. The set $\pi_2^{-1}(W_t)$ is therefore a proper closed subset of Γ, of dimension at most $\dim \Gamma - 1 = n - 3$. If W_t is nonempty, every fibre over it has dimension at least t, so

$$\dim W_t + t \leqslant \dim \pi_2^{-1}(W_t) \leqslant n - 3.$$

This proves (4.9), since if $W_t = \emptyset$ then $1 + \dim W_t = 0$.

Now assume $t \geqslant \max\{(n+1)/3, \dim \mathrm{Sing}\, X\}$, and take $[\boldsymbol{v}] \in B_t$. Choose an irreducible component Z of $\mathrm{Sing}(X \cap H_{\boldsymbol{v}})$ with $\dim Z \geqslant t$. If Z meets X_{sm}, then its dense open subset $Z \cap X_{\mathrm{sm}}$ lies in $\Gamma_{[\boldsymbol{v}]}$, so $[\boldsymbol{v}] \in W_t$. Now suppose $Z \subseteq \mathrm{Sing}\, X$. Then $\dim \mathrm{Sing}\, X \geqslant \dim Z \geqslant t$, where equality must hold in view of the assumption $t \geqslant \dim \mathrm{Sing}\, X$. Thus Z is one of the finitely many t-dimensional components of $\mathrm{Sing}\, X$. Its cone $\widehat{Z}$ cannot be linear: Lemma 3.10 would give $t + 1 = \dim \widehat{Z} \leqslant n/3$, contrary to $t > n/3$. Thus the linear span L of $\widehat{Z}$ has dimension at least $t + 2$. Since $Z \subseteq H_{\boldsymbol{v}}$ by definition, this constrains $\boldsymbol{v}$ to lie in the orthogonal complement $L^\perp$, where $\dim L^\perp \leqslant n - \dim L \leqslant n - t - 2$. Taking the finite union over all possible irreducible components $Z \subseteq \mathrm{Sing}\, X$ proves (4.10). □

Corollary 4.4. *Assume $n \geqslant 10$ and $X(\mathbb{Q}) = \emptyset$. Then*

$$\dim \operatorname{Sing}(X) \leqslant \lfloor (2n-6)/3 \rfloor \leqslant n-6. \tag{4.11}$$

The set $B_{n-6}(\mathbb{Q})$ is finite if $n = 10$, and is empty if $n \geqslant 11$. Moreover,

$$B_{n-5}(\mathbb{Q}) = \emptyset. \tag{4.12}$$

Geometrically, we have $B_{n-4} = \emptyset$ and

$$1 + \dim B_{n-6} \leqslant 4, \qquad 1 + \dim B_{n-5} \leqslant 3. \tag{4.13}$$

Proof. The inequality (4.11) follows from (3.5), Indeed, for $n \geqslant 12$ we have $(2n-6)/3 \leqslant n-6$, and for $n = 10, 11$ we have $\lfloor (2n-6)/3 \rfloor = n-6$. Since $n-6 \geqslant (n+1)/3$, taking $t = n-6, n-5$ in (4.10) proves (4.13). On the other hand, the claim $B_{n-4} = \emptyset$ follows from (3.5) and Theorem 2.5. Separately, Proposition 4.1 with $t = n-6, n-5$ gives

$$1 + \dim A_{n-6} \leqslant \max\left\{0, \left\lfloor \frac{12-n}{2} \right\rfloor\right\}, \qquad 1 + \dim A_{n-5} \leqslant \max\left\{0, \left\lfloor \frac{9-n}{2} \right\rfloor\right\}.$$

This proves the assertions about $\mathbb{Q}$-points. □

5. Pointwise bounds for exponential sums

Let $F \in \mathbb{Z}[x_1, \dots, x_n]$ be a cubic form, and let $X \subset \mathbb{P}^{n-1}_{\mathbb{Q}}$ be the hypersurface $F = 0$. In this section, we assume that $n \geqslant 10$ and $X(\mathbb{Q}) = \emptyset$. By Proposition 2.3, X is geometrically integral and is not a cone. Also, $\dim \operatorname{Sing} X \leqslant n-6$ by (4.11).

Recall the definition of the complete exponential sum $S_q(\boldsymbol{v})$ from (1.1). We will collect together various pointwise upper bounds for $S_q(\boldsymbol{v})$, particularly for prime powers $q = p^r$, in view of the multiplicativity property (1.2). As usual, any implied constants in our estimates are allowed to depend on F in any way.

5.1. **Prime modulus.** Let p be a prime and put $\pi_m(p) = \#\,\mathbb{P}^m(\mathbb{F}_p) = 1 + p + \cdots + p^m$, for an integer $m \geqslant 0$. Let $X_p \subset \mathbb{P}^{n-1}_{\mathbb{F}_p}$ denote the hypersurface $F = 0$ modulo p. For $\boldsymbol{v} \in \mathbb{Z}^n$, write $\overline{\boldsymbol{v}}$ for its reduction modulo p. In this section we shall be concerned with estimating

$$S_p(\boldsymbol{v}) = \sum_{a \in \mathbb{F}_p^*} \sum_{\boldsymbol{x} \in \mathbb{F}_p^n} e_p\big(aF(\boldsymbol{x}) + \boldsymbol{v} \cdot \boldsymbol{x}\big).$$

The following result, Proposition 5.1, provides crucial pointwise upper bounds for $S_p(\boldsymbol{v})$. Later, in Appendix A, we will provide a complementary set of stratified bounds for $S_p(\boldsymbol{v})$, which will be combined with Proposition 5.1 in §10.

Proposition 5.1. *We have the following estimates:*

(i) If $p \mid \boldsymbol{v}$ then $S_p(\boldsymbol{v}) = S_p(\boldsymbol{0}) \ll p^{n-3/2}$.
(ii) If $p \nmid \boldsymbol{v}$ and $n \geqslant 10$, then $S_p(\boldsymbol{v}) \ll p^{n-2}$.

We begin with exact identities that convert the complete exponential sum into point counting estimates for X_p and the hyperplane section $Y_{\boldsymbol{v}} = X_p \cap V(\overline{\boldsymbol{v}} \cdot \boldsymbol{x}) \subset \mathbb{P}^{n-2}_{\mathbb{F}_p}$. Arguing as in Hooley [Hoo86, Eq. (47)], we have

$$S_p(\boldsymbol{v}) = p^2\big(\#Y_{\boldsymbol{v}}(\mathbb{F}_p) - \pi_{n-3}(p)\big) - p\big(\#X_p(\mathbb{F}_p) - \pi_{n-2}(p)\big) \tag{5.1}$$

when $p \nmid \boldsymbol{v}$. If $p \mid \boldsymbol{v}$, on the other hand, then a simpler orthogonality argument yields

$$S_p(\boldsymbol{0}) = p(p-1)\big(\#X_p(\mathbb{F}_p) - \pi_{n-2}(p)\big). \tag{5.2}$$

The following result will allow us to deal with degenerate cases in which $Y_{\boldsymbol{v}}$ is either a cone or fails to be geometrically integral.

Lemma 5.2. *There is a finite set of primes, depending only on F, such that for every prime outside this set and every $\overline{\boldsymbol{v}} \neq \boldsymbol{0}$, the cubic hypersurface $Y_{\boldsymbol{v}}$ is geometrically integral. If $Y_{\boldsymbol{v}}$ is a cone over $\overline{\mathbb{F}}_p$, furthermore, with maximal vertex of projective dimension s, then*

$$s \leqslant 1. \tag{5.3}$$

Proof. After excluding finitely many primes, spreading out (4.11) gives

$$\dim \mathrm{Sing}(X_p) = \dim \mathrm{Sing}(X) \leqslant n-6. \tag{5.4}$$

Likewise, if $V_{2,p} = \{\boldsymbol{x} \in \mathbb{A}^n_{\overline{\mathbb{F}}_p} : \operatorname{rank} H(\boldsymbol{x}) \leqslant 2\}$, then spreading out Corollary 3.9 gives

$$\dim(V_{2,p} \cap \{F=0\}) = \dim(V_2 \cap \{F=0\}) \leqslant 2 \cdot 2 - 2 = 2. \tag{5.5}$$

We may also suppose that $p > 3$ and that X_p is geometrically integral of dimension $n-2$ and is not a cone, by Proposition 2.3. Indeed, it is well known that geometric integrality and dimension spread out, whereas the set of vertices of X spreads out by Remark 2.4.

The hyperplane section $Y_{\boldsymbol{v}}$ is geometrically integral by Theorem 2.5, because (5.4) gives

$$\dim X_p - \dim \mathrm{Sing}(X_p) \geqslant (n-2) - (n-6) = 4.$$

Suppose now that $Y_{\boldsymbol{v}}$ is a cone over $\overline{\mathbb{F}}_p$ with maximal vertex $\mathbb{P}^s$. Over $\overline{\mathbb{F}}_p$, choose coordinates $(\boldsymbol{u}, \boldsymbol{z}, t)$ such that $t=0$ cuts out $\overline{\boldsymbol{v}} \cdot \boldsymbol{x} = 0$, the variables $\boldsymbol{u} = (u_0, \dots, u_s)$ run along the vertex, and the minimal base is defined by a cubic form $C(\boldsymbol{z})$. Then

$$F(\boldsymbol{u}, \boldsymbol{z}, t) = C(\boldsymbol{z}) + tQ(\boldsymbol{u}, \boldsymbol{z}, t)$$

for a quadratic form Q. At every point $(\boldsymbol{u}, \boldsymbol{0}, 0)$ of the affine cone over the vertex, all entries of the Hessian matrix $H(\boldsymbol{u}, \boldsymbol{0}, 0)$ vanish except possibly those in the t-row and the t-column. Thus $\operatorname{rank} H(\boldsymbol{u}, \boldsymbol{0}, 0) \leqslant 2$ and so the affine $(s+1)$-plane over the vertex is contained in $V_{2,p}$. Since the vertex is trivially also contained in $\{F=0\}$, it follows from (5.5) that

$$s+1 \leqslant 2,$$

which completes the proof of the lemma. □

Proof of Proposition 5.1. It is clearly enough to work outside a finite set of primes depending on F. To begin with it follows from Lemma 2.1 that

$$|\#X_p(\mathbb{F}_p) - \pi_{n-2}(p)| \ll_n p^{(n-1+\sigma_p)/2} \leqslant p^{n-7/2}, \tag{5.6}$$

since $\sigma_p = \dim \mathrm{Sing}(X_p) \leqslant n-6$, by (5.4). Inserting this into (5.2) gives part (i).

Assume now that $\overline{\boldsymbol{v}} \neq \boldsymbol{0}$. By (5.1) and (5.6), we have

$$S_p(\boldsymbol{v}) = p^2\big(\#Y_{\boldsymbol{v}}(\mathbb{F}_p) - \pi_{n-3}(p)\big) + O(p^{n-5/2}). \tag{5.7}$$

It remains to count points on $Y_{\boldsymbol{v}}$. If $Y_{\boldsymbol{v}}$ is not a cone, the second estimate in Lemma 2.1 yields

$$|\#Y_{\boldsymbol{v}}(\mathbb{F}_p) - \pi_{n-3}(p)| \ll_n p^{n-4}.$$

Suppose now that $Y_{\boldsymbol{v}}$ is a cone. Its defining cubic form over $\mathbb{F}_p$ is absolutely irreducible by Lemma 5.2. Its maximal vertex is an s-plane defined over $\mathbb{F}_p$, where $s \geqslant 0$. After choosing an $\mathbb{F}_p$-linear complement its minimal base is a geometrically integral, non-conical cubic hypersurface $Z_{\boldsymbol{v}} \subset \mathbb{P}^{n-s-3}_{\mathbb{F}_p}$, defined by a cubic form in $n-s-2$ variables. Let

$$b = n-s-4.$$

Projection away from the maximal vertex shows that

$$\#Y_{\boldsymbol{v}}(\mathbb{F}_p) = \pi_s(p) + p^{s+1}\#Z_{\boldsymbol{v}}(\mathbb{F}_p),$$

and $\pi_{n-3}(p) = \pi_s(p) + p^{s+1}\pi_b(p)$. If $b \geqslant 2$, then Lemma 2.1 gives

$$|\#Z_{\boldsymbol{v}}(\mathbb{F}_p) - \pi_b(p)| \ll_n p^{b-1},$$

from which it follows that

$$|\#Y_{\boldsymbol{v}}(\mathbb{F}_p) - \pi_{n-3}(p)| \ll_n p^{s+1+b-1} = p^{n-4}. \tag{5.8}$$

Yet by (5.3) and Remark 2.4, we have $s \leqslant 1$ and $b = n-s-4 \geqslant 2$ (in fact $b \geqslant 5$) whenever $n \geqslant 10$. Combining (5.8) with (5.7) therefore proves part (ii), uniformly for every $\overline{\boldsymbol{v}} \neq \boldsymbol{0}$. □

5.2. **Prime squares.** For integers $t \geqslant 0$, recall the definitions of B_t and W_t from (4.1) and (4.6), respectively. By (4.8), we have $W_t \subseteq B_t$. Corollary 4.4 gives valuable information about $B_{n-6}(\mathbb{Q})$ and $B_{n-5}(\mathbb{Q})$.

The incidence correspondences S and Γ have natural analogues $\mathcal{S}_R$ and Γ_R over any base ring R in place of $\overline{\mathbb{Q}}$. It is clear from the definition of $\mathcal{S}$ in (4.2) that $\mathcal{S}_R = \mathcal{S}_{\mathbb{Z}} \otimes R$ for any ring R. Using the definition (4.5) of Γ as a scheme-theoretic closure, it can also be shown that $\Gamma_R = \Gamma_{\mathbb{Z}} \otimes R$ for $R \in \{\overline{\mathbb{Q}}, \mathbb{Q}, \mathbb{F}_p\}$ for all but finitely many primes p. Indeed, flat base change for scheme-theoretic closures [TSta, Tag 081I] proves this for $R = \mathbb{Q}$ and $R = \overline{\mathbb{Q}}$. Moreover, scheme-theoretic density spreads out [TSta, Tag 0573], proving the claim for $R = \mathbb{F}_p$.

Let $\mathcal{B}_{t,R}$ (resp. $\mathcal{W}_{t,R}$) denote the set of fibres of $\mathcal{S}_R$ (resp. Γ_R) of dimension $\geqslant t$. Then $\mathcal{B}_{t,R}$ and $\mathcal{W}_{t,R}$ are closed sets, by upper semicontinuity for proper morphisms. Let $C \otimes R$ denote the preimage of a set $C \subseteq \mathbb{P}^{n-1}_{\mathbb{Z}}$ along the map $\mathbb{P}^{n-1}_R \to \mathbb{P}^{n-1}_{\mathbb{Z}}$. Since fibre dimensions are preserved by base change [TSta, Tag 05F8], we have $\mathcal{B}_{t,R} = \mathcal{B}_{t,\mathbb{Z}} \otimes R$ for any ring R, and $\mathcal{W}_{t,R} = \mathcal{W}_{t,\mathbb{Z}} \otimes R$ for $R \in \{\overline{\mathbb{Q}}, \mathbb{Q}, \mathbb{F}_p\}$ for all but finitely many primes p.

Let $\mathcal{B}_{t,p}$, $\mathcal{W}_{t,p}$, $\mathcal{S}_p$, and Γ_p denote $\mathcal{B}_{t,\mathbb{F}_p}$, $\mathcal{W}_{t,\mathbb{F}_p}$, $\mathcal{S}_{\mathbb{F}_p}$, and $\Gamma_{\mathbb{F}_p}$, respectively.

Lemma 5.3. *Let p be any prime and let $t \geqslant 0$. If $p \nmid \boldsymbol{v}$ and $[\overline{\boldsymbol{v}}] \notin \mathcal{W}_{t,p}(\mathbb{F}_p)$, then*

$$S_{p^2}(\boldsymbol{v}) \ll_F p^{n+t+1}. \tag{5.9}$$

Moreover, for every prime p and every $\boldsymbol{v} \in \mathbb{Z}^n$,

$$S_{p^2}(\boldsymbol{v}) \ll_F p^{2n-3}. \tag{5.10}$$

Proof. It is enough to prove the estimates outside a fixed finite set of primes. We may therefore assume that $p > 3$, that X_p is geometrically integral, and that $\dim \operatorname{Sing}(X_p) \leqslant n-6$.

Writing $a = b + pc$ and $\boldsymbol{x} = \boldsymbol{y} + p\boldsymbol{z}$, and then summing over $c \in \mathbb{F}_p$ and $\boldsymbol{z} \in \mathbb{F}_p^n$, gives

$$S_{p^2}(\boldsymbol{v}) = \sum_{a \in (\mathbb{Z}/p^2\mathbb{Z})^*} \sum_{\boldsymbol{x} \in (\mathbb{Z}/p^2\mathbb{Z})^n} e_{p^2}\left(aF(\boldsymbol{x}) + \boldsymbol{v} \cdot \boldsymbol{x}\right) = p^{n+1} \sum_{b \in \mathbb{F}_p^*} \sum_{\substack{\boldsymbol{y} \in \mathbb{F}_p^n \\ F(\boldsymbol{y}) = 0 \\ b\nabla F(\boldsymbol{y}) = -\overline{\boldsymbol{v}}}} e_{p^2}\left(bF(\boldsymbol{y}) + \boldsymbol{v} \cdot \boldsymbol{y}\right),$$

since $aF(\boldsymbol{x}) + \boldsymbol{v} \cdot \boldsymbol{x}$ may be rewritten as

$$(b+pc)F(\boldsymbol{y} + p\boldsymbol{z}) + \boldsymbol{v} \cdot (\boldsymbol{y} + p\boldsymbol{z}) \equiv bF(\boldsymbol{y}) + \boldsymbol{v} \cdot \boldsymbol{y} + p\left(b\nabla F(\boldsymbol{y}) \cdot \boldsymbol{z} + cF(\boldsymbol{y}) + \boldsymbol{v} \cdot \boldsymbol{z}\right) \bmod p^2.$$

The final sum over b and $\boldsymbol{y}$ is well-defined: in view of the conditions $F(\boldsymbol{y}) = 0$ and $b\nabla F(\boldsymbol{y}) = -\overline{\boldsymbol{v}}$, each summand $e_{p^2}\left(bF(\boldsymbol{y}) + \boldsymbol{v} \cdot \boldsymbol{y}\right)$ only depends on $b, \boldsymbol{y} \bmod p$ and $\boldsymbol{v} \bmod p^2$.

By the triangle inequality and the Lang–Weil estimate,

$$|S_{p^2}(\boldsymbol{v})| \leqslant p^{n+1} \sum_{b \in \mathbb{F}_p^*} \sum_{\substack{\boldsymbol{y} \in \mathbb{F}_p^n \\ F(\boldsymbol{y}) = 0 \\ b\nabla F(\boldsymbol{y}) = -\overline{\boldsymbol{v}}}} 1 = p^{n+1}(p-1)\,\#V(\mathbb{F}_p) \ll_n p^{n+2+\dim V},$$

where $V = \operatorname{Sing}(X_{p,\mathrm{sm}} \cap \{\overline{\boldsymbol{v}} \cdot \boldsymbol{x} = 0\})$ if $p \nmid \boldsymbol{v}$, and $V = \operatorname{Sing}(\widehat{X}_p)$ if $p \mid \boldsymbol{v}$. If $p \nmid \boldsymbol{v}$ and $[\overline{\boldsymbol{v}}] \notin \mathcal{W}_{t,p}(\mathbb{F}_p)$, then $\dim V \leqslant \dim \Gamma_{p,[\overline{\boldsymbol{v}}]} \leqslant t-1$, by the analogue of (4.7) over $\mathbb{F}_p$. Thus (5.9) follows. Moreover Theorem 2.5 implies $\dim V \leqslant \dim \operatorname{Sing}(X_p \cap \{\overline{\boldsymbol{v}} \cdot \boldsymbol{x} = 0\}) \leqslant \dim \operatorname{Sing}(X_p) + 1 \leqslant n-5$ if $p \nmid \boldsymbol{v}$. If $p \mid \boldsymbol{v}$, then $\dim V = 1 + \dim \operatorname{Sing}(X_p) \leqslant n-5$. Either way, (5.10) follows. □

5.3. **Higher prime power modulus.** Using the goodness of F, it is possible to obtain a pointwise bound $S_{p^r}(\boldsymbol{v}) \ll p^{r(1+5n/6+\varepsilon)}$ by following the proof of [HB07, Eq. (7.1)], or Bernert's version in [Ber25, §3]. The van der Corput differencing process is insensitive to any linear or quadratic terms, and $S_{p^r}(\boldsymbol{v})$ has only a cubic term $F(\boldsymbol{x})$ and a linear term $\boldsymbol{v} \cdot \boldsymbol{x}$. However, for $\boldsymbol{v} \neq \boldsymbol{0}$, it turns out that it will be more useful for us to bound $S_{p^r}(\boldsymbol{v})$ on average over $\boldsymbol{v}$ when $r \geqslant 3$. Nonetheless, for $\boldsymbol{v} = \boldsymbol{0}$ and for parts of the averaging procedure over $\boldsymbol{v} \neq \boldsymbol{0}$, it will be convenient for us to draw on the recent work of Bernert [Ber25].

Theorem 5.4. *If $n \geqslant 10$ and F is good, then $\mathfrak{S} := \sum_{q \geqslant 1} q^{-n} S_q(\mathbf{0})$ converges absolutely to a positive real number, and satisfies*

$$\mathfrak{S} = \prod_p \chi_p,$$

where

$$\chi_p := \sum_{k \geqslant 0} p^{-kn} S_{p^k}(\mathbf{0}) = \lim_{j \to \infty} p^{-j(n-1)} \#\{\boldsymbol{x} \bmod p^j : F(\boldsymbol{x}) \equiv 0 \bmod p^j\} > 0.$$

Proof. This follows from [Ber25, §2, Theorem 1]. □

The main point in the proof of Theorem 5.4 is the absolute convergence

$$\sum_{q \geqslant 1} q^{-n} |S_q(\mathbf{0})| < \infty. \tag{5.11}$$

We remark that Bernert's result is stronger, in that he takes the absolute value *inside* the a-sum in $S_q(\mathbf{0})$, rather than *outside* the a-sum as in (5.11). Also, as Bernert remarks, Theorem 5.4 was already proven for $n \geqslant 11$ by Heath-Brown [HB07].

5.4. **Hensel lifting.** It will be useful for us to develop a flexible version of Hensel lifting. For integers $A, T \geqslant 1$ and vectors $\boldsymbol{y} \in \mathbb{Z}^n$, put

$$\mathcal{G}_{A,T}(\boldsymbol{y}) := \max_{\substack{\alpha \in (\mathbb{Z}/T\mathbb{Z})^* \\ \boldsymbol{d} \in (\mathbb{Z}/T\mathbb{Z})^n}} \left| \sum_{\boldsymbol{z} \in (\mathbb{Z}/T\mathbb{Z})^n} e_T(\boldsymbol{d} \cdot \boldsymbol{z} + \alpha Q_{\boldsymbol{y}}(\boldsymbol{z}) + \alpha A F(\boldsymbol{z})) \right|, \tag{5.12}$$

where $Q_{\boldsymbol{y}}(\boldsymbol{z}) := \frac{1}{2} \boldsymbol{z}^t H(\boldsymbol{y}) \boldsymbol{z}$ is an integral quadratic form defined in terms of F and $\boldsymbol{y}$. We proceed by proving the following three results.

Lemma 5.5. *The quantity $\mathcal{G}_{A,T}(\boldsymbol{y})$ is multiplicative in T. That is,*

$$\mathcal{G}_{A,T_1T_2}(\boldsymbol{y}) = \mathcal{G}_{A,T_1}(\boldsymbol{y}) \mathcal{G}_{A,T_2}(\boldsymbol{y}) \tag{5.13}$$

for all $A, T_1, T_2 \geqslant 1$ with $\gcd(T_1, T_2) = 1$.

Lemma 5.6. *As a function of $\boldsymbol{y}$, the quantity $\mathcal{G}_{A,T}(\boldsymbol{y})$ only depends on $\boldsymbol{y} \bmod \gcd(A, T)$.*

Lemma 5.7. *Let $A, T \geqslant 1$ and $q = A^2 T$. Then*

$$|S_q(\boldsymbol{v})| \leqslant A^{n+1} T \sum_{a \in (\mathbb{Z}/A\mathbb{Z})^*} \sum_{\substack{\boldsymbol{y} \bmod A \\ F(\boldsymbol{y}) \equiv 0 \bmod A \\ a \nabla F(\boldsymbol{y}) + \boldsymbol{v} \equiv \mathbf{0} \bmod A}} \mathcal{G}_{A,T}(\boldsymbol{y}). \tag{5.14}$$

Lemma 5.6 ensures that each summand on the right-hand side of (5.14) is well-defined.

Proof of Lemma 5.5. Let

$$\mathcal{S}_{A,T}(\boldsymbol{d}, \alpha, \boldsymbol{y}) := \sum_{\boldsymbol{z} \bmod T} e_T(\boldsymbol{d} \cdot \boldsymbol{z} + \alpha Q_{\boldsymbol{y}}(\boldsymbol{z}) + \alpha A F(\boldsymbol{z})) \tag{5.15}$$

be the $\boldsymbol{z}$-sum in (5.12). If $\gcd(T_1, T_2) = 1$, the Chinese remainder theorem implies

$$\mathcal{S}_{A,T_1T_2}(\boldsymbol{d}, \alpha, \boldsymbol{y}) = \mathcal{S}_{A,T_1}(\overline{T_2}\boldsymbol{d}, \overline{T_2}\alpha, \boldsymbol{y}) \mathcal{S}_{A,T_2}(\overline{T_1}\boldsymbol{d}, \overline{T_1}\alpha, \boldsymbol{y}).$$

By the Chinese remainder theorem, the pairs $(\overline{T_2}\boldsymbol{d}, \overline{T_2}\alpha) \bmod T_1$ and $(\overline{T_1}\boldsymbol{d}, \overline{T_1}\alpha) \bmod T_2$ range independently over their respective parameter sets. Thus the two local maxima can be attained simultaneously, and taking the maximum of the absolute value gives the claimed equality. □

Proof of Lemma 5.6. A priori, $\mathcal{G}_{A,T}(\boldsymbol{y})$ depends only on $\boldsymbol{y} \bmod T$. That is, $\mathcal{G}_{A,T}(\boldsymbol{y})$ is T-periodic in $\boldsymbol{y}$. It remains to show that it is A-periodic as well. First write $e_T(x) = e_{A^2T}(A^2x)$ to get

$$\begin{aligned}\mathcal{S}_{A,T}(\boldsymbol{d},\alpha,\boldsymbol{y}) &= \sum_{\boldsymbol{z} \bmod T} e_{A^2T}(A^2\boldsymbol{d}\cdot\boldsymbol{z} + \alpha A^2 Q_{\boldsymbol{y}}(\boldsymbol{z}) + \alpha A^3 F(\boldsymbol{z}))\\ &= e_{A^2T}(-\alpha F(\boldsymbol{y})) \sum_{\boldsymbol{z} \bmod T} e_{A^2T}(A^2\boldsymbol{d}\cdot\boldsymbol{z} + \alpha F(\boldsymbol{y}+A\boldsymbol{z}) - \alpha A\nabla F(\boldsymbol{y})\cdot\boldsymbol{z}),\end{aligned} \tag{5.16}$$

by Taylor expansion of $F(\boldsymbol{y}+A\boldsymbol{z})$. Replacing $(\boldsymbol{y},\boldsymbol{z})$ with $(\boldsymbol{y}+A\boldsymbol{h},\boldsymbol{z}-\boldsymbol{h})$, and taking absolute values, gives

$$\begin{aligned}|\mathcal{S}_{A,T}(\boldsymbol{d},\alpha,\boldsymbol{y}+A\boldsymbol{h})| &= \Big|\sum_{\boldsymbol{z} \bmod T} e_{A^2T}(A^2\boldsymbol{d}\cdot\boldsymbol{z} + \alpha F(\boldsymbol{y}+A\boldsymbol{z}) - \alpha A\nabla F(\boldsymbol{y}+A\boldsymbol{h})\cdot\boldsymbol{z})\Big|\\ &= \Big|\sum_{\boldsymbol{z} \bmod T} e_{A^2T}(A^2\boldsymbol{d}'\cdot\boldsymbol{z} + \alpha F(\boldsymbol{y}+A\boldsymbol{z}) - \alpha A\nabla F(\boldsymbol{y})\cdot\boldsymbol{z})\Big|\end{aligned}$$

by Taylor expansion of $\nabla F(\boldsymbol{y}+A\boldsymbol{h})$, where $\boldsymbol{d}' := \boldsymbol{d} - \alpha H(\boldsymbol{y})\boldsymbol{h} - \alpha A\nabla F(\boldsymbol{h})$. By (5.16) with $\boldsymbol{d}'$ in place of $\boldsymbol{d}$, we get $|\mathcal{S}_{A,T}(\boldsymbol{d},\alpha,\boldsymbol{y}+A\boldsymbol{h})| = |\mathcal{S}_{A,T}(\boldsymbol{d}',\alpha,\boldsymbol{y})|$. Since the map $(\alpha,\boldsymbol{d}) \mapsto (\alpha,\boldsymbol{d}')$ is a bijection, it follows from (5.12) that $\mathcal{G}_{A,T}(\boldsymbol{y}+A\boldsymbol{h}) = \mathcal{G}_{A,T}(\boldsymbol{y})$, which is what we wanted. □

Proof of Lemma 5.7. Let $M := AT$. Then $q = A^2T = AM$, $\mathrm{rad}(q) = \mathrm{rad}(AT) = \mathrm{rad}(M)$, and $M^2 = A^2T^2 = qT \equiv 0 \bmod q$. For $a, \boldsymbol{x}$ as in (1.1), write

$$\boldsymbol{x} = \boldsymbol{x}_0 + M\boldsymbol{h}_0, \qquad a = a_0 + Me,$$

with $0 \leqslant \boldsymbol{x}_0 < M$ and $0 \leqslant a_0 < M$ running over a complete set of representatives modulo M, and similarly with $0 \leqslant \boldsymbol{h}_0 < A$ and $0 \leqslant e < A$. Since $\mathrm{rad}(M) = \mathrm{rad}(q)$, the condition $\gcd(a,q) = 1$ is equivalent to $\gcd(a_0, M) = 1$. Taylor expansion of $F(\boldsymbol{x}_0 + M\boldsymbol{h}_0)$ gives

$$aF(\boldsymbol{x}) + \boldsymbol{v}\cdot\boldsymbol{x} \equiv (a_0+Me)F(\boldsymbol{x}_0) + a_0M\nabla F(\boldsymbol{x}_0)\cdot\boldsymbol{h}_0 + \boldsymbol{v}\cdot(\boldsymbol{x}_0 + M\boldsymbol{h}_0) \bmod q,$$

since $M^2 \equiv 0 \bmod q$. Orthogonality in $e, \boldsymbol{h}_0 \bmod A$ gives

$$S_q(\boldsymbol{v}) = A^{n+1} \sum_{\substack{0\leqslant a_0<M\\ \gcd(a_0,M)=1}} \sum_{\substack{0\leqslant \boldsymbol{x}_0<M\\ F(\boldsymbol{x}_0)\equiv 0 \bmod A\\ a_0\nabla F(\boldsymbol{x}_0)+\boldsymbol{v}\equiv\boldsymbol{0} \bmod A}} e_q(a_0F(\boldsymbol{x}_0) + \boldsymbol{v}\cdot\boldsymbol{x}_0). \tag{5.17}$$

For all $\boldsymbol{x}_0$ in the support of (5.17), write $\boldsymbol{x}_0 = \boldsymbol{y} + A\boldsymbol{z}$, with $0 \leqslant \boldsymbol{y} < A$ and $0 \leqslant \boldsymbol{z} < T$. Then

$$\boldsymbol{d} := \frac{a_0\nabla F(\boldsymbol{y}) + \boldsymbol{v}}{A} \in \mathbb{Z}^n,$$

since $a_0\nabla F(\boldsymbol{x}_0) + \boldsymbol{v} \equiv \boldsymbol{0} \bmod A$. Taylor expansion of $F(\boldsymbol{y}+A\boldsymbol{z})$ gives

$$\begin{aligned}\frac{e_q(a_0F(\boldsymbol{x}_0) + \boldsymbol{v}\cdot\boldsymbol{x}_0)}{e_q(a_0F(\boldsymbol{y}) + \boldsymbol{v}\cdot\boldsymbol{y})} &= e_q(a_0A\nabla F(\boldsymbol{y})\cdot\boldsymbol{z} + a_0A^2Q_{\boldsymbol{y}}(\boldsymbol{z}) + a_0A^3F(\boldsymbol{z}) + A\boldsymbol{v}\cdot\boldsymbol{z})\\ &= e_T(\boldsymbol{d}\cdot\boldsymbol{z} + a_0Q_{\boldsymbol{y}}(\boldsymbol{z}) + a_0AF(\boldsymbol{z}))\end{aligned}$$

in view of the identity $e_q(A^2x) = e_{A^2T}(A^2x) = e_T(x)$. By the triangle inequality on (5.17),

$$|S_q(\boldsymbol{v})| \leqslant A^{n+1} \sum_{\substack{0\leqslant a_0<M\\ \gcd(a_0,M)=1}} \sum_{\substack{0\leqslant \boldsymbol{y}<A\\ F(\boldsymbol{y})\equiv 0 \bmod A\\ a_0\nabla F(\boldsymbol{y})+\boldsymbol{v}\equiv\boldsymbol{0} \bmod A}} |\mathcal{S}_{A,T}(\boldsymbol{d},a_0,\boldsymbol{y})|,$$

where $\mathcal{S}_{A,T}$ was defined in (5.15). By definition, $|\mathcal{S}_{A,T}| \leqslant \mathcal{G}_{A,T}(\boldsymbol{y})$. Since $\mathcal{G}_{A,T}(\boldsymbol{y})$ is independent of a_0, and the condition $a_0\nabla F(\boldsymbol{y}) + \boldsymbol{v} \equiv \boldsymbol{0} \bmod A$ depends only on $a_0 \bmod A$, the desired inequality (5.14) follows from the fact that each fibre of the map $(\mathbb{Z}/M\mathbb{Z})^* \to (\mathbb{Z}/A\mathbb{Z})^*$ has cardinality $\varphi(M)/\varphi(A) = \varphi(AT)/\varphi(A) \leqslant T$. □

5.5. **Gauss sums.** Given $\boldsymbol{a} \in \mathbb{Z}^n$ and a positive integer d, the quantity

$$N_d(\boldsymbol{a}) = \#\{\boldsymbol{y} \in (\mathbb{Z}/d\mathbb{Z})^n : H(\boldsymbol{a})\boldsymbol{y} \equiv \boldsymbol{0} \bmod d\} \tag{5.18}$$

will play a pivotal role in estimating (5.12), where $H(\boldsymbol{a})$ is the Hessian matrix associated to F.

Lemma 5.8. *Let $r(\boldsymbol{a})$ be the rank of $H(\boldsymbol{a})$ over $\mathbb{Q}$. Then $N_d(\boldsymbol{a}) \ll_F d^{n-r(\boldsymbol{a})}|\boldsymbol{a}|^{r(\boldsymbol{a})}$.*

Proof. This is [Ber25, Eq. (3.1)], applied to the cubic form $C := 6F$. It does not require goodness. For $C = 6F$, Bernert's notation $M, \boldsymbol{h}, q$ corresponds to our notation $H, \boldsymbol{a}, d$, and his bilinear forms $B_i(\boldsymbol{x}, \boldsymbol{y})$ are simply the entries of the vector $M(\boldsymbol{x})\boldsymbol{y} = H(\boldsymbol{x})\boldsymbol{y}$. □

Lemma 5.9. *If $A, T \geqslant 1$ are integers with $T \mid A$, then*

$$\mathcal{G}_{A,T}(\boldsymbol{a}) \leqslant T^{n/2} N_T(\boldsymbol{a})^{1/2}.$$

Proof. Since $T \mid A$, the definition of $\mathcal{G}_{A,T}$ in (5.12) simplifies to

$$\mathcal{G}_{A,T}(\boldsymbol{a}) = \max_{\substack{\alpha \in (\mathbb{Z}/T\mathbb{Z})^* \\ \boldsymbol{d} \in (\mathbb{Z}/T\mathbb{Z})^n}} \left| \sum_{\boldsymbol{z} \in (\mathbb{Z}/T\mathbb{Z})^n} e_T(\boldsymbol{d} \cdot \boldsymbol{z} + \alpha Q_{\boldsymbol{a}}(\boldsymbol{z})) \right|. \tag{5.19}$$

where $Q_{\boldsymbol{a}}(\boldsymbol{z}) := \frac{1}{2}\boldsymbol{z}^t H(\boldsymbol{a})\boldsymbol{z}$. On squaring and differencing the $\boldsymbol{z}$-sum, exactly as in the close of the proof of [BHB09, Lemma 11], we are led to the desired conclusion. □

6. Squarefull exponential sums on average

Throughout this section assume $n \geqslant 10$ and $X(\mathbb{Q}) = \emptyset$; in particular, F is good.

6.1. **Squarefull exponential sums: averaging over $\boldsymbol{v}$.** Any squarefull integer r admits a factorisation $r = c^2 d$ for square-free d such that $d \mid c$. Let $V \geqslant 1$. In this section we consider the averages

$$\sum_{\boldsymbol{v} \in \mathcal{V}} |S_r(\boldsymbol{v})|,$$

for an arbitrary set $\mathcal{V} \subseteq \mathbb{Z}^n \cap (\boldsymbol{v}_0 + [-V, V]^n)$, uniformly in $\boldsymbol{v}_0 \in \mathbb{R}^n$. Let

$$d_1 = \prod_{\substack{p \mid d \\ v_p(c) = 1}} p, \qquad d_2 = \frac{d}{d_1}. \tag{6.1}$$

In particular, we have $d_1 d_2^2 \mid c$. Thus we can write $c = d_1 c_2$, where $\gcd(d_1, c_2) = 1$ and $d_2^2 \mid c_2$. We begin by proving the following estimate.

Lemma 6.1. *Let c, d be positive integers such that d is square-free and $d \mid c$. Let $\boldsymbol{k} \in \mathbb{Z}^n$ such that $d \mid F(\boldsymbol{k})$. Then*

$$\#\left\{\boldsymbol{x} \in (\mathbb{Z}/c\mathbb{Z})^n : \begin{array}{l} \boldsymbol{x} \equiv \boldsymbol{k} \bmod d \\ F(\boldsymbol{x}) \equiv 0 \bmod c \end{array}\right\} \ll \left(\frac{c}{d}\right)^{n-1+\varepsilon} \gcd(d_2, \nabla F(\boldsymbol{k})) \gcd(d_2, \boldsymbol{k}),$$

for any $\varepsilon > 0$, where $d_2 = d/d_1$ and d_1 is defined in (6.1).

Proof. Let $N(c, d)$ be the quantity to be counted in the lemma. By the Chinese remainder theorem, it suffices to treat the case where $c = p^\ell$ and $d = p^j$, where $j \in \{0, 1\}$ and $\ell \geqslant j$. If $j = 0$, additive orthogonality gives

$$N(p^\ell, 1) = p^{-\ell} \sum_{1 \leqslant a \leqslant p^\ell} \sum_{\boldsymbol{x} \bmod p^\ell} e_{p^\ell}(aF(\boldsymbol{x})) = p^{-\ell} \sum_{h=0}^{\ell} p^{(\ell-h)n} S_{p^h}(\boldsymbol{0}) = p^{(n-1)\ell} \sum_{h=0}^{\ell} p^{-hn} S_{p^h}(\boldsymbol{0})$$

where $h = \ell - v_p(a)$. Thus $N(p^\ell, 1) \leqslant p^{(n-1)\ell} \sum_{q \geqslant 1} q^{-n} |S_q(\boldsymbol{0})| \ll_F p^{(n-1)\ell}$ by (5.11).

Suppose $j = 1$. For $\ell = 1$ we have $N(p, p) = 1$. Now suppose $\ell \geqslant 2$. For any fixed prime p we have $N(p^\ell, p) \leqslant N(p^\ell, 1) \ll_F p^{(n-1)\ell} \ll_p p^{(n-1)(\ell-1)}$. At every other prime we have $p > 3$ and

$$F(\boldsymbol{u}) = 0, \quad \boldsymbol{u} \in \overline{\mathbb{F}}_p^n \setminus \{\boldsymbol{0}\} \quad \Longrightarrow \quad \operatorname{rank} H(\boldsymbol{u}) \geqslant 2, \tag{6.2}$$

by spreading out the claim $V_1 \cap \widehat{X} = \{\mathbf{0}\}$ of Corollary 3.9.

If $p \mid \boldsymbol{k}$, homogeneity gives $N(p^2, p) = p^n = p^{(\ell-1)(n-1)+1}$ for $\ell = 2$, and

$$N(p^\ell, p) = p^{2n} N(p^{\ell-3}, 1) \ll_F p^{2+2(n-1)} p^{(\ell-3)(n-1)} = p^{(\ell-1)(n-1)+2}$$

for $\ell \geqslant 3$. If $p \nmid \boldsymbol{k}$ and $p \nmid \nabla F(\boldsymbol{k})$, Hensel lifting gives $N(p^\ell, p) = p^{(\ell-1)(n-1)}$.

It remains to consider the case where $p \nmid \boldsymbol{k}$ and $p \mid \nabla F(\boldsymbol{k})$. Write $\boldsymbol{x} = \boldsymbol{k} + p\boldsymbol{y}$. If $p^2 \nmid F(\boldsymbol{k})$, then $N(p^\ell, p) = 0$ for all $\ell \geqslant 2$. Now suppose $p^2 \mid F(\boldsymbol{k})$. Then

$$p^{-2} F(\boldsymbol{k} + p\boldsymbol{y}) = p^{-2} F(\boldsymbol{k}) + p^{-1} \nabla F(\boldsymbol{k}) \cdot \boldsymbol{y} + \tfrac{1}{2} \boldsymbol{y}^t H(\boldsymbol{k}) \boldsymbol{y} + pF(\boldsymbol{y}) \in \mathbb{Z}_p[\boldsymbol{y}].$$

Here $\operatorname{rank} H(\boldsymbol{k} \bmod p) \geqslant 2$ by (6.2). Applying Lemma 6.2 with $s = \ell - 2 \geqslant 0$ therefore gives

$$N(p^\ell, p) = p^n \#\{\boldsymbol{y} \bmod p^s : p^{-2} F(\boldsymbol{k} + p\boldsymbol{y}) \equiv 0 \bmod p^s\} \leqslant p^n (\ell - 1) p^{(\ell-2)(n-1)}.$$

This is satisfactory, since $\ell - 1 \ll p^{(\ell-1)\varepsilon}$ and $p^n p^{(\ell-2)(n-1)} = p^{(\ell-1)(n-1)+1}$. □

Lemma 6.2. *Suppose p is odd. Let $f_i \in \mathbb{Z}_p[\boldsymbol{y}]$ be homogeneous of degree i, for $0 \leqslant i \leqslant 3$. Suppose* $\operatorname{rank}(f_2 \bmod p) \geqslant 2$. *Let $g := f_0 + f_1 + f_2 + pf_3$. Then, for every $s \geqslant 0$,*

$$\#\{\boldsymbol{y} \bmod p^s : g(\boldsymbol{y}) \equiv 0 \bmod p^s\} \leqslant (s+1) p^{s(n-1)}. \tag{6.3}$$

Proof. Since $p \neq 2$, we may diagonalise the quadratic form f_2 over $\mathbb{Z}_p$. By permuting coordinates, assume the coefficients are listed in increasing order of valuation. Then $f_2 = \sum_{1 \leqslant i \leqslant n} c_i y_i^2$ with $n \geqslant 2$ and $v_p(c_1) = v_p(c_2) = 0$, say. Fix $y_3, \dots, y_n$ modulo p^s. After collecting homogeneous terms in y_1, y_2, the resulting binary polynomial again has the form $h_0 + h_1 + h_2 + ph_3$, with each h_i homogeneous of degree i, and $h_2 \equiv c_1 y_1^2 + c_2 y_2^2 \bmod p$. Thus its quadratic part has rank two modulo p. The binary case of the lemma gives at most $(s+1)p^s$ solutions for each of the $p^{s(n-2)}$ choices of the remaining variables, proving the required bound.

For $n = 2$, we induct on $s \geqslant 0$. If $s \leqslant 1$ the bound (6.3) is well known, so assume $s \geqslant 2$. Suppose again that $f_2 = c_1 y_1^2 + c_2 y_2^2$. By completing the square, we may assume that $p \mid f_1$. If $p \nmid f_0$, then $g \equiv 0 \bmod p$ is a smooth affine conic, so $\#\{\boldsymbol{y} \in \mathbb{F}_p^2 : g = 0\} \leqslant p + 1 \leqslant (s+1)p$, and (6.3) holds by Hensel's lemma. If $p \mid f_0$, then $g \equiv 0 \bmod p$ is smooth away from the origin. The contribution to (6.3) from $\boldsymbol{y} \equiv \mathbf{0} \bmod p$ is zero unless $p^2 \mid f_0$, in which case it is at most $p^2(s-1)p^{s-2}$, by the $s - 2$ case of (6.3) for the polynomial

$$p^{-2} g(p\boldsymbol{z}) = p^{-2} f_0 + p^{-1} f_1(\boldsymbol{z}) + f_2(\boldsymbol{z}) + p^2 f_3(\boldsymbol{z}).$$

The contribution to (6.3) from $\boldsymbol{y} \not\equiv \mathbf{0} \bmod p$ is at most $p^{s-1}(1+1)p$, by Hensel's lemma and the $s = 1$ case of (6.3). Adding the two contributions gives (6.3). □

We will apply Lemma 6.1 twice in this paper, each time together with the same auxiliary bound, Lemma 6.3 below, whose statement requires some notation. For $0 \leqslant \varrho \leqslant n$, let

$$\delta_n(\varrho) := \begin{cases} 0, & 0 \leqslant \varrho \leqslant 1, \\ \min\{2\varrho - 2, \lfloor n/3 \rfloor + \varrho - 1, n - 2\}, & 2 \leqslant \varrho < 2\lceil n/3 \rceil, \\ n - 1, & \varrho \geqslant 2\lceil n/3 \rceil. \end{cases} \tag{6.4}$$

Then for all primes p,

$$\#\{\boldsymbol{x} \in \mathbb{F}_p^n : F(\boldsymbol{x}) = 0,\ \operatorname{rank} H(\boldsymbol{x}) = \varrho\} \ll_F p^{\delta_n(\varrho)}, \tag{6.5}$$

by the Lang–Weil estimate and Corollary 3.9. Let

$$\gamma_n(\varrho) := \mathbf{1}_{\delta_n(\varrho) \leqslant n-5} + \mathbf{1}_{\varrho = 0}. \tag{6.6}$$

Lemma 6.3. *If $0 \leqslant \varrho \leqslant n$, then*

$$\sum_{\substack{\boldsymbol{u} \in \mathbb{F}_p^n \\ F(\boldsymbol{u}) = 0 \\ \operatorname{rank} H(\boldsymbol{u}) = \varrho}} \gcd(p, \nabla F(\boldsymbol{u})) \gcd(p, \boldsymbol{u}) \ll_F p^{\delta_n(\varrho) + \gamma_n(\varrho)}.$$

Proof. The case $\varrho = 0$ is immediate from (6.5), since $\delta_n(0) = 0 \leqslant n-5$ and $\gamma_n(0) = 2$. Now suppose $\varrho \geqslant 1$. Then $H(\boldsymbol{u})$ is nonzero, so $\boldsymbol{u} \not\equiv \mathbf{0} \bmod p$ and $\gcd(p, \boldsymbol{u}) = 1$. The case $\gamma_n(\varrho) \geqslant 1$ is then immediate from (6.5). Finally, suppose $\gamma_n(\varrho) = 0$. Then $\delta_n(\varrho) \geqslant n-4$ by (6.6). In view of (4.11), the Lang–Weil estimate gives

$$\#\{\boldsymbol{u} \bmod p : \nabla F(\boldsymbol{u}) \equiv 0 \bmod p\} \ll_F p^{n-5} \leqslant p^{\delta_n(\varrho)-1}.$$

On the other hand, there are $\ll p^{\delta_n(\varrho)}$ smooth residues $\boldsymbol{u}$ at hand, by (6.5). After multiplying by $\gcd(p, \nabla F(\boldsymbol{u}))$, we find that both the singular and smooth contributions are satisfactory. □

Let us now record an initial upper bound for $S_r(\boldsymbol{v})$, in which we recall the factorisation $r = c^2 d$ of any squarefull r, with square-free d such that $d \mid c$.

Lemma 6.4. *Let $r = c^2 d$ be squarefull. Then*

$$|S_r(\boldsymbol{v})| \leqslant c^{n+1} d^{n/2+1} \sum_{a \in (\mathbb{Z}/c\mathbb{Z})^*} \sum_{\substack{\boldsymbol{a} \in (\mathbb{Z}/c\mathbb{Z})^n \\ c \mid F(\boldsymbol{a}) \\ c \mid (a\nabla F(\boldsymbol{a}) + \boldsymbol{v})}} N_d(\boldsymbol{a})^{1/2},$$

where $N_d(\boldsymbol{a})$ is given by (5.18).

Proof. Apply Lemmas 5.7 and 5.9 with $A := c$ and $T := d$, using the fact that $d \mid c$. □

Lemma 6.5. *We have*

$$\sum_{\substack{\boldsymbol{u} \in \mathbb{F}_p^n \\ F(\boldsymbol{u}) = 0}} N_p(\boldsymbol{u})^{1/2} \gcd(p, \nabla F(\boldsymbol{u})) \gcd(p, \boldsymbol{u}) \ll_F p^{3n/2 - \lceil n/3 \rceil - 1}. \tag{6.7}$$

Proof. Lemma 6.3 bounds the left-hand side by $\sum_{0 \leqslant \varrho \leqslant n} p^{\delta_n(\varrho) + \gamma_n(\varrho)} p^{(n-\varrho)/2}$. For $\varrho \geqslant 2\lceil n/3 \rceil$, we have $\delta_n(\varrho) = n-1$ and $\gamma_n(\varrho) = 0$, so

$$\delta_n(\varrho) + \gamma_n(\varrho) + \frac{n-\varrho}{2} = n - 1 + \frac{n-\varrho}{2} \leqslant \frac{3n}{2} - \lceil n/3 \rceil - 1.$$

For $\varrho = 0$, we have $\delta_n(\varrho) = 0$ and $\gamma_n(\varrho) = 2$, so $\delta_n(\varrho) + \gamma_n(\varrho) + (n-\varrho)/2 = 2 + n/2 \leqslant 3n/2 - \lceil n/3 \rceil - 1$, since $n \geqslant 10$. For $1 \leqslant \varrho < 2\lceil n/3 \rceil$, definitions (6.4) and (6.6) give

$$\delta_n(\varrho) - \varrho \leqslant \lfloor n/3 \rfloor - 1, \qquad \delta_n(\varrho) + 2\gamma_n(\varrho) = \delta_n(\varrho) + 2 \cdot \mathbf{1}_{\delta_n(\varrho) \leqslant n-5} \leqslant n - 2.$$

Consequently,

$$\begin{aligned} \delta_n(\varrho) + \gamma_n(\varrho) + \frac{n-\varrho}{2} &= \frac{n}{2} + \frac{\delta_n(\varrho) - \varrho}{2} + \frac{\delta_n(\varrho) + 2\gamma_n(\varrho)}{2} \\ &\leqslant n + \frac{\lfloor n/3 \rfloor - 3}{2} \\ &\leqslant \frac{3n}{2} - \lceil n/3 \rceil - 1, \end{aligned}$$

where the last inequality follows from $\lfloor n/3 \rfloor + 2\lceil n/3 \rceil \leqslant n+1$. □

In what follows it will be convenient to set

$$E_c(\mathcal{V}) = \max_{\boldsymbol{k} \in (\mathbb{Z}/c\mathbb{Z})^n} \#\{\boldsymbol{v} \in \mathcal{V} : \boldsymbol{v} \equiv \boldsymbol{k} \bmod c\}, \tag{6.8}$$

for any integer $c > 0$, for which we have the trivial bound $E_c(\mathcal{V}) \ll (1 + V/c)^n$. Let

$$L_j(c, d, \mathcal{V}) = \sum_{\boldsymbol{v} \in \mathcal{V}} c^{n+1} d^{n/2+1} \sum_{a \in (\mathbb{Z}/c\mathbb{Z})^*} \sum_{\substack{\boldsymbol{a} \in (\mathbb{Z}/c\mathbb{Z})^n \\ c \mid F(\boldsymbol{a}) \\ c \mid (a\nabla F(\boldsymbol{a}) + \boldsymbol{v})}} N_d(\boldsymbol{a})^{j/2}, \tag{6.9}$$

for integer $j \geqslant 0$. It follows from Lemma 6.4 that

$$\sum_{\boldsymbol{v} \in \mathcal{V}} |S_r(\boldsymbol{v})| \leqslant L_1(c, d, \mathcal{V}). \tag{6.10}$$

We are now ready to prove the following result.

Lemma 6.6. *Let $d \mid c$ with d square-free. We have*

$$L_1(c,d,\mathcal{V}) \ll_{F,\varepsilon} c^{2n+1+\varepsilon} d^{n-\lceil n/3\rceil+1} E_c(\mathcal{V}).$$

Proof. Interchanging the sums in (6.9) gives

$$L_1(c,d,\mathcal{V}) \leqslant c^{n+2} d^{n/2+1} E_c(\mathcal{V}) \sum_{\substack{\boldsymbol{a} \bmod c \\ c \mid F(\boldsymbol{a})}} N_d(\boldsymbol{a})^{1/2}.$$

Split the last sum into residue classes modulo d and apply Lemma 6.1. It is at most

$$\ll_{F,\varepsilon} (c/d)^{n-1+\varepsilon} \sum_{\substack{\boldsymbol{u} \bmod d \\ d \mid F(\boldsymbol{u})}} N_d(\boldsymbol{u})^{1/2} \gcd(d_2, \nabla F(\boldsymbol{u})) \gcd(d_2, \boldsymbol{u}).$$

By the Chinese remainder theorem and Lemma 6.5, the last sum is $\ll_{F,\varepsilon} d^{3n/2-\lceil n/3\rceil-1+\varepsilon}$. Combining these estimates proves the assertion. □

We now provide an alternative estimate based on Poisson summation. Put

$$a_n = \frac{n}{2} - \lceil n/3 \rceil - 1, \qquad b_n = \frac{\lfloor n/3 \rfloor - 1}{2}. \tag{6.11}$$

For $n \geqslant 10$ we have $0 \leqslant a_n \leqslant b_n$. The factors d_1, d_2 retain the meaning of (6.1).

Lemma 6.7. *Let $r = c^2 d$, where $d \mid c$ is square-free, and let $\mathcal{V} \subseteq \mathbb{Z}^n \cap (\boldsymbol{v}_0 + [-V,V]^n)$, with $V \geqslant 1$. Uniformly in $\boldsymbol{v}_0 \in \mathbb{R}^n$, we have*

$$\sum_{\boldsymbol{v} \in \mathcal{V}} |S_r(\boldsymbol{v})| \leqslant L_1(c,d,\mathcal{V}) \ll_{F,\varepsilon} r^{n/2+1+\varepsilon} d_1^{a_n} d_2^{b_n} (V + r^{1/3})^n.$$

Proof. The first inequality is (6.10). On replacing the box radius by $V + r^{1/3}$, the second inequality reduces to the claim

$$L_1(c,d,\mathcal{V}) \ll_{F,\varepsilon} r^{n/2+1+\varepsilon} d_1^{a_n} d_2^{b_n} V^n \qquad (V \geqslant r^{1/3}).$$

We henceforth assume $V \geqslant r^{1/3}$. If $c \leqslant V$, then $E_c(\mathcal{V}) \ll (V/c)^n$, and Lemma 6.6 gives

$$\begin{aligned} L_1(c,d,\mathcal{V}) &\ll_{F,\varepsilon} c^{n+1+\varepsilon} d^{n-\lceil n/3\rceil+1} V^n \\ &= r^{n/2+1} c^{\varepsilon} \frac{d}{c} d^{a_n} V^n \leqslant r^{n/2+1+\varepsilon} d_1^{a_n} d_2^{b_n} V^n, \end{aligned}$$

since $d/c \leqslant 1$, $d = d_1 d_2$, and $a_n \leqslant b_n$. We may therefore assume $c > V$.

Two auxiliary results will be useful. First, for integers $m \geqslant 1$ and real $R \geqslant 1$, Lemma 5.8 and goodness (Definition 3.3) give

$$\sum_{|\boldsymbol{h}| \leqslant R} N_m(\boldsymbol{h})^{1/2} \ll_{F,\varepsilon} m^{n/2} R^{\varepsilon} \sum_{j=0}^{n} (R^3/m)^{j/2} \ll_n R^{\varepsilon} m^{n/2} (1 + R^3/m)^{n/2}. \tag{6.12}$$

Second, on recalling the definition (5.18) of $N_p(\boldsymbol{u})$, it follows from combining the incidence bound (3.7) with the Lang–Weil estimate, that

$$\sum_{\boldsymbol{u} \in \mathbb{F}_p^n} N_p(\boldsymbol{u}) \ll_F p^{n+\lfloor n/3 \rfloor - 1}. \tag{6.13}$$

In the nonnegative sum (6.9) defining $L_1(c,d,\mathcal{V})$, weaken $c \mid F(\boldsymbol{a})$ to $d_1 \mid F(\boldsymbol{a})$, and majorise the box $\boldsymbol{v}_0 + [-V,V]^n$ by the smooth weight function $\exp(-\|\boldsymbol{v} - \boldsymbol{v}_0\|_2^2/V^2)$. For $a \in (\mathbb{Z}/c\mathbb{Z})^*$, put

$$\sigma(\boldsymbol{h}; a) = \sum_{\substack{\boldsymbol{a} \bmod c \\ d_1 \mid F(\boldsymbol{a})}} N_d(\boldsymbol{a})^{1/2} e_c(a\boldsymbol{h} \cdot \nabla F(\boldsymbol{a})).$$

Poisson summation in the class $\boldsymbol{v} \equiv -a\nabla F(\boldsymbol{a}) \bmod c$ gives

$$L_1(c,d,\mathcal{V}) \ll_n c^{n+1} d^{n/2+1} (V/c)^n \sum_{a \in (\mathbb{Z}/c\mathbb{Z})^*} \sum_{\boldsymbol{h} \in \mathbb{Z}^n} |\sigma(\boldsymbol{h};a)| \exp(-\pi^2 \|\boldsymbol{h}\|_2^2 V^2/c^2). \tag{6.14}$$

Here the Fourier factor $e_c(\boldsymbol{h}\cdot\boldsymbol{v}_0)$ has absolute value one, so all estimates remain uniform in $\boldsymbol{v}_0$.

Recall that $c = d_1c_2$, $\gcd(d_1,c_2)=1$, and $d_2^2 \mid c_2$. Write $\overline{c_2}$ for the inverse of c_2 modulo d_1, and $\overline{d_1}$ for the inverse of d_1 modulo c_2. The Chinese remainder theorem gives

$$\sigma(\boldsymbol{h};a) = \left(\sum_{\substack{\boldsymbol{u} \bmod d_1 \\ d_1 \mid F(\boldsymbol{u})}} N_{d_1}(\boldsymbol{u})^{1/2} e_{d_1}(a\overline{c_2}\boldsymbol{h}\cdot\nabla F(\boldsymbol{u})) \right) \left(\sum_{\boldsymbol{u} \bmod c_2} N_{d_2}(\boldsymbol{u})^{1/2} e_{c_2}(a\overline{d_1}\boldsymbol{h}\cdot\nabla F(\boldsymbol{u})) \right).$$

By Lemma 6.5, with its gcd factors omitted, the first factor is

$$\ll_{F,\varepsilon} d_1^{3n/2-\lceil n/3\rceil-1+\varepsilon} = d_1^{n+a_n+\varepsilon}.$$

For the second factor, Cauchy's inequality over $\boldsymbol{u} \bmod d_2$, followed by quadratic differencing, gives

$$\left| \sum_{\boldsymbol{u} \bmod c_2} N_{d_2}(\boldsymbol{u})^{1/2} e_{c_2}(a\overline{d_1}\boldsymbol{h}\cdot\nabla F(\boldsymbol{u})) \right|^2 \leqslant \left(\sum_{\boldsymbol{b} \bmod d_2} N_{d_2}(\boldsymbol{b}) \right) c_2^n N_{c_2/d_2}(\boldsymbol{h}).$$

This is the Cauchy–Schwarz and quadratic-differencing calculation following [BHB09, Eq. (5.14)]; no nonsingularity assumption is used in that calculation.

Since d_2 is square-free, combining the Chinese remainder theorem with (6.13) gives

$$\sum_{\boldsymbol{b} \bmod d_2} N_{d_2}(\boldsymbol{b}) \ll_{F,\varepsilon} d_2^{n+\lfloor n/3\rfloor-1+\varepsilon}.$$

Combining the two factors, and using $c/d = (d_1c_2)/(d_1d_2) = c_2/d_2$, yields

$$|\sigma(\boldsymbol{h};a)| \ll_{F,\varepsilon} d_1^{n+a_n+\varepsilon} c_2^{n/2} d_2^{n/2+b_n+\varepsilon} N_{c/d}(\boldsymbol{h})^{1/2} = (cd)^{n/2} d^{\varepsilon} d_1^{a_n} d_2^{b_n} N_{c/d}(\boldsymbol{h})^{1/2}. \tag{6.15}$$

Substitute (6.15) into (6.14) and truncate the Gaussian at $|\boldsymbol{h}| \ll (c/V)\log(2c)$; its tail is negligible by the trivial bound for $\sigma(\boldsymbol{h};a)$. Applying (6.12) with $m = c/d$ gives

$$\begin{aligned} L_1(c,d,\mathcal{V}) &\ll_{F,\varepsilon} c^{n/2+2+\varepsilon} d^{n+1} d_1^{a_n} d_2^{b_n} V^n \sum_{|\boldsymbol{h}| \ll (c/V)\log(2c)} N_{c/d}(\boldsymbol{h})^{1/2} \\ &\ll_{F,\varepsilon} r^{n/2+1+\varepsilon} d_1^{a_n} d_2^{b_n} V^n (1+r/V^3)^{n/2} \\ &\ll_{F,\varepsilon} r^{n/2+1+\varepsilon} d_1^{a_n} d_2^{b_n} V^n. \end{aligned}$$

In the last step we used $V \geqslant r^{1/3}$. This proves the result. □

6.2. **Squarefull exponential sums: further averaging over moduli.** For $R, V \geqslant 1$ and a set $\mathcal{V} \subseteq \mathbb{Z}^n \cap (\boldsymbol{v}_0 + [-V,V]^n)$, put

$$\Sigma(R,\mathcal{V}) = \sum_{\substack{r \asymp R \\ r \text{ squarefull}}} \sum_{\boldsymbol{v} \in \mathcal{V}} |S_r(\boldsymbol{v})|. \tag{6.16}$$

Retain a_n, b_n from (6.11), and set

$$\theta_n^{(2)} = \max\left\{ \frac{1}{2}, \frac{n-2\lceil n/3\rceil}{6} \right\}, \qquad \theta_n^{(3)} = \max\left\{ \frac{2}{5}, \frac{n-2\lceil n/3\rceil}{6} \right\}. \tag{6.17}$$

Lemma 6.8. *For every $\varepsilon > 0$, uniformly in $\boldsymbol{v}_0 \in \mathbb{R}^n$, we have*

$$\Sigma(R,\mathcal{V}) \ll_{F,\varepsilon} R^{n/2+1+\theta_n^{(2)}+\varepsilon} (V + R^{1/3})^n. \tag{6.18}$$

For cubefull moduli, we have the stronger estimate

$$\sum_{\substack{r\asymp R\\ r \text{ cubefull}}}\sum_{\boldsymbol{v}\in\mathcal{V}}|S_r(\boldsymbol{v})| \ll_{F,\varepsilon} R^{n/2+1+\theta_n^{(3)}+\varepsilon}(V+R^{1/3})^n. \tag{6.19}$$

In particular, for $n = 10$ the powers of R in these two bounds are $13/2+\varepsilon$ and $32/5+\varepsilon$, respectively. Both estimates include the zero frequency $\boldsymbol{v}=\mathbf{0}$.

Proof. For a squarefull integer $r = c^2d$, put $w_n(r) = d_1^{a_n}d_2^{b_n}$, with d_1, d_2 as in (6.1). Then w_n is multiplicative on squarefull integers, and

$$w_n(p^\nu) = \begin{cases} 1, & \nu \geqslant 2 \text{ even},\\ p^{a_n}, & \nu = 3,\\ p^{b_n}, & \nu \geqslant 5 \text{ odd}. \end{cases}$$

For real s sufficiently large, it follows that

$$\sum_{r \text{ squarefull}} \frac{w_n(r)}{r^s} = \prod_p \left(1 + p^{a_n-3s} + \frac{p^{-2s}+p^{b_n-5s}}{1-p^{-2s}}\right),$$

$$\sum_{r \text{ cubefull}} \frac{w_n(r)}{r^s} = \prod_p \left(1 + p^{a_n-3s} + \frac{p^{-4s}+p^{b_n-5s}}{1-p^{-2s}}\right).$$

The first product converges for

$$s > \max\left\{\frac{1}{2}, \frac{a_n+1}{3}, \frac{b_n+1}{5}\right\},$$

and the second for

$$s > \max\left\{\frac{1}{4}, \frac{a_n+1}{3}, \frac{b_n+1}{5}\right\}.$$

Now

$$\frac{a_n+1}{3} = \frac{n-2\lceil n/3\rceil}{6}, \qquad \frac{b_n+1}{5} = \frac{\lfloor n/3\rfloor+1}{10}.$$

For $n \geqslant 11$, is it then easy to check that $(a_n+1)/3 \geqslant \max\{1/2, (b_n+1)/5\}$. For $n = 10$, we have $(a_n+1)/3 = 1/3$ and $(b_n+1)/5 = 2/5$. Thus for any $n \geqslant 10$, the two convergence thresholds are exactly $\theta_n^{(2)}$ and $\theta_n^{(3)}$ from (6.17). Rankin's trick gives, for every $\eta > 0$,

$$\sum_{\substack{r\asymp R\\ r \text{ squarefull}}} w_n(r) \ll_{n,\eta} R^{\theta_n^{(2)}+\eta}, \qquad \sum_{\substack{r\asymp R\\ r \text{ cubefull}}} w_n(r) \ll_{n,\eta} R^{\theta_n^{(3)}+\eta}.$$

Finally, combining these estimates with Lemma 6.7 completes the proof. □

7. Lifting Smithwork

As hinted at in §1, the Poisson summation based estimate of §6 is strong for some collections of moduli q and vectors $\boldsymbol{v}$. However, for other q and $\boldsymbol{v}$, it does not fit well with the stratification estimates that we will provide in §10. To resolve this issue, it will be useful to modify the choice of the auxiliary lifting modulus A in the factorisation $q = A^2T$ of Lemma 5.7. Very roughly speaking, in §6 we took $A \approx q^{1/2}$ for a squarefull modulus q, whereas in the present section we will eventually take $A \approx q^{1/3}$ for a cubefull modulus q. The reduction in the size of A will slightly weaken our estimates on average over $\boldsymbol{v}$, while also making the estimates more applicable to relatively thin sets of vectors $\boldsymbol{v}$. The restriction to cubefull moduli will also force us to treat arbitrary cube-free moduli separately later, in §10. The precise manner in which we combine cube-free and cubefull estimates is somewhat complicated, and is done in §11.

The main result of this section is Lemma 7.7, concerning cubefull averages for $n = 10$. In fact, Lemma 7.7 is restricted to cubefull moduli whose prime factors are all sufficiently large, with the complementary case being deferred to §8. However, we begin more generally, assuming only that $n \geqslant 10$ and $X(\mathbb{Q}) = \emptyset$, and that q is an arbitrary positive integer. Our results are based

on the fact that the bound for $S_q(\boldsymbol{v})$ given in (5.14) depends only on $\boldsymbol{v} \bmod A$. Thus (5.14) controls $S_q(\boldsymbol{v})$ uniformly along a given residue class $\boldsymbol{v} \bmod A$.

Lemma 7.1. *Let $q = A^2T$, where $A, T \geqslant 1$. Then*

$$\sum_{\boldsymbol{a} \bmod A} \max_{\substack{\boldsymbol{v} \bmod q \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod A}} |S_q(\boldsymbol{v})| \leqslant A^{n+2}T \sum_{\substack{\boldsymbol{y} \bmod A \\ F(\boldsymbol{y}) \equiv 0 \bmod A}} \mathcal{G}_{A,T}(\boldsymbol{y}), \tag{7.1}$$

where $\mathcal{G}_{A,T}(\boldsymbol{y})$ is defined in (5.12).

Proof. Plug (5.14) into the left-hand side of (7.1). Switching the order of summation of the variables $\boldsymbol{a}, \boldsymbol{y} \bmod A$ gives (7.1), since $\sum_{a \in (\mathbb{Z}/A\mathbb{Z})^*} A^{n+1}T \leqslant A^{n+2}T$. □

Henceforth suppose that

$$\operatorname{rad}(T) \mid \operatorname{rad}(A). \tag{7.2}$$

For $a \geqslant 1$, $t \geqslant 0$, and $0 \leqslant i < a$, let

$$\lambda_i(a,t) := \begin{cases} \frac{1}{2}\mathbf{1}_{i<t}, & t \leqslant a, \\ \frac{1}{2}, & t > a, \quad i < 2a - t, \\ 1, & t > a, \quad i \geqslant 2a - t. \end{cases} \tag{7.3}$$

Lemma 7.2. *Suppose $v_p(A) = a \geqslant 1$ and $T = p^t$, where $t \geqslant 0$. Let the Smith normal form of $H(\boldsymbol{y}) \bmod p^a$ be*

$$\operatorname{diag}(p^{j_1}, \dots, p^{j_n}), \qquad 0 \leqslant j_\nu \leqslant a,$$

and put

$$c_i := \#\{\nu : j_\nu \leqslant i\} \qquad (0 \leqslant i < a).$$

If either $t \leqslant a$ or $p \neq 2$, then

$$\mathcal{G}_{A,T}(\boldsymbol{y}) \leqslant p^{nt - \sum_{i=0}^{a-1} \lambda_i(a,t) c_i}. \tag{7.4}$$

Proof. By (5.12) and (5.15), it suffices to prove the bound in (7.4) for $|\mathcal{S}_{A,T}(\boldsymbol{d}, \alpha, \boldsymbol{y})|$, uniformly in $\boldsymbol{d} \in (\mathbb{Z}/T\mathbb{Z})^n$ and $\alpha \in (\mathbb{Z}/T\mathbb{Z})^*$. Recall the definition of $N_d(\boldsymbol{y})$ from (5.18). If $t \leqslant a$, then by Lemma 5.9 we have

$$|\mathcal{S}_{A,T}(\boldsymbol{d}, \alpha, \boldsymbol{y})|^2 \leqslant T^n N_T(\boldsymbol{y}) = p^{nt} p^{\sum_{\nu=1}^n \min(j_\nu, t)}.$$

Here

$$\begin{aligned} \sum_{\nu=1}^n \min(j_\nu, t) &= \sum_{\nu=1}^n \#\{0 \leqslant i < t : i < j_\nu\} \\ &= \sum_{i=0}^{t-1} \#\{\nu : j_\nu > i\} = nt - \sum_{i=0}^{t-1} c_i = nt - 2\sum_{i=0}^{a-1} \lambda_i(a,t) c_i, \end{aligned}$$

by the $t \leqslant a$ case of (7.3). Thus (7.4) holds for $t \leqslant a$.

Now suppose $p \neq 2$ and $t > a$. The cubic term in (5.15) depends only on $\boldsymbol{z} \bmod p^{t-a}$. Writing $\boldsymbol{z} = \boldsymbol{v} + p^{t-a}\boldsymbol{x}$ gives

$$|\mathcal{S}_{A,T}(\boldsymbol{d}, \alpha, \boldsymbol{y})| \leqslant \sum_{\boldsymbol{v} \bmod p^{t-a}} \left| \sum_{\boldsymbol{x} \bmod p^a} e_{p^a}((\boldsymbol{d} + \alpha H(\boldsymbol{y})\boldsymbol{v}) \cdot \boldsymbol{x} + \alpha p^{t-a} Q_{\boldsymbol{y}}(\boldsymbol{x})) \right|$$

Translating $\boldsymbol{x}$ by $\boldsymbol{h} \in \ker(p^{t-a}H(\boldsymbol{y}) \bmod p^a)$ and averaging over $\boldsymbol{h}$, noting that $p^{t-a}Q_{\boldsymbol{y}}(\boldsymbol{h}) \equiv \overline{2}p^{t-a}H(\boldsymbol{y})\boldsymbol{h} \cdot \boldsymbol{h} \equiv 0 \bmod p^a$ since p is odd, shows that the $\boldsymbol{x}$-sum vanishes unless

$$\boldsymbol{d} + \alpha H(\boldsymbol{y})\boldsymbol{v} \in \ker(p^{t-a}H(\boldsymbol{y}) \bmod p^a)^\perp = \operatorname{Im}(p^{t-a}H(\boldsymbol{y}) \bmod p^a),$$

where we recall that $\ker(M)^\perp = \operatorname{Im}(M)$ holds for every symmetric matrix $M \bmod p^a$. After writing $H(\boldsymbol{y})$ in Smith normal form, we find that the number of admissible vectors $\boldsymbol{v} \bmod p^{t-a}$ is

either zero or $\prod_{\nu=1}^{n} p^{t-a-\min(t-a,a-j_\nu)} = \prod_{\nu=1}^{n} p^{\max(0,t-2a+j_\nu)}$. Lemma 5.9 bounds the remaining sum over $\boldsymbol{x} \bmod p^a$ for each admissible $\boldsymbol{v}$, giving in total

$$|\mathcal{S}_{A,T}(\boldsymbol{d},\alpha,\boldsymbol{y})| \leqslant p^{\sum_{\nu=1}^{n}\max(0,t-2a+j_\nu)}(p^{na}N_{p^a}(p^{t-a}\boldsymbol{y}))^{1/2}.$$

Put $h = \max\{2a-t,0\}$, so that $0 \leqslant h < a$. By the definition of c_i, we have

$$\sum_{\nu=1}^{n}\max(0,t-2a+j_\nu) = n(t-a) - \sum_{i=h}^{a-1} c_i,$$

and

$$N_{p^a}(p^{t-a}\boldsymbol{y}) = p^{\sum_{\nu=1}^{n}\min(t-a+j_\nu,a)} = p^{na-\sum_{i=0}^{h-1}c_i}.$$

Substituting into the preceding bound gives

$$|\mathcal{S}_{A,T}(\boldsymbol{d},\alpha,\boldsymbol{y})| \leqslant p^{nt-\frac{1}{2}\sum_{i=0}^{h-1}c_i-\sum_{i=h}^{a-1}c_i}.$$

For $t > a$, the weights $\lambda_i(a,t)$ are $1/2$ for $i < h$ and 1 for $i \geqslant h$. Maximising over $\boldsymbol{d}$ and α therefore gives the required bound. □

For $0 \leqslant \varrho \leqslant n$, let $\delta_n(\varrho)$ be as in (6.4) and let $\gamma_n(\varrho)$ be as in (6.6).

Lemma 7.3. *For $a \geqslant 1$ and $0 \leqslant \varrho \leqslant n$,*

$$\#\{\boldsymbol{x} \bmod p^a : F(\boldsymbol{x}) \equiv 0 \bmod p^a,\ \operatorname{rank} H(\boldsymbol{x} \bmod p) = \varrho\} \ll_{\varepsilon,F} p^{(a-1)(n-1+\varepsilon)+\delta_n(\varrho)+\gamma_n(\varrho)\mathbf{1}_{a\geqslant 2}}.$$

Proof. If $a = 1$, this is trivial by (6.5). If $a \geqslant 2$, we apply Lemma 6.1 with $c = p^a$ and $d = p$. Lemma 6.3 then completes the proof. □

For $j \geqslant 0$, let

$$\tau_n(j) := \begin{cases} 0, & j = 0, \\ \min\{n-1, \lfloor n/3 \rfloor + j - 1\}, & 1 \leqslant j \leqslant n-1, \\ n, & j \geqslant n. \end{cases} \tag{7.5}$$

Lemma 7.4. *For every prime p and matrix $\mathcal{T} \in \operatorname{Mat}_{n\times n}(\mathbb{F}_p)$,*

$$\#\{\boldsymbol{y} \in \mathbb{F}_p^n : \operatorname{rank}(\mathcal{T} + H(\boldsymbol{y})) \leqslant j\} \ll_F p^{\tau_n(j)}.$$

Proof. The case $j \geqslant n$ is trivial, since the rank condition is automatic and $\tau_n(j) = n$. Hence suppose $0 \leqslant j < n$. Let p be large in terms of F, so that Corollary 3.9 spreads out to the prime p. Consider the projective variety $W_{\mathcal{T},j} := \{[x_0 : \boldsymbol{y}] \in \mathbb{P}^n : \operatorname{rank}(x_0\mathcal{T} + H(\boldsymbol{y})) \leqslant j\}$, and identify $\mathbb{A}^n$ with the chart $x_0 \neq 0$ in $\mathbb{P}^n$ via $\boldsymbol{y} \mapsto [1 : \boldsymbol{y}]$. Then $W_{\mathcal{T},j} \cap \mathbb{A}^n = \{\boldsymbol{y} \in \mathbb{A}^n : \operatorname{rank}(\mathcal{T} + H(\boldsymbol{y})) \leqslant j\}$, and we have

$$\dim(W_{\mathcal{T},j} \cap \mathbb{A}^n) \leqslant \dim(W_{\mathcal{T},j}) \leqslant 1 + \dim(W_{\mathcal{T},j} \cap \{x_0 = 0\}) = \dim(V_j),$$

where V_j is defined in (3.2). But $\dim(V_j) \leqslant \tau_n(j)$, by Corollary 3.9. The Lang–Weil estimate completes the proof, uniformly in $\mathcal{T}$ and p, since the locus $\operatorname{rank}(\mathcal{T} + H(\boldsymbol{y})) \leqslant j$ is cut out by minors of uniformly bounded number and degree. The finitely many exceptional primes are absorbed into the implied constant. □

For $a \geqslant 1$ and any sequence $\boldsymbol{c} = (c_0, \dots, c_{a-1}) \in \mathbb{Z}^a$ with $0 \leqslant c_0 \leqslant \cdots \leqslant c_{a-1} \leqslant n$, set

$$\mathcal{C}_{n,a}(\boldsymbol{c}) := \min\left\{(a-1)(n-1) + \gamma_n(c_0)\mathbf{1}_{a\geqslant 2},\ \sum_{i=1}^{a-1}\tau_n(c_{i-1}+c_i)\right\}. \tag{7.6}$$

Lemma 7.5. *Let $a \geqslant 1$ and $\boldsymbol{c} = (c_0, \dots, c_{a-1}) \in \mathbb{Z}^a$ with $0 \leqslant c_0 \leqslant \cdots \leqslant c_{a-1} \leqslant n$. Let $0 \leqslant j_1 \leqslant \cdots \leqslant j_n \leqslant a$ be the unique integers satisfying*

$$\#\{\nu : j_\nu \leqslant i\} = c_i \qquad (0 \leqslant i < a).$$

For a prime p, let $\mathcal{P}_p(\boldsymbol{c})$ be the set of $\boldsymbol{x} \bmod p^a$ such that $F(\boldsymbol{x}) \equiv 0 \bmod p^a$ and $H(\boldsymbol{x}) \bmod p^a$ has Smith normal form $\operatorname{diag}(p^{j_1}, \dots, p^{j_n})$. Then, for every $\varepsilon > 0$ and all sufficiently large primes p in terms of F and ε, we have

$$\#\mathcal{P}_p(\boldsymbol{c}) \ll_{\varepsilon,F} p^{\delta_n(c_0)+\mathcal{C}_{n,a}(\boldsymbol{c})+\varepsilon a}.$$

Proof. Every $\boldsymbol{x} \in \mathcal{P}_p(\boldsymbol{c})$ satisfies $\operatorname{rank} H(\boldsymbol{x} \bmod p) = c_0$. Discarding the remaining Smith conditions and applying Lemma 7.3 with $\varrho = c_0$ gives the bound corresponding to the first entry in (7.6). For the second entry, we will prove by induction on $a \geqslant 1$ that

$$\#\mathcal{P}_p(\boldsymbol{c}) \leqslant e^{O_F(a)} p^{\delta_n(c_0)+\sum_{i=1}^{a-1} \tau_n(c_{i-1}+c_i)}$$

for all p. This will then imply Lemma 7.5 by taking $p^\varepsilon \geqslant e^{O_F(1)}$.

The base case $a = 1$ is simply (6.5). Now suppose $a \geqslant 2$. Assume $j_1 \leqslant \dots \leqslant j_n$ for convenience. Let $\boldsymbol{y} \bmod p^{a-1}$ be a vector in $\mathcal{P}_p(c_0, \dots, c_{a-2})$. Fix a lift $\boldsymbol{y} \bmod p^a$, invertible matrices $A, B \bmod p^a$, and a matrix $M \bmod p$, for which

$$A(H(\boldsymbol{y}) - p^{a-1}M)B \equiv \operatorname{diag}(p^{j_1}, \dots, p^{j_{c_{a-2}}}, 0, \dots, 0) \bmod p^a.$$

This is possible by the Smith normal form of $H(\boldsymbol{y}) \bmod p^{a-1}$. Our task is now to give an upper bound for the number of vectors $\boldsymbol{x} \equiv \boldsymbol{y} + p^{a-1}\boldsymbol{z} \bmod p^a$, with $\boldsymbol{z} \bmod p$, for which the matrix

$$AH(\boldsymbol{x})B \equiv \operatorname{diag}(p^{j_1}, \dots, p^{j_{c_{a-2}}}, 0, \dots, 0) + p^{a-1}A(M + H(\boldsymbol{z}))B \bmod p^a \tag{7.7}$$

has Smith normal form $\operatorname{diag}(p^{j_1}, \dots, p^{j_{c_{a-1}}}, 0, \dots, 0) \bmod p^a$. By elimination in the first c_{a-2} rows and columns, using the fact that $p^{a-1-j_i}p^{a-1} \equiv 0 \bmod p^a$ for all $1 \leqslant i \leqslant c_{a-2}$, we find that the right-hand side of (7.7) has the same Smith normal form as the matrix

$$\operatorname{diag}(p^{j_1}, \dots, p^{j_{c_{a-2}}}, p^{a-1}\pi_{n-c_{a-2}}),$$

where $\pi_{n-c_{a-2}}$ is the bottom-right $(n - c_{a-2}) \times (n - c_{a-2})$ block of $A(M + H(\boldsymbol{z}))B$. Thus

$$\operatorname{rank}(\pi_{n-c_{a-2}} \bmod p) = c_{a-1} - c_{a-2}.$$

Adding a row or a column to a matrix over $\mathbb{F}_p$ increases its rank by at most one, so

$$\begin{aligned}\operatorname{rank}(M + H(\boldsymbol{z}) \bmod p) &= \operatorname{rank}(A(M + H(\boldsymbol{z}))B \bmod p)\\ &\leqslant 2c_{a-2} + \operatorname{rank}(\pi_{n-c_{a-2}} \bmod p) = c_{a-1} + c_{a-2}.\end{aligned}$$

Lemma 7.4 leaves $O_F(p^{\tau_n(c_{a-1}+c_{a-2})})$ possibilities for $\boldsymbol{z}$. This completes the induction. □

For any sequence $\boldsymbol{\lambda} = (\lambda_0, \dots, \lambda_{a-1}) \in \mathbb{R}^a_{\geqslant 0}$, put

$$\mathfrak{M}_{n,a}(\boldsymbol{\lambda}) := \max_{\substack{\boldsymbol{c}=(c_0,\dots,c_{a-1})\in\mathbb{Z}^a\\ 0\leqslant c_0\leqslant\dots\leqslant c_{a-1}\leqslant n}} \left\{ \delta_n(c_0) + \mathcal{C}_{n,a}(\boldsymbol{c}) - \sum_{i=0}^{a-1} \lambda_i c_i \right\}. \tag{7.8}$$

For $a \geqslant 1$ and $t \geqslant 0$, recall $\lambda_i(a,t)$ from (7.3) and let

$$\Phi_n(a,t) := \mathfrak{M}_{n,a}\big((\lambda_i(a,t))_{0\leqslant i<a}\big), \qquad D_n(a,t) := (n+2)a + (n+1)t + \Phi_n(a,t). \tag{7.9}$$

Proposition 7.6. *Let $q \geqslant 1$ be squarefull. Suppose every prime $p \mid q$ is large enough in terms of F and ε. Choose $1 \leqslant a_p \leqslant \lfloor v_p(q)/2 \rfloor$ and let $t_p := v_p(q) - 2a_p$. If $A := \prod_{p|q} p^{a_p}$ and $T := \prod_{p|q} p^{t_p}$, then the condition (7.2) holds and*

$$\sum_{\boldsymbol{a} \bmod A} \max_{\substack{\boldsymbol{v} \bmod q\\ \boldsymbol{v}\equiv\boldsymbol{a} \bmod A}} |S_q(\boldsymbol{v})| \ll_{\varepsilon,F} q^\varepsilon \prod_{p|q} p^{D_n(a_p,t_p)}. \tag{7.10}$$

Proof. Plugging Lemma 7.2 into Lemma 7.1 using the T-multiplicativity (5.13) gives

$$\sum_{\boldsymbol{a} \bmod A} \max_{\substack{\boldsymbol{v} \bmod q\\ \boldsymbol{v}\equiv\boldsymbol{a} \bmod A}} |S_q(\boldsymbol{v})| \leqslant A^{n+2}T \prod_{p|q} \sum_{\substack{\boldsymbol{y} \bmod p^{a_p}\\ F(\boldsymbol{y})\equiv 0 \bmod p^{a_p}}} p^{nt_p - \sum_{i=0}^{a_p-1} \lambda_i(a_p,t_p)c_i}, \tag{7.11}$$

where $\boldsymbol{c} \in \mathbb{Z}^{a_p}$ is such that $\boldsymbol{y} \in \mathcal{P}_p(\boldsymbol{c})$ in the notation of Lemma 7.5. The number of possible vectors $\boldsymbol{c}$ is $O((a_p+1)^n)$, since each $\boldsymbol{c}$ corresponds to a unique Smith normal form modulo p^{a_p}. Therefore, by Lemma 7.5 and the divisor bound on q,

$$\sum_{\boldsymbol{a} \bmod A} \max_{\substack{\boldsymbol{v} \bmod q \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod A}} |S_q(\boldsymbol{v})| \ll_\varepsilon A^{n+2} T q^\varepsilon T^n \prod_{p \mid q} p^{\Phi_n(a_p, t_p)}. \tag{7.12}$$

The definition of $D_n(a,t)$ in (7.9) accounts for the powers of A and T. □

Lemma 7.7. *Assume $n = 10$. Let $r \geqslant 1$ be cubefull. Suppose every prime $p \mid r$ is large enough in terms of F and ε. Let*

$$b := \prod_{v_p(r) > 0} p^{\lfloor v_p(r)/3 \rfloor}, \qquad z_i := \prod_{\substack{v_p(r) > 0 \\ v_p(r) \equiv i \bmod 3}} p, \tag{7.13}$$

for $0 \leqslant i \leqslant 2$, so that $r = b^3 z_1 z_2^2$ and $z_0 z_1 z_2 \mid b$. Define

$$A := b z_2, \quad T := b z_1.$$

Then $r = A^2 T$, the condition (7.2) holds, and

$$\sum_{\boldsymbol{a} \bmod A} \max_{\substack{\boldsymbol{v} \bmod r \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod A}} |S_r(\boldsymbol{v})| \ll_{F,\varepsilon} r^{10+\varepsilon} z_0^{-2} z_1^{-4} z_2^{-1}. \tag{7.14}$$

Proof. It suffices to prove

$$\Phi_{10}(e,e) \leqslant 7e-2, \qquad \Phi_{10}(e,e+1) \leqslant 7e-5, \qquad \Phi_{10}(e+1,e) \leqslant 7e+7$$

for every integer $e \geqslant 1$. Indeed, if $p \mid r$ and $v_p(r) = 3e+y$ with $0 \leqslant y \leqslant 2$, then the pair $(v_p(A), v_p(T))$ is respectively (e,e), $(e,e+1)$ or $(e+1,e)$ according as $y = 0, 1$ or 2. Writing $a = v_p(A)$ and $t = v_p(T)$, it then follows from the $n = 10$ case of (7.9) that

$$D_{10}(a,t) \leqslant \begin{cases} 30e-2 = v_p(r^{10} z_0^{-2}), & y = 0, \\ 30e+6 = v_p(r^{10} z_1^{-4}), & y = 1, \\ 30e+19 = v_p(r^{10} z_2^{-1}), & y = 2. \end{cases}$$

Together with Proposition 7.6, this implies (7.14).

Write $\delta = \delta_{10}$, $\gamma = \gamma_{10}$ and $\tau = \tau_{10}$, for the functions defined in (6.4), (6.6) and (7.5), respectively. Their values at integers $0 \leqslant x \leqslant 10$ are given in the following table:

x	0	1	2	3	4	5	6	7	8	9	10
$\delta(x)$	0	0	2	4	6	7	8	8	9	9	9
$\gamma(x)$	2	1	1	1	0	0	0	0	0	0	0
$\tau(x)$	0	3	4	5	6	7	8	9	9	9	10.

Moreover, $\tau(x) = 10$ for every integer $x \geqslant 10$.

For an integer sequence $0 \leqslant c_0 \leqslant \cdots \leqslant c_{a-1} \leqslant 10$, put

$$x = c_0, \qquad P = \sum_{i=0}^{a-1} \lambda_i(a,t) c_i = \frac{1}{2} \sum_{i=0}^{e-1} c_i + \frac{1}{2} c_{e-1} \mathbf{1}_{y=1},$$

by (7.3), where the extra term $\frac{1}{2} c_{e-1}$ appears only when $(a,t) = (e, e+1)$. Define

$$U = \delta(x) + 9(a-1) + \gamma(x) - P, \qquad V = \delta(x) + \sum_{i=1}^{a-1} \tau(c_{i-1} + c_i) - P.$$

The definitions (7.9), (7.8) and (7.6) give

$$\Phi_{10}(a,t) \leqslant \max_{\boldsymbol{c}} \min\{U, V\}.$$

Here we have replaced $\gamma(x) \mathbf{1}_{a \geqslant 2}$ in (7.6) by the larger quantity $\gamma(x)$. Indeed, the argument below will handle all cases $a \geqslant 1$ uniformly.

First suppose $x \geqslant 4$. Then $\gamma(x) = 0$, and from the table of values of δ we find that

$$\delta(x) - x/2 \leqslant 5, \qquad \delta(x) - x \leqslant 2, \qquad 9 - x/2 \leqslant 7.$$

Since every $c_i \geqslant x$, we have $P \geqslant (e + \mathbf{1}_{y=1})x/2$ and these inequalities imply

(a,t)	upper bound for U	linear combination used
(e,e)	$5 + 7(e-1) = 7e - 2$	$(\delta(x) - x/2) + (e-1)(9 - x/2)$
$(e,e+1)$	$2 + 7(e-1) = 7e - 5$	$(\delta(x) - x) + (e-1)(9 - x/2)$
$(e+1,e)$	$14 + 7(e-1) = 7e + 7$	$(\delta(x) - x/2) + 9 + (e-1)(9 - x/2)$,

where the third column records the decomposition of U used to obtain the second column.

Now suppose $x \leqslant 3$. Put

$$R = \delta(x) - x/2 + \gamma(x)/2.$$

Checking the values of δ and γ gives $R \leqslant 3$ and $R - x/2 \leqslant 2$. Moreover, τ satisfies

$$\tau(c_{i-1} + c_i) \leqslant \tau(2c_i) \leqslant \min\{c_i + 5, 10\},$$

because $c_{i-1} \leqslant c_i$ and $c_i \geqslant 0$. From our formulas for U, V, P we obtain

$$(U+V)/2 = R + 9(a-1)/2 + \sum_{i=1}^{a-1} \tau(c_{i-1} + c_i)/2 - \sum_{i=1}^{e-1} c_i/2 - \mathbf{1}_{y=1} c_{e-1}/2.$$

Now plug in $\tau(c_{i-1} + c_i) \leqslant c_i + 5$ for $i \leqslant e-1$, and $\tau(c_{i-1} + c_i) \leqslant 5 + 5$ for $i = e$ if $a = e+1$. Since $9(a-1)/2 + 5(a-1)/2 = 7(a-1)$ and $\mathbf{1}_{y=1} c_{e-1}/2 \geqslant \mathbf{1}_{y=1} x/2$, we thus obtain

(a,t)	upper bound for $(U+V)/2$
(e,e)	$R + 7(e-1) \leqslant 7e - 4$
$(e,e+1)$	$R - x/2 + 7(e-1) \leqslant 7e - 5$
$(e+1,e)$	$R + 7e + \frac{5}{2} \leqslant 7e + \frac{11}{2} \leqslant 7e + 7.$

Since $\min\{U,V\} \leqslant U$ and $\min\{U,V\} \leqslant (U+V)/2$, the two cases establish all three required bounds for Φ_{10}. □

8. Localisation at small primes

In the last few results of §7, we restricted attention to moduli $q \geqslant 1$ supported on sufficiently large primes p, in order to accommodate the factor of $e^{O_F(\Omega(q))}$ that arose from the Smith-based Hensel lifting analysis in the proof of Lemma 7.5. Here, $\Omega(q)$ denotes the total number of prime factors of q (with multiplicity). The factor $e^{O_F(\Omega(q))}$ is perhaps unexpected, and certainly less commonly seen in Diophantine contexts compared to $e^{O_F(\omega(q))}$, where $\omega(q)$ denotes the number of distinct prime factors of q (without multiplicity). Nonetheless, the proof we have given naturally leads to the factor $e^{O_F(\Omega(q))}$ and so we are forced to confront small primes p separately, or else the factor $e^{O_F(\Omega(q))}$ would contribute a fixed power of q in our estimates whenever q is a large power of a fixed prime.

The goal of the present section is precisely to provide satisfactory estimates at any fixed prime $p = p_0$. It is easiest to do this after first localising the counting function in the circle method to convenient p-adic congruence classes and real boxes, as we record below in the formula (8.7). In the case of a real box this is done by choosing a suitable real weight function, a standard procedure that we will recall in §12. In the p-adic case, the analogue of choosing a weight function is simply to restrict to a union of congruence classes. Throughout this section, we assume $n \geqslant 10$ and $X(\mathbb{Q}) = \emptyset$. Then X is non-conical by Proposition 2.3.

Lemma 8.1. *Let p be a prime. Then there is a primitive vector $\boldsymbol{\xi}_p \in \mathbb{Z}_p^n$ such that*

$$F(\boldsymbol{\xi}_p) = 0, \qquad \nabla F(\boldsymbol{\xi}_p) \neq \mathbf{0}, \qquad \operatorname{rank}_{\mathbb{Q}_p} H(\boldsymbol{\xi}_p) = r(F) \geqslant 2\lceil n/3 \rceil.$$

Proof. It is known that every non-conical cubic form in $\geqslant 10$ variables has a smooth p-adic zero for all $p \leqslant \infty$; see for instance Theorem 2 of [Ple71]. By definition of $r(F)$, the locus on $X_{\mathbb{Q}_p}$ where $\operatorname{rank} H = r(F)$ is a nonempty Zariski-open subset. Since X is geometrically integral, this open subset meets the Zariski-dense set $X_{\mathrm{sm}}(\mathbb{Q}_p)$. Finally, $r(F) \geqslant 2\lceil n/3 \rceil$ by Lemma 3.4. □

Since $H(\boldsymbol{\xi}_p)$ is symmetric of rank $r(F)$ over the characteristic-zero field $\mathbb{Q}_p$, we may assume after permuting coordinates that its leading $r(F)\times r(F)$ principal minor is nonsingular. Write

$$\Delta_p(\boldsymbol{x}) := \det\left(\frac{\partial^2 F}{\partial x_i \partial x_j}(\boldsymbol{x})\right)_{1\leqslant i,j\leqslant r(F)}.$$

Choose $M_p \geqslant 1$ sufficiently large that, on the ball

$$B_p := \boldsymbol{\xi}_p + p^{M_p}\mathbb{Z}_p^n, \tag{8.1}$$

all points are primitive, one fixed component of ∇F is nonzero, and

$$v_p(\Delta_p(\boldsymbol{x})) = v_p(\Delta_p(\boldsymbol{\xi}_p)) \qquad (\boldsymbol{x}\in B_p). \tag{8.2}$$

To preserve the multiplicativity of $S_q(\boldsymbol{v})$ later on in Lemma 8.2, consider the unit orbit

$$\Omega_p := \mathbb{Z}_p^* B_p = \bigcup_{\lambda\in\mathbb{Z}_p^*} \lambda B_p. \tag{8.3}$$

By scalar symmetry, the smoothness and Hessian-rank properties on B_p extend to Ω_p.

For $k\geqslant 1$, let $\Omega_p[k]$ denote the image of Ω_p in $(\mathbb{Z}/p^k\mathbb{Z})^n$. For $\boldsymbol{v}\in\mathbb{Z}^n$, put

$$S_{p^k}^{\Omega_p}(\boldsymbol{v}) := \sum_{\alpha\in(\mathbb{Z}/p^k\mathbb{Z})^*}\sum_{\boldsymbol{x}\in\Omega_p[k]} e_{p^k}\big(\alpha F(\boldsymbol{x}) + \boldsymbol{v}\cdot\boldsymbol{x}\big). \tag{8.4}$$

For $\lfloor k/2\rfloor \geqslant M_p$, the proof of Lemma 6.4, with $c = p^{\lfloor k/2\rfloor}$, shows that

$$S_{p^k}^{\Omega_p}(\boldsymbol{v}) \ll_p c^{n+1}\sum_{a\in(\mathbb{Z}/c\mathbb{Z})^*} \#\{\boldsymbol{x} \bmod c : \boldsymbol{x}\in\Omega_p,\ a\nabla F(\boldsymbol{x}) \equiv -\boldsymbol{v} \bmod c\}.$$

By the p-adic implicit function theorem, there are only $O_p(1)$ choices for $(x_1,\dots,x_\rho)$ mod c once the residues $a, x_{\rho+1},\dots,x_n$ mod c are fixed, where $\rho = r(F)$. Thus

$$S_{p^k}^{\Omega_p}(\boldsymbol{v}) \ll_p c^{n+1}c^{1+n-r(F)} \ll_p p^{k(n+1-r(F)/2)}. \tag{8.5}$$

The finitely many remaining values of k are absorbed into the implied constant in (8.5).

We proceed to sketch how to restrict the circle method to appropriate residue classes at a finite set of primes. Let $\mathcal{P}_{\text{small}}$ be a finite set of primes, choose Ω_p as in (8.3), and put

$$W := \prod_{p\in\mathcal{P}_{\text{small}}} p^{M_p}. \tag{8.6}$$

The Chinese remainder theorem defines a subset $\Omega_H\subset(\mathbb{Z}/H\mathbb{Z})^n$ for any divisor $H\mid W$. In §12 we will use the circle method to study the counting function

$$N(P) := \sum_{\substack{\boldsymbol{x}\in\mathbb{Z}^n\\ F(\boldsymbol{x})=0\\ \boldsymbol{x} \bmod W\in\Omega_W}} w(\boldsymbol{x}/P), \tag{8.7}$$

for a suitable fixed compactly supported weight function $w\in C_c^\infty(\mathbb{R}^n)$. Proving a lower bound for (8.7) is enough for the existence of an integral zero. To this end, observe that the restricted local density at each $p\mid W$ is positive. Since every Ω_p contains a neighbourhood of the smooth zero $\boldsymbol{\xi}_p$, the p-adic implicit function theorem gives

$$\liminf_{j\to\infty} p^{-j(n-1)}\#\left\{\boldsymbol{x} \bmod p^j : \boldsymbol{x}\in\Omega_p[j],\ F(\boldsymbol{x})\equiv 0 \bmod p^j\right\} > 0. \tag{8.8}$$

For any integer $H\mid W$, let

$$S_{q,H}(\boldsymbol{v}) := \sum_{a\in(\mathbb{Z}/q\mathbb{Z})^*}\sum_{\substack{\boldsymbol{x} \bmod \operatorname{lcm}(q,H)\\ \boldsymbol{x} \bmod H\in\Omega_H}} e_q\big(aF(\boldsymbol{x})\big)e_{\operatorname{lcm}(q,H)}(\boldsymbol{v}\cdot\boldsymbol{x}). \tag{8.9}$$

The following lemma records the necessary multiplicativity properties of $S_{q,W}(\boldsymbol{v})$.

Lemma 8.2. *Let $q_0, q_1 \geqslant 1$. If $\gcd(Wq_0, q_1) = 1$, then*

$$S_{q_0q_1,W}(\boldsymbol{v}) = S_{q_0,W}(\boldsymbol{v})S_{q_1}(\boldsymbol{v}).$$

Now suppose $p \mid q_0 \Rightarrow p \mid W$. Then

$$S_{q_0,W}(\boldsymbol{v}) = \prod_{p \mid W} S_{p^{v_p(q_0)},p^{M_p}}(\boldsymbol{v}).$$

Moreover, for all $p \mid W$ with $v_p(q_0) \geqslant M_p$, we have

$$S_{p^{v_p(q_0)},p^{M_p}}(\boldsymbol{v}) = S^{\Omega_p}_{p^{v_p(q_0)}}(\boldsymbol{v}).$$

Finally, for all $p \mid W$ with $v_p(q_0) < M_p$, we have

$$S_{p^{v_p(q_0)},p^{M_p}}(\boldsymbol{v}) \ll_p 1.$$

Proof. Use the Chinese remainder theorem. Unit twists are removed using scalar symmetry in the definitions (8.3) and (8.9). □

Define the localised singular series

$$\mathfrak{S}_\Omega := \sum_{q \geqslant 1} \operatorname{lcm}(q, W)^{-n} S_{q,W}(\mathbf{0}). \tag{8.10}$$

Lemma 8.3. *The series $\mathfrak{S}_\Omega$ converges absolutely to a positive real number.*

Proof. In view of Theorem 5.4 and Lemma 8.2, it suffices to show that for each $p \mid W$ the Euler factor $\sum_{k \geqslant 0} \operatorname{lcm}(p^k, p^{M_p})^{-n} S_{p^k,p^{M_p}}(\mathbf{0})$ converges absolutely to a positive real number. The absolute convergence itself is clear from Lemma 8.2 and (8.5), since $r(F) \geqslant 2n/3 \geqslant 3$. It remains to exhibit positivity. First observe that if $j \geqslant M_p$ then

$$\#\left\{\boldsymbol{x} \bmod p^j : \boldsymbol{x} \in \Omega_p[j],\ F(\boldsymbol{x}) \equiv 0 \bmod p^j\right\} = \sum_{\substack{\boldsymbol{x} \bmod p^j \\ \boldsymbol{x} \bmod p^{M_p} \in \Omega_p[M_p]}} p^{-j} \sum_{1 \leqslant a \leqslant p^j} e_{p^j}(aF(\boldsymbol{x})).$$

Let $0 \leqslant k \leqslant j$. The total contribution to the right-hand side from a with $v_p(a) = j - k$ is

$$\sum_{\substack{\boldsymbol{x} \bmod p^j \\ \boldsymbol{x} \bmod p^{M_p} \in \Omega_p[M_p]}} p^{-j} \sum_{\substack{1 \leqslant b \leqslant p^k \\ \gcd(b,p^k)=1}} e_{p^k}(bF(\boldsymbol{x})) = p^{-j} p^{n(j-\max(k,M_p))} S_{p^k,p^{M_p}}(\mathbf{0}),$$

by counting lifts modulo p^j of residues modulo $p^{\max(k,M_p)}$. It follows that

$$\sum_{0 \leqslant k \leqslant j} \operatorname{lcm}(p^k, p^{M_p})^{-n} S_{p^k,p^{M_p}}(\mathbf{0}) = p^{-j(n-1)} \#\left\{\boldsymbol{x} \bmod p^j : \boldsymbol{x} \in \Omega_p[j],\ F(\boldsymbol{x}) \equiv 0 \bmod p^j\right\}.$$

Taking $j \to \infty$ completes the proof, in view of the lower bound (8.8). □

The following variant of Proposition 7.6 will also be useful in the circle method.

Lemma 8.4. *Fix a prime p, and let Ω_p be the neighbourhood chosen in (8.3). Let $k \geqslant 0$ and $a := \lceil k/3 \rceil$. Then*

$$\sum_{\boldsymbol{a} \bmod p^a} \max_{\substack{\boldsymbol{v} \bmod p^{\max(k,M_p)} \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod p^a}} |S_{p^k,p^{M_p}}(\boldsymbol{v})| \ll_{F,p} p^{k(n+(4-r(F))/6)}. \tag{8.11}$$

Proof. For any fixed k, this is trivial since $S_{p^k,p^{M_p}}(\boldsymbol{v}) \ll_{p,k} 1$. Thus suppose $k \geqslant 3M_p$, and let $t := k - 2a$. On Ω_p, a fixed $r(F) \times r(F)$ Hessian minor has bounded valuation. Therefore, if $\boldsymbol{y} \in \Omega_p$, then $H(\boldsymbol{y})$ has at least $r(F)$ Smith invariant factors p^j with $j \ll_p 1$. Since $0 \leqslant t \leqslant a$, it follows from Lemma 7.2 that $\mathcal{G}_{p^a,p^t}(\boldsymbol{y}) \ll_{F,p} p^{nt - r(F)t/2}$. Moreover, the p-adic implicit function theorem gives

$$\#\{\boldsymbol{y} \bmod p^a : \boldsymbol{y} \in \Omega_p,\ F(\boldsymbol{y}) \equiv 0 \bmod p^a\} \ll_{F,p} p^{(n-1)a}.$$

The proof of Lemma 7.1, with the restriction to Ω_p retained, now gives

$$\sum_{\boldsymbol{a} \bmod p^a} \max_{\substack{\boldsymbol{v} \bmod p^k \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod p^a}} |S_{p^k}^{\Omega_p}(\boldsymbol{v})| \ll_{F,p} p^{(n+2)a} p^t p^{(n-1)a} p^{nt - r(F)t/2}.$$

Here $p^a, p^t \asymp_p p^{k/3}$, so (8.11) follows. □

9. Second moments over linear spaces

Throughout this section suppose that $n \geqslant 10$ and $X(\mathbb{Q}) = \emptyset$. In particular, F is absolutely irreducible by Proposition 2.3. For a prime power $q = p^a$, a nontrivial additive character ψ_q of $\mathbb{F}_q$, and a vector $\boldsymbol{v} \in \mathbb{F}_q^n$, put

$$T_q(\boldsymbol{v}) = T(\psi_q, F; \boldsymbol{v}) := \sum_{\substack{\boldsymbol{x} \in \mathbb{F}_q^n \\ F(\boldsymbol{x}) = 0}} \psi_q(\boldsymbol{x} \cdot \boldsymbol{v}) - q^{n-1} \mathbf{1}_{\boldsymbol{v} = \mathbf{0}}. \tag{9.1}$$

When $q = p$ and $\psi_p = e_p$, orthogonality in the outer sum of (1.1) gives

$$S_p(\boldsymbol{v}) = p T_p(\boldsymbol{v}), \tag{9.2}$$

even when $\boldsymbol{v} = \mathbf{0}$. We will be interested in the second moment of $T_q(\boldsymbol{v})$ over certain linear spaces.

Given a linear space L over $\mathbb{Q}$, we may define $L(\mathbb{F}_{p^a})$ by choosing an integral model of L. For any two fixed choices of integral models of L, the sets $L(\mathbb{F}_{p^a})$ then coincide for all sufficiently large primes p. For brevity, we thus omit any further discussion of integral models of L.

Lemma 9.1. *Let $L \subset \mathbb{Q}^n$ be a linear space of dimension $n - 2$. For every sufficiently large prime p, every $q = p^a$, and every nontrivial ψ_q, one has*

$$\sum_{\boldsymbol{v} \in L(\mathbb{F}_q)} |T_q(\boldsymbol{v})|^2 \ll_{F,L} q^{2n-3}. \tag{9.3}$$

Proof. Let $X_1, \dots, X_n$ and $V_1, \dots, V_n$ be the generic $\boldsymbol{x}$ and $\boldsymbol{v}$ variables, respectively. After a change of variables defined over $\mathbb{Q}$, with the corresponding dual change in the $\boldsymbol{x}$-variables, we may assume that $L = \{V_1 = V_2 = 0\}$. Moreover, observe that for every $\boldsymbol{v} = (0, 0, \boldsymbol{v}') \in L(\mathbb{F}_q)$ we can rewrite

$$\begin{aligned} T_q(\boldsymbol{v}) &= \sum_{\substack{\boldsymbol{x} \in \mathbb{F}_q^n \\ F(\boldsymbol{x}) = 0}} \psi_q(\boldsymbol{x} \cdot \boldsymbol{v}) - q^{n-1} \mathbf{1}_{\boldsymbol{v} = \mathbf{0}} \\ &= \sum_{\boldsymbol{x}' \in \mathbb{F}_q^{n-2}} \psi_q(\boldsymbol{v}' \cdot \boldsymbol{x}') \sum_{\substack{(x_1, x_2) \in \mathbb{F}_q^2 \\ F(x_1, x_2, \boldsymbol{x}') = 0}} 1 - q^{n-1} \mathbf{1}_{\boldsymbol{v} = \mathbf{0}} \\ &= \sum_{\boldsymbol{x}' \in \mathbb{F}_q^{n-2}} \big(\nu_F(\boldsymbol{x}') - q\big) \psi_q(\boldsymbol{v}' \cdot \boldsymbol{x}'), \end{aligned}$$

where

$$\nu_F(\boldsymbol{x}') = \#\{(x_1, x_2) \in \mathbb{F}_q^2 : F(x_1, x_2, \boldsymbol{x}') = 0\},$$

and $\boldsymbol{x}' = (x_3, \dots, x_n)$. Hence

$$\begin{aligned} \sum_{\substack{\boldsymbol{v} \in \mathbb{F}_q^n \\ v_1 = v_2 = 0}} |T_q(\boldsymbol{v})|^2 &= \sum_{\substack{\boldsymbol{v} \in \mathbb{F}_q^n \\ v_1 = v_2 = 0}} \sum_{(\boldsymbol{x}', \boldsymbol{y}') \in \mathbb{F}_q^{2n-4}} \big(\nu_F(\boldsymbol{x}') - q\big)\big(\nu_F(\boldsymbol{y}') - q\big) \psi_q\big(\boldsymbol{v}' \cdot (\boldsymbol{x}' - \boldsymbol{y}')\big) \\ &= q^{n-2} \sum_{\boldsymbol{x}' \in \mathbb{F}_q^{n-2}} \big(\nu_F(\boldsymbol{x}') - q\big)^2, \end{aligned}$$

by averaging over $\boldsymbol{v}$.

Since $X(\mathbb{Q}) = \emptyset$, we have $F(X_1, X_2, 0, \dots, 0) \neq 0$, so $\deg F(X_1, X_2, \boldsymbol{x}') = 3$ for all $\boldsymbol{x}' \in \mathbb{F}_q^{n-2}$ if p is sufficiently large. By Noether's lemma on absolute irreducibility, as in [Sch76, Ch. V, Thm. 2A], there exists a polynomial $G_{F,L}(\mathbf{X}') \in \mathbb{Z}[X_3, \dots, X_n]$ such that

$$G_{F,L}(\boldsymbol{x}') \neq 0 \quad \Longrightarrow \quad F(X_1, X_2, \boldsymbol{x}') \text{ is irreducible over } \overline{\mathbb{F}}_q,$$

for every sufficiently large prime p and every $q = p^a$, where $G_{F,L}(\boldsymbol{x}')$ is evaluated in $\mathbb{F}_q$. Moreover, $G_{F,L}$ can be chosen nonzero because the generic specialization is absolutely irreducible, as we show in Lemma 9.2 below.

For sufficiently large p, the reduction of $G_{F,L}$ is nonzero, so there are $O_{F,L}(q^{n-3})$ vectors $\boldsymbol{x}' \in \mathbb{F}_q^{n-2}$ at which $G_{F,L}$ vanishes. For each such $\boldsymbol{x}'$, the polynomial $F(X_1, X_2, \boldsymbol{x}')$ has leading cubic part $F(X_1, X_2, 0, \dots, 0) \neq 0$, provided that p is large enough. Thus $\nu_F(\boldsymbol{x}') \ll q$ trivially. For the remaining vectors $\boldsymbol{x}'$, the Lang–Weil bound gives $|\nu_F(\boldsymbol{x}') - q| \ll q^{1/2}$. Hence

$$\begin{aligned}\sum_{\boldsymbol{v} \in L(\mathbb{F}_q)} |T_q(\boldsymbol{v})|^2 &\ll_{F,L} q^{n-2} \sum_{\substack{\boldsymbol{x}' \in \mathbb{F}_q^{n-2} \\ G_{F,L}(\boldsymbol{x}') = 0}} q^2 + q^{n-2} \sum_{\substack{\boldsymbol{x}' \in \mathbb{F}_q^{n-2} \\ G_{F,L}(\boldsymbol{x}') \neq 0}} q \\ &\ll_{F,L} q^{2n-3},\end{aligned}$$

as claimed. □

Lemma 9.2. *Let $K = \overline{\mathbb{Q}}(T_3, \dots, T_n)$. Then $F(X_1, X_2, T_3, \dots, T_n)$ is irreducible in $\overline{K}[X_1, X_2]$.*

Proof. Since $F(X_1, X_2, T_3, \dots, T_n)$ is irreducible in $\overline{\mathbb{Q}}[T_3, \dots, T_n][X_1, X_2]$, it is irreducible in $K[X_1, X_2]$ by Gauss's lemma. If $F(X_1, X_2, T_3, \dots, T_n)$ were reducible in $\overline{K}[X_1, X_2]$, then by Galois theory it would have three $\mathrm{Gal}(\overline{K}/K)$-conjugate linear factors $F_i = X_1 + b_i X_2 + c_i \in \overline{K}[X_1, X_2]$, say, possibly after switching the variables X_1 and X_2. Equating top-degree terms in X_1, X_2 gives $F(X_1, X_2, 0, \dots, 0) \in \overline{K}^* \prod_{i=1}^3 (X_1 + b_i X_2)$. Factoring $F(X_1, X_2, 0, \dots, 0)$ over $\overline{\mathbb{Q}}$ gives $b_i \in \overline{\mathbb{Q}} \subset K$. Thus $\mathrm{Gal}(\overline{K}/K)$ fixes each b_i, whence $b_1 = b_2 = b_3$. However, $X(\mathbb{Q}) = \emptyset$, so the binary cubic form $F(X_1, X_2, 0, \dots, 0) \in \mathbb{Q}[X_1, X_2]$ is irreducible, and hence square-free. This is a contradiction. The statement of the lemma follows. □

We next record precisely how the moment will be used. Fix a prime $\ell \neq p$ and a field isomorphism $\iota : \overline{\mathbb{Q}}_\ell \to \mathbb{C}$. Let $i : \widehat{X} = \{F = 0\} \hookrightarrow \mathbb{A}^n$, and let π_1, π_2 be the projections of $\mathbb{A}^n_{\boldsymbol{x}} \times \mathbb{A}^n_{\boldsymbol{v}}$. For a nontrivial additive character ψ of $\mathbb{F}_p$, put

$$K_{p,\psi} := R\pi_{2!}\big(\pi_1^* i_* \overline{\mathbb{Q}}_\ell \otimes \mathcal{L}_\psi(\boldsymbol{x} \cdot \boldsymbol{v})\big), \tag{9.4}$$

where we use ι to identify the values of ψ with elements of $\overline{\mathbb{Q}}_\ell$ when defining the Artin–Schreier sheaf $\mathcal{L}_\psi$. Here $\mathcal{L}_\psi(\boldsymbol{x} \cdot \boldsymbol{v})$ denotes the pullback of $\mathcal{L}_\psi$ along the map $(\boldsymbol{x}, \boldsymbol{v}) \mapsto \boldsymbol{x} \cdot \boldsymbol{v}$.

For every $q = p^a$, $a \geqslant 1$, and every $\boldsymbol{v} \in \mathbb{F}_q^n \setminus \{\mathbf{0}\}$, its trace is $T_q(\boldsymbol{v})$, where $\psi_q = \psi \circ \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}$. More precisely, by the construction of $K_{p,\psi}$ and the Grothendieck–Lefschetz trace formula,

$$T_q(\boldsymbol{v}) = t_{K_{p,\psi}}(\boldsymbol{v}; q) = \sum_i (-1)^i \iota \left(\mathrm{tr}\left(\mathrm{Fr}_{q,\boldsymbol{v}} \mid \mathcal{H}^i(K_{p,\psi})_{\overline{\boldsymbol{v}}}\right)\right), \tag{9.5}$$

where $\mathrm{Fr}_{q,\boldsymbol{v}}$ denotes geometric Frobenius. The sheaf inside $R\pi_{2!}$ in (9.4) is punctually pure of weight 0. Consequently, by the compact-support weight theorem [Del80, Theorem 3.3.1], each cohomology sheaf $\mathcal{H}^i(K_{p,\psi})$ is constructible and mixed with integer weights $\leqslant i$.

It will be useful to have the following *uniform* constructibility result as p varies.

Lemma 9.3. *The complement of the origin has a finite partition into smooth homogeneous locally closed sets defined over $\mathbb{Q}$, compatible with any prescribed finite collection of homogeneous closed sets, such that the cohomology sheaves of $K_{p,\psi}$ are lisse on every stratum for all sufficiently large p and every nontrivial ψ. Their total ranks are bounded independently of p and ψ.*

Proof. The arithmetic stratification and uniform rank bound are [FK01, Theorem 2.1], applied to the projection from $\widehat{X} \times \mathbb{A}^n$ and the function $(\boldsymbol{x}, \boldsymbol{v}) \mapsto \boldsymbol{x} \cdot \boldsymbol{v}$. Intersect first with the prescribed finite collection of closed sets. On each irreducible closure take a smooth dense open subset

on which all the cohomology sheaves are lisse. Since scalar multiplication on $\widehat{X}$ gives scalar equivariance of the Fourier complex, the union of the scalar multiples of this open subset has the same property and is homogeneous. Its complement has smaller dimension. Repeat on that complement. Dimension decreases at every step, so this gives finitely many smooth homogeneous strata. Refining a lisse stratum does not increase its stalk ranks. □

Remark 9.4. Since F is homogeneous, there exists an alternative sheaf-theoretic interpretation of $T_q(\boldsymbol{v})$ that involves constant coefficients instead of Artin–Schreier coefficients. On the other hand, the Artin–Schreier perspective is more general, so we have opted for that here.

Proposition 9.5. *Let $V \subset \mathbb{A}^n$ be a geometrically irreducible affine cone of codimension $r+1$ and degree d, defined over $\mathbb{Q}$, and consider the restriction $K_{p,\psi}|_{V_{\mathbb{F}_p}}$. Let $U \subset V \setminus \{\mathbf{0}\}$ be a nonempty smooth open set, defined over $\mathbb{Q}$, such that, for all but finitely many primes p, the sheaves $\mathcal{H}^i(K_{p,\psi})|_{U_{\mathbb{F}_p}}$ are lisse and the sum of their ranks is bounded independently of p. Then either*

$$|T_p(\boldsymbol{v})| \ll_{F,V,U} p^{(n+r-2)/2} \qquad \big(\boldsymbol{v} \in U(\mathbb{F}_p)\big)$$

for all sufficiently large primes p, or $d \geqslant r+1$.

Proof. Fix a sufficiently large prime p. From (9.5), we know that the restriction of T_{p^a} to $U(\mathbb{F}_{p^a})$ is an alternating sum of trace functions of lisse sheaves that are mixed with integer weights. Hence, an application of [Xu20, Theorem 3.5 and Remark 3.6] with $w = n+r-2$ gives that either

$$|T_{p^a}(\boldsymbol{v})| \leqslant C p^{a(n+r-2)/2} \qquad \big(a \geqslant 1,\ \boldsymbol{v} \in U(\mathbb{F}_{p^a})\big),$$

where C is a uniform upper bound for the sum of the ranks in the statement, or

$$\limsup_{a\to\infty} \frac{1}{p^{a(2n-2)}} \sum_{\boldsymbol{v}\in U(\mathbb{F}_{p^a})} |T_{p^a}(\boldsymbol{v})|^2 \geqslant 1,$$

since

$$\dim U + w + 1 = (n-r-1) + (n+r-2) + 1 = 2n-2.$$

In the first case, taking $a=1$ gives the desired bound. Let us therefore assume that the second alternative holds and, by contradiction, that $d \leqslant r$. Consider the projective variety associated to V, say $Y \subset \mathbb{P}^{n-1}$. Since Y has codimension $r+1$ in $\mathbb{P}^{n-1}$ and degree $d \leqslant r$, by applying [Swi73, Lemma 3 and Corollary] twice, it follows that Y is contained in a linear subspace $P \subset \mathbb{P}^{n-1}$ of codimension two. Moreover, since Y is defined over $\mathbb{Q}$, its linear span is also defined over $\mathbb{Q}$, and we may choose P to be defined over $\mathbb{Q}$. Let $L \subset \mathbb{A}^n$ be the linear subspace associated to P. Then $V \subset L$ and $\dim L = n-2$.

By Lemma 9.1, for every $a \geqslant 1$ we have

$$\sum_{\substack{\boldsymbol{v}\in L(\mathbb{F}_{p^a})\\ \boldsymbol{v}\neq\mathbf{0}}} |T_{p^a}(\boldsymbol{v})|^2 \ll_{F,L} p^{a(2n-3)}.$$

On the other hand, since $U \subset L \setminus \{\mathbf{0}\}$, positivity gives

$$\begin{aligned}
1 &\leqslant \limsup_{a\to\infty} \frac{1}{p^{a(2n-2)}} \sum_{\boldsymbol{v}\in U(\mathbb{F}_{p^a})} |T_{p^a}(\boldsymbol{v})|^2 \\
&\leqslant \limsup_{a\to\infty} \frac{1}{p^{a(2n-2)}} \sum_{\substack{\boldsymbol{v}\in L(\mathbb{F}_{p^a})\\ \boldsymbol{v}\neq\mathbf{0}}} |T_{p^a}(\boldsymbol{v})|^2 \\
&\ll \lim_{a\to\infty} p^{-a} \\
&= 0.
\end{aligned}$$

This gives a contradiction. □

10. Hybrid cube-free stratifications

Throughout this section $n \geqslant 10$ and $X(\mathbb{Q}) = \emptyset$.

10.1. **A fixed partition of the rational frequencies.** We first obtain a stratification using Proposition A.1. Recall from (9.2) that $S_p(\boldsymbol{v}) = pT_p(\boldsymbol{v})$. Combining Proposition A.1 with Corollary 4.4 and Proposition 5.1, we obtain a stratification

$$\mathbb{A}^n = Z_0 \supseteq Z_1 \supseteq \cdots \supseteq Z_{n-3}$$

of homogeneous closed sets Z_j (affine cones of projective varieties), for $0 \leqslant j \leqslant n-3$, with the following properties:

i) For $0 \leqslant j \leqslant n-4$ we have $\widehat{W}_j \subseteq Z_{j+1} \subseteq \widehat{B}_j$, in the notation of (4.1) and (4.6).

ii) For every $j = 0, \dots, n-3$, the variety Z_j has codimension at least j in $\mathbb{A}^n$. Moreover, $\dim Z_{n-5} \leqslant 4$, $\dim Z_{n-4} \leqslant 3$ and $Z_{n-3} = \{\mathbf{0}\}$.

iii) Fix integral models of Z_j. If $0 \leqslant j \leqslant n-6$, then for every $\boldsymbol{v} \in (Z_j \setminus Z_{j+1})(\mathbb{F}_p)$, we have

$$|S_p(\boldsymbol{v})| \ll_F p^{\frac{n+1+j}{2}}. \tag{10.1}$$

If $j = n-5$, then (10.1) holds for all $\boldsymbol{v} \in (Z_j \setminus Z_{j+2})(\mathbb{F}_p)$. If $j = n-4$, then (10.1) holds for all $\boldsymbol{v} \in Z_j(\mathbb{F}_p)$, including $\boldsymbol{v} = \mathbf{0}$.

iv) For every $j = 1, \dots, n-4$, if R_j is the Zariski closure of $Z_j(\mathbb{Q})$ in Z_j, then

$$Z_j(\mathbb{Q}) = R_j(\mathbb{Q}) \tag{10.2}$$

and $\dim R_j \leqslant n-j-1$. Moreover, $\dim R_{n-5} \leqslant 1$ and $R_{n-4} = \{\mathbf{0}\}$.

In particular, combining the inclusion $\widehat{W}_j \subseteq Z_{j+1}$ with Lemma 5.3, we find that if $0 \leqslant j \leqslant n-5$ then for every $\boldsymbol{v} \in \mathbb{Z}^n$ with $\boldsymbol{v} \bmod p \in (Z_j \setminus Z_{j+1})(\mathbb{F}_p)$, we have

$$|S_{p^2}(\boldsymbol{v})| \ll_F p^{n+1+j}. \tag{10.3}$$

Moreover, if $j = n-4$, then (10.3) holds for all $\boldsymbol{v} \in \mathbb{Z}^n$, including $\boldsymbol{v} \in p\mathbb{Z}^n$.

We now modify the stratification for square-free moduli so that it will be more convenient to work with. It will be more convenient for us to rearrange the vectors $\boldsymbol{v} \in \mathbb{Q}^n$ according to the weight of the exponential sum on each component of Z_j. For example, suppose that $L \subseteq Z_j$ is a linear space of dimension $n-j-1$, defined over $\mathbb{Q}$, with $L \not\subseteq Z_{j+1}$, where $1 \leqslant j \leqslant n-6$. By Lemma 9.3, there exists a smooth dense open subset

$$U \subseteq L \setminus Z_{j+1},$$

defined over $\mathbb{Q}$, on which the cohomology sheaves of $K_{p,\psi}$ are lisse for all sufficiently large primes p, with uniformly bounded total rank. Proposition 9.5 then gives

$$|S_p(\boldsymbol{v})| \ll_{F,L,U} p^{\frac{n+j}{2}} \qquad (\boldsymbol{v} \in U(\mathbb{F}_p)).$$

Hence we can assign $U(\mathbb{Q})$ to level $j-1$ in the rearranged partition, while leaving the lower-dimensional complement $L \setminus U$ in Z_j to be treated separately.

With this example in mind, for every $j = 1, \dots, n-6$, consider

$$L_j := \bigcup_{\substack{Y \text{ a geometric irreducible component of } R_j \\ Y \text{ defined over } \mathbb{Q}, \quad Y \not\subseteq Z_{j+1} \\ \dim Y = n-j-1, \quad \deg Y \leqslant j}} Y. \tag{10.4}$$

By Lemma 9.3, we can choose a smooth homogeneous dense open subset

$$U_j \subseteq L_j \setminus Z_{j+1}, \tag{10.5}$$

defined over $\mathbb{Q}$, such that the restrictions to U_j of the cohomology sheaves of $K_{p,\psi}$ are lisse for all sufficiently large primes p and every nontrivial ψ, with uniformly bounded total rank. If $L_j = \emptyset$, take $U_j = \emptyset$.

Recall the convention $Z_0 = \mathbb{A}^n$ from above. With the additional conventions

$$U_0 = U_{n-5} := \emptyset,$$

we define, for every $0 \leqslant j \leqslant n-6$,

$$P_j := \big(\big((Z_j \setminus Z_{j+1}) \setminus U_j\big) \cup U_{j+1}\big)\,(\mathbb{Q}) \setminus \{\mathbf{0}\}, \tag{10.6}$$

and put

$$P_{n-5} := Z_{n-5}(\mathbb{Q}) \setminus \{\mathbf{0}\}, \quad P_{n-4} = \{\mathbf{0}\}.$$

The sets $(Z_j \setminus Z_{j+1})(\mathbb{Q})$ for $0 \leqslant j \leqslant n-6$, together with the sets P_{n-5} and P_{n-4}, certainly partition $\mathbb{Q}^n$. Therefore, $\{P_j\}_{j=0}^{n-4}$ is also a partition of $\mathbb{Q}^n$. Indeed, each $U_j(\mathbb{Q})$, for $1 \leqslant j \leqslant n-6$, is removed from the set $(Z_j \setminus Z_{j+1})(\mathbb{Q})$ and assigned to the preceding set P_{j-1}. We record now some properties of this partition which will be used extensively later. For any set $Y \subseteq \mathbb{Q}^n$, let

$$N(Y, L, \boldsymbol{v}_0, m, \boldsymbol{b}) := \#\{\boldsymbol{v} \in Y \cap \mathbb{Z}^n : |\boldsymbol{v} - \boldsymbol{v}_0| \leqslant L,\ \boldsymbol{v} \equiv \boldsymbol{b} \bmod m\}.$$

Proposition 10.1. *Let $\{P_j\}_j$ be the partition of $\mathbb{Q}^n$ defined above. Then the following hold.*

i) For every $0 \leqslant j \leqslant n-5$ and every $\boldsymbol{v} \in P_j \cap \mathbb{Z}^n$, there exists a positive integer

$$\Delta(\boldsymbol{v}) \ll_F (2 + |\boldsymbol{v}|)^{O_F(1)}$$

such that

$$p \nmid \Delta(\boldsymbol{v}) \Rightarrow |S_p(\boldsymbol{v})| \ll_F p^{\frac{n+j+1}{2}}.$$

ii) Set

$$T = 1 + \frac{L}{m}.$$

Then, for every $m, L \geqslant 1$, $\boldsymbol{v}_0 \in \mathbb{R}^n$, and every class $\boldsymbol{b} \bmod m$, one has

$$N(P_j, L, \boldsymbol{v}_0, m, \boldsymbol{b}) \ll_{F,\varepsilon} \begin{cases} T^n & \text{if } j = 0, \\ T^{n-j-2+\frac{1}{j+1}+\varepsilon} & \text{if } 1 \leqslant j \leqslant 2, \\ T^{n-j-2+\varepsilon} & \text{if } 3 \leqslant j \leqslant n-6, \\ T & \text{if } j = n-5. \end{cases} \tag{10.7}$$

Proof. We start by establishing part (i). Assume that $\boldsymbol{v} \in P_j \cap \mathbb{Z}^n$, initially with $0 \leqslant j \leqslant n-6$. Fix an integer $N \geqslant 1$, depending only on F, such that all the chosen models and local bounds are valid for $p \nmid N$. By combining the bound (10.1) with Proposition 9.5, we obtain

$$|S_p(\boldsymbol{v})| \ll_F p^{\frac{n+j+1}{2}}$$

whenever $p \nmid N$, the reduction of $\boldsymbol{v}$ is nonzero, and

$$\boldsymbol{v} \bmod p \notin (Z_{j+1} \setminus U_{j+1})(\mathbb{F}_p). \tag{10.8}$$

By the definition of P_j, either $\boldsymbol{v} \notin Z_{j+1}$ or $\boldsymbol{v} \in U_{j+1}$. In the first case, choose g from a fixed finite family of integral defining equations for Z_{j+1} such that $g(\boldsymbol{v}) \neq 0$. Then, for $p \nmid N$,

$$p \nmid g(\boldsymbol{v}) \quad \Longrightarrow \quad \boldsymbol{v} \bmod p \notin Z_{j+1}(\mathbb{F}_p).$$

In the second case, U_{j+1} is open in L_{j+1}, so $L_{j+1} \setminus U_{j+1}$ is closed. Choose g from a fixed finite family of integral defining equations for this complement such that $g(\boldsymbol{v}) \neq 0$. Since $\boldsymbol{v} \in L_{j+1}$, its reduction belongs to $L_{j+1}(\mathbb{F}_p)$ for $p \nmid N$. Hence

$$p \nmid N g(\boldsymbol{v}) \quad \Longrightarrow \quad \boldsymbol{v} \bmod p \in U_{j+1}(\mathbb{F}_p).$$

In both cases define $\Delta(\boldsymbol{v}) := N\,|g(\boldsymbol{v})|$. Then $p \nmid \Delta(\boldsymbol{v})$ ensures that the reduction of $\boldsymbol{v}$ is nonzero and lies outside $(Z_{j+1} \setminus U_{j+1})(\mathbb{F}_p)$, so the required bound follows. Since g is chosen from a fixed finite family of polynomials depending only on F, we have

$$1 \leqslant \Delta(\boldsymbol{v}) \ll_F (2 + |\boldsymbol{v}|)^{O_F(1)}.$$

Finally, if $j = n-5$, take $\Delta(\boldsymbol{v}) := N|\boldsymbol{v}|$. As noted after (10.1), Proposition 5.1 then gives

$$|S_p(\boldsymbol{v})| \ll_F p^{n-2} = p^{\frac{n+j+1}{2}} \qquad (p \nmid \Delta(\boldsymbol{v})).$$

This completes the proof of part (i).

Let us turn our attention to part (*ii*). When $j = 0$, the result is trivial since $P_0 \subseteq \mathbb{Q}^n$. Let us deal with the cases $j = 1, 2$. Since $Z_j(\mathbb{Q}) = R_j(\mathbb{Q})$ by (10.2), the definition of P_j gives

$$\begin{aligned} N(P_j, L, \boldsymbol{v}_0, m, \boldsymbol{b}) &\leqslant N\big(R_j \setminus (Z_{j+1} \cup U_j), L, \boldsymbol{v}_0, m, \boldsymbol{b}\big) + N(Z_{j+1}, L, \boldsymbol{v}_0, m, \boldsymbol{b}) \\ &= N\big(R_j \setminus (Z_{j+1} \cup U_j), L, \boldsymbol{v}_0, m, \boldsymbol{b}\big) + N(R_{j+1}, L, \boldsymbol{v}_0, m, \boldsymbol{b}). \end{aligned}$$

Since $\operatorname{codim} R_{j+1} \geqslant j + 2$, Lemma 2.7 implies that

$$N(R_{j+1}, L, \boldsymbol{v}_0, m, \boldsymbol{b}) \ll_F T^{n-j-2},$$

which is acceptable. Let Y be a geometric irreducible component of

$$\overline{R_j \setminus (Z_{j+1} \cup U_j)}^{\mathrm{Zar}}.$$

Based on the definition of L_j and the density of U_j in L_j (see (10.4) and (10.5)), we distinguish three cases.

i) The component Y is defined over $\mathbb{Q}$ and has $\deg Y \geqslant j + 1$. Hence, by the uniform Bombieri–Pila bound recorded in [Pil95, Theorem A], we have

$$N(Y, L, \boldsymbol{v}_0, m, \boldsymbol{b}) \ll_{F,\varepsilon} T^{n-j-2+\frac{1}{j+1}+\varepsilon}.$$

ii) The component Y is not defined over $\mathbb{Q}$. In this case, there exists a subvariety $W \subseteq Y$, defined over $\mathbb{Q}$, of codimension at least 1 in Y, such that $Y(\mathbb{Q}) \subseteq W(\mathbb{Q})$. Hence, by Lemma 2.7, we have

$$N(Y, L, \boldsymbol{v}_0, m, \boldsymbol{b}) \ll_F T^{n-j-2}.$$

iii) The component Y has $\deg Y \leqslant j$ and is defined over $\mathbb{Q}$. Then necessarily $\dim Y \leqslant n - j - 2$, or else Y would be contained in L_j by (10.4). Hence, by Lemma 2.7, we have

$$N(Y, L, \boldsymbol{v}_0, m, \boldsymbol{b}) \ll_F T^{n-j-2}.$$

Summing over the finitely many components, the bound follows. For $3 \leqslant j \leqslant n-6$, the argument is almost the same, except for the treatment of the geometrically irreducible components of

$$\overline{R_j \setminus (Z_{j+1} \cup U_j)}^{\mathrm{Zar}}$$

that are defined over $\mathbb{Q}$ and have degree at least $j + 1 \geqslant 4$. If Y is such a component, then we have

$$N(Y, L, \boldsymbol{v}_0, m, \boldsymbol{b}) \ll_{F,\varepsilon} T^{n-j-2+\varepsilon},$$

by [Ver24, Theorem 1.2], provided that Y is not cylindrical over a curve.

We claim that Y is not cylindrical over a curve. Since R_j, Z_{j+1} and U_j are homogeneous, Y is an affine cone associated to a projective variety. Put $r := \dim Y$. If Y were cylindrical over a curve, there would exist a surjective $\mathbb{Q}$-linear map

$$\pi : \mathbb{A}^n \longrightarrow \mathbb{A}^{n-r+1}$$

and an affine curve C such that $Y = \pi^{-1}(C)$. Since π is linear and Y is homogeneous, C is also homogeneous. Moreover, C is geometrically irreducible, since Y is geometrically irreducible. Hence C must be a line. It follows that $Y = \pi^{-1}(C)$ is a linear space, contradicting $\deg Y \geqslant 4$. The case $j = n - 5$ follows from the fact that R_{n-5} has dimension $\leqslant 1$ and $R_{n-4} = \{\mathbf{0}\}$. □

For prime squares, recall the bounds in (10.3). Hence we obtain a partition $\{Q_j\}_{j=0}^{n-4}$ of $\mathbb{Q}^n$, where

$$Q_j = (Z_j \setminus Z_{j+1})(\mathbb{Q}) \qquad (0 \leqslant j \leqslant n-5), \qquad Q_{n-4} = \{\mathbf{0}\},$$

where we recall the convention that $Z_0 = \mathbb{A}^n$. Here we use the fact that $Z_{n-4}(\mathbb{Q}) = \{\mathbf{0}\}$.

As for the partition associated with prime moduli, we record the following proposition.

Proposition 10.2. *Let $\{Q_j\}_{j=0}^{n-4}$ be the partition of $\mathbb{Q}^n$ defined above. Then the following hold.*

i) For every $0 \leqslant j \leqslant n-4$ and every $\boldsymbol{v} \in Q_j \cap \mathbb{Z}^n$, there exists a positive integer

$$\Theta(\boldsymbol{v}) \ll_F (2+|\boldsymbol{v}|)^{O_F(1)}$$

such that

$$p \nmid \Theta(\boldsymbol{v}) \quad \Longrightarrow \quad |S_{p^2}(\boldsymbol{v})| \ll_F p^{n+j+1}.$$

ii) Set

$$T = 1 + \frac{L}{m}.$$

Then, for every $m, L \geqslant 1$, $\boldsymbol{v}_0 \in \mathbb{R}^n$, and every class $\boldsymbol{b} \bmod m$, one has

$$N(Q_j, L, \boldsymbol{v}_0, m, \boldsymbol{b}) \ll_F \begin{cases} T^n, & \text{if } j = 0, \\ T^{n-j-1}, & \text{if } 1 \leqslant j \leqslant n-6, \\ T, & \text{if } j = n-5, \\ 1, & \text{if } j = n-4. \end{cases} \tag{10.9}$$

In particular, for $1 \leqslant j \leqslant n-4$ we have $N(Q_j, L, \boldsymbol{v}_0, m, \boldsymbol{b}) \ll_F T^{n-j-1}$.

Proof. The arguments are analogous to the ones of Proposition 10.1, except that the proof of (10.9) only requires Lemma 2.7, rather than any progress towards the dimension growth conjecture. □

10.2. **The cube-free averages.** In this section we are going to combine the two partitions of $\mathbb{Q}^n$ we have constructed. Since $\{P_j\}_j$ and $\{Q_k\}_k$ are two partitions of $\mathbb{Q}^n$, the set $\{P_j \cap Q_k\}_{j,k=0}^{n-4}$ is also a partition of $\mathbb{Q}^n$. It will be useful to record the following fact: observe that by definition of our partitions, together with the definition of U_j in (10.5), we have

$$P_j \cap Q_k = \begin{cases} \big((Z_j \setminus Z_{j+1}) \setminus U_j\big)(\mathbb{Q}) \setminus \{\mathbf{0}\}, & 0 \leqslant j \leqslant n-6, \quad k = j, \\ U_{j+1}(\mathbb{Q}), & 0 \leqslant j \leqslant n-6, \quad k = j+1, \\ Z_{n-5}(\mathbb{Q}) \setminus \{\mathbf{0}\}, & j = k = n-5, \\ \{\mathbf{0}\}, & j = k = n-4, \\ \emptyset, & \text{otherwise.} \end{cases} \tag{10.10}$$

Definition 10.3. Let N_F be the product of finitely many primes depending only on F. Let $\boldsymbol{v} \in \mathbb{Z}^n$. For $p \nmid N_F$, let $\overline{\boldsymbol{v}}$ denote the reduction of $\boldsymbol{v}$ modulo p, and put

$$e(\boldsymbol{v}; p) = \min\left(\{0 \leqslant h \leqslant n-5 : \overline{\boldsymbol{v}} \notin (Z_{h+1,p} \setminus U_{h+1,p})(\mathbb{F}_p)\} \cup \{n-4\}\right),$$
$$f(\boldsymbol{v}; p) = \min\left(\{0 \leqslant h \leqslant n-5 : \overline{\boldsymbol{v}} \notin Z_{h+1,p}(\mathbb{F}_p)\} \cup \{n-4\}\right),$$

with the convention $U_{h,p} = \emptyset$ for $h \geqslant n-5$. For $p \mid N_F$, put $e(\boldsymbol{v}; p) = f(\boldsymbol{v}; p) = 0$.

The definitions of $e(\boldsymbol{v}; p)$ and $f(\boldsymbol{v}; p)$ are motivated by (10.8) and (10.3), respectively. For cube-free moduli d, we now define further key invariants based on the factorisation type of d.

Definition 10.4. Every positive cube-free integer d can be written uniquely as $d = d_1 d_2^2$, where d_1, d_2 are coprime positive square-free integers. Suppose now that $\boldsymbol{v} \in P_j \cap Q_k \cap \mathbb{Z}^n$, so that necessarily

$$0 \leqslant j \leqslant k \leqslant n-4.$$

Then let

$$K(d, \boldsymbol{v}) = \prod_{\substack{p \mid d_1 \\ e(\boldsymbol{v};p) > j+2}} p^{\frac{e(\boldsymbol{v};p)-j}{2}-1} \prod_{\substack{p \mid d_2 \\ f(\boldsymbol{v};p) > k+2}} p^{f(\boldsymbol{v};p)-k-2}. \tag{10.11}$$

Moreover, for all integers $h \geqslant j+1$ and $\ell \geqslant k+1$ define the formal infinite products

$$q_{\boldsymbol{v},h} = \prod_{\substack{p \\ e(\boldsymbol{v};p) \geqslant h}} p, \qquad s_{\boldsymbol{v},\ell} = \prod_{\substack{p \\ f(\boldsymbol{v};p) \geqslant \ell}} p,$$

and let

$$R_{h,\ell}(d,\boldsymbol{v}) = \gcd(d_1, q_{\boldsymbol{v},h}) \gcd(d_2, s_{\boldsymbol{v},\ell}). \tag{10.12}$$

We start with the following lemma based on the invariants $K(d,\boldsymbol{v})$ and $R_{j+2,k+2}(d,\boldsymbol{v})$.

Lemma 10.5. *Let $0 \leqslant j,k \leqslant n-5$ and $\boldsymbol{v} \in (P_j \cap Q_k) \cap \mathbb{Z}^n$. If $D_1, D_2 \geqslant 1/2$, then for any two square-free integers $d_1 \sim D_1$ and $d_2 \sim D_2$ with $\gcd(d_1,d_2)=1$, and for every $\varepsilon > 0$, we have*

$$|S_{d_1 d_2^2}(\boldsymbol{v})| \ll D^{\varepsilon} D_1^{\frac{n+j+3}{2}} D_2^{n+k+3} K(d_1 d_2^2, \boldsymbol{v}), \tag{10.13}$$

where $D = D_1 D_2^2$. In fact, if $K_j \geqslant 1/2$ then

$$\sideset{}{^*}\sum_{d_1 \sim D_1} \sideset{}{^*}\sum_{\substack{d_2 \sim D_2 \\ \gcd(d_1,d_2)=1 \\ K(d_1 d_2^2, \boldsymbol{v}) \sim K_j}} |S_{d_1 d_2^2}(\boldsymbol{v})| \ll_{F,\varepsilon} [D(2+|\boldsymbol{v}|)]^{\varepsilon} D_1^{\frac{n+j+3}{2}} D_2^{n+k+3} K_j, \tag{10.14}$$

where every starred sum is restricted to square-free integers. Moreover, if $j+1 \leqslant h \leqslant n-3$ and $k+1 \leqslant \ell \leqslant n-3$, then

$$\sideset{}{^*}\sum_{d_1 \sim D_1} \sideset{}{^*}\sum_{\substack{d_2 \sim D_2 \\ \gcd(d_1,d_2)=1 \\ R_{h,\ell}(d_1 d_2^2, \boldsymbol{v})=1}} |S_{d_1 d_2^2}(\boldsymbol{v})| \ll_{F,\varepsilon} [D(2+|\boldsymbol{v}|)]^{\varepsilon} D_1^{\frac{n+\max\{j+3,h\}}{2}} D_2^{n+\max\{k+2,\ell\}}. \tag{10.15}$$

Proof. In the notation of Propositions 10.1 and 10.2, we have

$$\begin{aligned} p \nmid \Delta(\boldsymbol{v}) \quad &\Longrightarrow \quad |S_p(\boldsymbol{v})| \ll_F p^{\frac{n+j+1}{2}}, \\ p \nmid \Theta(\boldsymbol{v}) \quad &\Longrightarrow \quad |S_{p^2}(\boldsymbol{v})| \ll_F p^{n+k+1}. \end{aligned}$$

For $q = \gcd(d_1, q_{\boldsymbol{v},j+1}, \Delta(\boldsymbol{v}))$ and $s = \gcd(d_2, s_{\boldsymbol{v},k+1}, \Theta(\boldsymbol{v}))$, multiplicativity and the local bounds (10.8) and (10.3) give

$$|S_{d_1 d_2^2}(\boldsymbol{v})| \ll_{F,\varepsilon} D^{\varepsilon} d_1^{\frac{n+j+1}{2}} d_2^{n+k+1} \prod_{p \mid q} p^{(e(\boldsymbol{v};p)-j)/2} \prod_{p \mid s} p^{f(\boldsymbol{v};p)-k}. \tag{10.16}$$

Here the product of the local constants is absorbed into D^{ε}. By the definition of K, we have

$$\prod_{p \mid q} p^{(e(\boldsymbol{v};p)-j)/2} \prod_{p \mid s} p^{f(\boldsymbol{v};p)-k} \leqslant K(qs^2, \boldsymbol{v})\, qs^2.$$

To conclude the proof of (10.13), we observe that $qs^2 \leqslant d_1 d_2^2$ and $K(qs^2,\boldsymbol{v}) \leqslant K(d_1 d_2^2, \boldsymbol{v})$, since the exponents in the definition (10.11) are positive.

For fixed q, s, there are $O(D_1 D_2/(qs))$ choices for d_1, d_2 satisfying $q \mid d_1$ and $s \mid d_2$. Consequently, the left-hand side of (10.14) is

$$\ll_{F,\varepsilon} D^{\varepsilon} D_1^{\frac{n+j+1}{2}} D_2^{n+k+1} \sum_{\substack{q \mid \gcd(q_{\boldsymbol{v},j+1}, \Delta(\boldsymbol{v})) \\ q \leqslant 2D_1}} \sum_{\substack{s \mid \gcd(s_{\boldsymbol{v},k+1}, \Theta(\boldsymbol{v})) \\ s \leqslant 2D_2 \\ \gcd(q,s)=1}} K_j q s^2 \frac{D_1 D_2}{qs}.$$

Using $s \leqslant 2D_2$ and the divisor bound on the positive integers $\Delta(\boldsymbol{v}), \Theta(\boldsymbol{v}) \ll (2+|\boldsymbol{v}|)^{O(1)}$, the claim (10.14) follows.

Finally, if $R_{h,\ell}(d_1 d_2^2, \boldsymbol{v}) = 1$, then $e(\boldsymbol{v};p) \leqslant h-1$ and $f(\boldsymbol{v};p) \leqslant \ell - 1$ in the inequality (10.16), so

$$|S_{d_1 d_2^2}(\boldsymbol{v})| \ll_{F,\varepsilon} D^{\varepsilon} d_1^{\frac{n+j+1}{2}} d_2^{n+k+1} q^{(h-1-j)/2} s^{\ell-1-k}.$$

Proceeding as we did for (10.14), we find that the left-hand side of (10.15) is

$$\ll_{F,\varepsilon} D^{\varepsilon} D_1^{\frac{n+j+1}{2}} D_2^{n+k+1} \sum_{\substack{q \mid \gcd(q_{\boldsymbol{v},j+1}, \Delta(\boldsymbol{v})) \\ q \leqslant 2D_1}} \sum_{\substack{s \mid \gcd(s_{\boldsymbol{v},k+1}, \Theta(\boldsymbol{v})) \\ s \leqslant 2D_2 \\ \gcd(q,s)=1}} q^{(h-1-j)/2} s^{\ell-1-k} \frac{D_1 D_2}{qs}.$$

The divisor bound yields (10.15) on optimising over q and s, since $\frac{n+j+3}{2}+\max\{0,(h-3-j)/2\} = \frac{n+\max\{j+3,h\}}{2}$ and $n+k+2+\max\{0,\ell-2-k\} = n+\max\{k+2,\ell\}$. □

Let $D \geqslant 1/2$, $L \geqslant 1$, $m \in \mathbb{Z}_{\geqslant 1}$, $\boldsymbol{v}_0 \in \mathbb{R}^n$, and $\boldsymbol{b} \in \mathbb{Z}^n$. For the statements of the next three lemmas, we write

$$\begin{aligned} &T = 1 + L/m, \qquad H = 2 + |\boldsymbol{v}_0| + L + m, \\ &\mathcal{V} = \{\boldsymbol{v} \in \mathbb{Z}^n : |\boldsymbol{v} - \boldsymbol{v}_0| \leqslant L,\ \boldsymbol{v} \equiv \boldsymbol{b} \bmod m\}. \end{aligned} \tag{10.17}$$

All constants below will be uniform in the parameters of (10.17).

Lemma 10.6. *Let notation be as in (10.17). For every $\varepsilon > 0$,*

$$\sum_{\substack{d \sim D \\ d \text{ cube-free} \\ \gcd(d,m)=1}} \sum_{\boldsymbol{v} \in (P_{n-5} \cap Q_{n-5}) \cap \mathcal{V}} |S_d(\boldsymbol{v})| \ll_{F,\varepsilon} (DH)^{\varepsilon} D^{n-1} T.$$

Proof. Let $j = k = n-5$ and $h = \ell = n-3$. If $\boldsymbol{v} \in P_j \cap Q_k \cap \mathbb{Z}^n$, then by (10.12) we have $R_{h,\ell}(d, \boldsymbol{v}) = 1$ for every d. Hence, by combining (10.15) with the bound (10.7) for P_{n-5}, the lemma follows. □

Henceforth we introduce the convenient indicator notation

$$\delta_S(j) = \begin{cases} 1 & \text{if } j \in S, \\ 0 & \text{otherwise,} \end{cases}$$

for any finite list of integers S. We also let $\delta_0(j,k) = \mathbf{1}_{j=k=0}$. By (10.10), we know that if $P_j \cap Q_k$ contains a nonzero vector $\boldsymbol{v}$, then $0 \leqslant j \leqslant k \leqslant n-5$ and $k \in \{j, j+1\}$. Having studied the case $j = n-5$ in Lemma 10.6, we turn our attention to the remaining cases $0 \leqslant j \leqslant n-6$. It will be convenient to make a further case division based on the size of the integer $R_{j+2,k+2}(d, \boldsymbol{v})$. We first treat the situation where $R_{j+2,k+2}(d, \boldsymbol{v}) \leqslant T$.

Lemma 10.7. *Let notation be as in (10.17). Suppose $0 \leqslant j \leqslant n-6$ and $k \in \{j, j+1\}$. For every $K_j \geqslant 1/2$ and $\varepsilon > 0$,*

$$\sum_{\substack{d \sim D \\ d \text{ cube-free} \\ \gcd(d,m)=1}} \sum_{\substack{\boldsymbol{v} \in (P_j \cap Q_k) \cap \mathcal{V} \\ R_{j+2,k+2}(d,\boldsymbol{v}) \leqslant T \\ K(d,\boldsymbol{v}) \sim K_j}} |S_d(\boldsymbol{v})| \ll_{F,\varepsilon} (DH)^{\varepsilon} T^{\theta_n(j,k)} D^{\frac{n+j+3}{2}} K_j^{-\alpha_n \delta_0(j,k)}, \tag{10.18}$$

where $\theta_n(j,k) = n - j - 1 + \delta_0(j,k) - \delta_{n-7,n-6}(j)$ and $\alpha_n = 1/(n-8)$.

Proof. Let

$$r = R_{j+2,k+2}(d, \boldsymbol{v}), \qquad c = \gcd(d, r^2), \qquad b = d/c.$$

Then $\operatorname{rad}(c) = r \leqslant T$, $\gcd(b,c) = 1$, $R_{j+2,k+2}(b, \boldsymbol{v}) = 1$, and $K(b, \boldsymbol{v}) = 1$, so $K(d, \boldsymbol{v}) = K(c, \boldsymbol{v})$. This follows from the definitions (10.12) and (10.11). For fixed $c, \boldsymbol{v}$, we have

$$\sum_{\substack{b \sim D/c \\ R_{j+2,k+2}(b,\boldsymbol{v})=1}} |S_b(\boldsymbol{v})| \ll_{F,\varepsilon} (DH)^{\varepsilon} (D/c)^{(n+j+3)/2}$$

by (10.15), since $k \leqslant j+1$. Consequently the left-hand side $\mathcal{L}$ of (10.18) satisfies

$$\mathcal{L} \ll_{F,\varepsilon} (DH)^{\varepsilon} D^{(n+j+3)/2} \sum_{\substack{r \leqslant T \\ r \text{ square-free} \\ \gcd(r,m)=1}} \sum_{\substack{\operatorname{rad}(c)=r \\ c \text{ cube-free} \\ c \leqslant 2D}} \sum_{\substack{\boldsymbol{v} \in (P_j \cap Q_k) \cap \mathcal{V} \\ R_{j+2,k+2}(c,\boldsymbol{v})=r \\ K(c,\boldsymbol{v}) \sim K_j}} \frac{|S_c(\boldsymbol{v})|}{c^{(n+j+3)/2}}. \tag{10.19}$$

Now write $c = d_1 d_2^2$, so that $r = d_1 d_2$. By (10.13), we have

$$\frac{S_c(\boldsymbol{v})}{c^{(n+j+3)/2}} \ll (DH)^{\varepsilon} d_2^{k-j} K(d_1 d_2^2, \boldsymbol{v}) = (DH)^{\varepsilon} d_2^{k-j} \prod_{p \mid d_1} p^{\frac{e(\boldsymbol{v};p)-j}{2} - 1} \prod_{p \mid d_2} p^{f(\boldsymbol{v};p)-k-2},$$

by (10.11), since $e(\boldsymbol{v};p) \geqslant j+2$ and $f(\boldsymbol{v};p) \geqslant k+2$. Put

$$B_{j,k} = \left(\frac{\delta_{1,2}(j)}{j+1} + 2\delta_0(j)\right)\mathbf{1}_{k=j}.$$

By (10.7) if $k=j$, and by (10.9) if $k=j+1$, a fixed residue class modulo r contains

$$\ll_{F,\varepsilon} (T/r)^{n-j-2+B_{j,k}+\varepsilon}$$

vectors $\boldsymbol{v} \in (P_j \cap Q_k) \cap \mathcal{V}$. Moreover, by the Chinese remainder theorem and the Lang–Weil bound on the varieties Z_h for $j+2 \leqslant h \leqslant n-4$, the number of residues $\boldsymbol{v} \bmod r$ for which $e(\boldsymbol{v};p) = h_p$ for all $p \mid d_1$ and $f(\boldsymbol{v};p) = \ell_p$ for all $p \mid d_2$ is

$$\ll_{F,\varepsilon} r^{\varepsilon} \prod_{p\mid d_1} p^{n-h_p-\delta_{n-5,n-4}(h_p)} \prod_{p\mid d_2} p^{n-\ell_p-\delta_{n-5,n-4}(\ell_p)}.$$

When $j=k=0$, it will also be convenient to note that

$$1 \ll K_j^{-\alpha_n\delta_0(j,k)} K(c,\boldsymbol{v})^{\alpha_n\delta_0(j,k)},$$

since $K(c,\boldsymbol{v}) \gg K_j$. Plugging the bounds above into (10.19) gives

$$\mathcal{L} \ll_{F,\varepsilon} (DH)^{\varepsilon} D^{(n+j+3)/2} T^{n-j-2+B_{j,k}} K_j^{-\alpha_n\delta_0(j,k)} \sum_{\substack{d_1d_2\leqslant T\\ d_1d_2 \text{ square-free}\\ \gcd(d_1d_2,m)=1}} \sup_{\substack{j+2\leqslant h_p\leqslant n-4\\ k+2\leqslant \ell_p\leqslant n-4}} \prod_{p\mid d_1} p^{a_p} \prod_{p\mid d_2} p^{b_p},$$

where

$$\begin{aligned} a_p &= (\tfrac{h_p-j}{2}-1)(1+\alpha_n\delta_0(j,k)) - (n-j-2+B_{j,k}) + (n-h_p-\delta_{n-5,n-4}(h_p)) \\ &\leqslant (h_p-j-2)(1+\alpha_n\delta_0(j,k)) - (n-j-2+B_{j,k}) + (n-h_p-\delta_{n-5,n-4}(h_p)) \\ &= (h_p-j-2)\alpha_n\delta_0(j,k) - B_{j,k} - \delta_{n-5,n-4}(h_p), \end{aligned}$$

since $h_p \geqslant j+2$, and where

$$\begin{aligned} b_p &= (k-j) + (\ell_p-k-2)(1+\alpha_n\delta_0(j,k)) - (n-j-2+B_{j,k}) + (n-\ell_p-\delta_{n-5,n-4}(\ell_p)) \\ &= (\ell_p-k-2)\alpha_n\delta_0(j,k) - B_{j,k} - \delta_{n-5,n-4}(\ell_p). \end{aligned}$$

In order to analyse the exponents a_p and b_p, let

$$f_{j,k}(h) := (h-j-2)\alpha_n\delta_0(j,k) - B_{j,k} - \delta_{n-5,n-4}(h)$$

for $j+2 \leqslant h \leqslant n-4$. Then $a_p \leqslant f_{j,k}(h_p)$ and $b_p \leqslant f_{j,k}(\ell_p)$, since $k \geqslant j$. If $n-7 \leqslant j \leqslant n-6$, then $j \geqslant 3$ and $n-5 \leqslant h \leqslant n-4$, so $B_{j,k}=0$ and $f_{j,k}(h) = -1$, whence $a_p, b_p \leqslant -1$ and

$$\mathcal{L} \ll_{F,\varepsilon} (DH)^{\varepsilon} D^{(n+j+3)/2} T^{n-j-2} \sum_{d_1d_2\leqslant T} (d_1d_2)^{-1} \ll_{F,\varepsilon} (DH)^{2\varepsilon} D^{(n+j+3)/2} T^{n-j-2},$$

which is satisfactory since $\delta_0(j,k)=0$ and $\delta_{n-7,n-6}(j)=1$. If $0 \leqslant j \leqslant n-8$ and $k \geqslant 1$, then $f_{j,k}(h) \leqslant -B_{j,k}$, where $0 \leqslant B_{j,k} \leqslant 1$, so $a_p, b_p \leqslant -B_{j,k}$ and

$$\mathcal{L} \ll_{F,\varepsilon} (DH)^{\varepsilon} D^{(n+j+3)/2} T^{n-j-2+B_{j,k}} \sum_{d_1d_2\leqslant T} (d_1d_2)^{-B_{j,k}} \ll_{F,\varepsilon} (DH)^{2\varepsilon} D^{(n+j+3)/2} T^{n-j-1},$$

which is satisfactory since $\delta_0(j,k) = \delta_{n-7,n-6}(j) = 0$. Finally, if $j=k=0$, then $B_{j,k}=2$ and

$$f_{0,0}(h) = \frac{h-2}{n-8} - 2 - \delta_{n-5,n-4}(h) \leqslant -1 \qquad (2 \leqslant h \leqslant n-4).$$

Indeed, $f_{0,0}(h)$ is maximised either at $h=n-6$, where $f_{0,0}(n-6) = -1$, or at $h=n-4$, where $f_{0,0}(n-4) = (n-6)/(n-8) - 3 \leqslant -1$ since $n \geqslant 10$. Thus $a_p, b_p \leqslant -1$ and

$$\mathcal{L} \ll_{F,\varepsilon} (DH)^{\varepsilon} D^{(n+j+3)/2} T^{n-j} K_j^{-\alpha_n} \sum_{d_1d_2\leqslant T} (d_1d_2)^{-1} \ll_{F,\varepsilon} (DH)^{2\varepsilon} D^{(n+j+3)/2} T^{n-j} K_j^{-\alpha_n},$$

which is satisfactory since $\delta_0(j,k)=1$ and $\delta_{n-7,n-6}(j)=0$. □

We will also need the following complementary estimate for $R_{j+2,k+2}(d,\boldsymbol{v}) \geqslant T$.

Lemma 10.8. *Let notation be as in (10.17). Suppose $0 \leqslant j \leqslant n-6$ and $k \in \{j, j+1\}$. Then*

$$\sum_{\substack{d \sim D \\ d \text{ cube-free} \\ \gcd(d,m)=1}} \sum_{\substack{\boldsymbol{v} \in (P_j \cap Q_k) \cap \mathcal{V} \\ R_{j+2,k+2}(d,\boldsymbol{v}) \geqslant T}} |S_d(\boldsymbol{v})| \ll_{F,\varepsilon} (DH)^{\varepsilon} \sum_{\ell=j+2}^{n-4} T^{n-\ell+1-\delta_{n-5,n-4}(\ell)} D^{\frac{n+\ell+1}{2}}. \tag{10.20}$$

Proof. Let $\mathcal{L}$ be the left-hand side of (10.20). As in the proof of Lemma 10.7, write

$$r = R_{j+2,k+2}(d, \boldsymbol{v}), \qquad c = \gcd(d, r^2), \qquad b = d/c.$$

Then $\operatorname{rad}(c) = r \geqslant T$, $\gcd(b, c) = 1$ and $R_{j+2,k+2}(b, \boldsymbol{v}) = 1$. For $j+2 \leqslant h \leqslant n-4$, let c_h be the product of all prime powers $p^\nu \parallel c$ of depth h, where the depth of a prime power is defined to be $e(\boldsymbol{v}; p)$ if $\nu = 1$ and $f(\boldsymbol{v}; p)$ if $\nu = 2$. Put $r_h = \operatorname{rad}(c_h)$. Then

$$c = \prod_h c_h, \qquad r = \prod_h r_h \geqslant T, \qquad r_h \leqslant c_h \leqslant r_h^2, \qquad c \leqslant 2D.$$

If $c = d_1 d_2^2$, then (10.13) gives

$$\frac{S_c(\boldsymbol{v})}{c^{(n+j+3)/2}} \ll (DH)^{\varepsilon} d_2^{k-j} K(d_1 d_2^2, \boldsymbol{v}) \leqslant (DH)^{\varepsilon} \prod_{h=j+2}^{n-4} c_h^{(h-j-2)/2},$$

by the definition of $K(d_1 d_2^2, \boldsymbol{v})$ in (10.11), since $k \geqslant j$. Yet for fixed $c, \boldsymbol{v}$, we have

$$\sum_{\substack{b \sim D/c \\ R_{j+2,k+2}(b,\boldsymbol{v})=1}} |S_b(\boldsymbol{v})| \ll (DH)^{\varepsilon} (D/c)^{(n+j+3)/2}$$

by (10.15), exactly as in the proof of Lemma 10.7. Therefore,

$$\mathcal{L} \ll \sum_{c_h, r_h} \sum_{\substack{\boldsymbol{v} \in (P_j \cap Q_k) \cap \mathcal{V} \\ \boldsymbol{v} \bmod r_h \in Z_h(\mathbb{Z}/r_h\mathbb{Z})}} (DH)^{\varepsilon} D^{(n+j+3)/2} \prod_{h=j+2}^{n-4} c_h^{(h-j-2)/2}.$$

Let $\mathcal{L}'$ be the contribution from $c_h \sim C_h$ and $r_h \sim R_h$. Then $\mathcal{L}' = 0$ unless

$$\prod_h C_h \ll D, \qquad \prod_h R_h \gg T, \qquad R_h \ll C_h \ll R_h^2.$$

For fixed r_h, there are at most $2^{\omega(r_h)} \ll_{\varepsilon} D^{\varepsilon}$ possible choices for c_h. Hence

$$\mathcal{L}' \ll \mathcal{N}(\boldsymbol{R}) (DH)^{\varepsilon} D^{(n+j+3)/2} \prod_{h=j+2}^{n-4} C_h^{(h-j-2)/2},$$

where $\mathcal{N}(\boldsymbol{R})$ is the quantity in Proposition 2.9 with $Y_0 = \mathbb{A}^n_{\mathbb{Z}}$, $\mathcal{U} = (P_j \cap Q_k) \cap \mathbb{Z}^n$, $Y_i = Z_{j+1+i}$ for $1 \leqslant i \leqslant s = n-5-j$, and with dyadic parameters R_{j+1+i} in place of R_i. Indeed, the set $\mathcal{U}$ lies outside all the varieties Z_h with $h \geqslant j+2$. In fact, $P_j \cap Z_{j+2} = \emptyset$ by (10.6), since $U_{j+1} \cap Z_{j+2} = \emptyset$ by (10.5).

By (10.7), the hypothesis (2.2) of Proposition 2.9 certainly holds with $\alpha = n - j$. Moreover, $\dim Z_h \leqslant f(h)$, where $f(h) := n - h - \delta_{n-5,n-4}(h)$. Since f is (strictly) decreasing on the interval $j+2 \leqslant h \leqslant n-4$, Proposition 2.9 implies that

$$\mathcal{N}(\boldsymbol{R}) \ll_{F,\varepsilon} (DH)^{\varepsilon} \left(1 + \frac{T^{n-j}}{\prod_{h=j+2}^{n-4} R_h^{n-j-f(h)-1}} + \sum_{\ell=j+2}^{n-4} \frac{T^{f(\ell)+1}}{\prod_{h>\ell} R_h^{f(\ell)-f(h)}} \right).$$

Let A_0 be the middle term in the parentheses, and let A_ℓ be the summand indexed by ℓ. Then

$$\mathcal{L}' \ll A (DH)^{\varepsilon} D^{(n+j+3)/2} \prod_{h=j+2}^{n-4} C_h^{(h-j-2)/2},$$

for some $A \in \{1, A_0, A_\ell\}$. If $A = 1$, then using $\prod_h C_h \ll D$ we find that

$$\mathcal{L}' \ll (DH)^\varepsilon D^{(n+j+3)/2} D^{(n-4-j-2)/2} = (DH)^\varepsilon D^{n-3/2} \ll (DH)^\varepsilon T^4 D^{n-3/2},$$

which is the $\ell = n-4$ term of (10.20). If $A = A_0$, then using $C_h \ll R_h^2$ we find that

$$\mathcal{L}' \ll (DH)^\varepsilon D^{(n+j+3)/2} T^{n-j} \prod_{h=j+2}^{n-4} \frac{R_h^{h-j-2}}{R_h^{n-j-f(h)-1}}.$$

The product over h equals $\prod_{h=j+2}^{n-4} R_h^{-1-\delta_{n-5,n-4}(h)} \ll T^{-1-\delta_{n-5,n-4}(j+2)}$, since $\prod_h R_h \gg T$ and $\delta_{n-5,n-4}(h) \geqslant \delta_{n-5,n-4}(j+2)$ for all $j+2 \leqslant h \leqslant n-4$. Thus

$$\mathcal{L}' \ll (DH)^\varepsilon D^{(n+j+3)/2} T^{n-j-1-\delta_{n-5,n-4}(j+2)},$$

which is the $\ell = j+2$ term of (10.20). Finally, suppose $A = A_\ell$, where $j+2 \leqslant \ell \leqslant n-4$. Then

$$\mathcal{L}' \ll (DH)^\varepsilon D^{(n+j+3)/2} T^{f(\ell)+1} \prod_{h=j+2}^{\ell} C_h^{(h-j-2)/2} \prod_{h=\ell+1}^{n-4} \frac{C_h^{(h-j-2)/2}}{R_h^{f(\ell)-f(h)}}.$$

Here $C_h^{(h-j-2)/2} \ll C_h^{(\ell-j-2)/2}$ for $h \leqslant \ell$. Also, $R_h^{f(\ell)-f(h)} \gg R_h^{h-\ell} \gg C_h^{(h-\ell)/2}$ for $\ell < h \leqslant n-4$, since f is strictly decreasing on this interval. It follows that

$$\mathcal{L}' \ll (DH)^\varepsilon D^{(n+j+3)/2} T^{f(\ell)+1} \prod_{h=j+2}^{n-4} C_h^{(\ell-j-2)/2} \ll (DH)^\varepsilon D^{(n+\ell+1)/2} T^{f(\ell)+1}$$

since $\prod_h C_h \ll D$. This matches the ℓ term of (10.20), so we are done. □

For the remainder of this section put $n = 10$. We conclude this section with the following result which will be useful for combining the cube-free average with the cubefull one.

Proposition 10.9. *Let $D \geqslant 1/2$, $L \geqslant 1$, $m \in \mathbb{Z}_{\geqslant 1}$, and $\boldsymbol{v}_0 \in \mathbb{R}^{10}$. Let $\mathcal{V}$ be any subset of $\{\boldsymbol{v} \in \mathbb{Z}^{10} \setminus \{\mathbf{0}\} : |\boldsymbol{v} - \boldsymbol{v}_0| \leqslant L\}$. Suppose $w : \mathcal{V} \to \mathbb{R}_{\geqslant 0}$ and for every $\boldsymbol{a} \bmod m$ define*

$$P(\boldsymbol{a}) := \sup_{\substack{\boldsymbol{v} \in \mathcal{V} \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod m}} |w(\boldsymbol{v})|,$$

with the convention that $\sup_\emptyset = 0$ if the set in the supremum is empty. Put

$$T = 1 + L/m, \qquad H = 2 + |\boldsymbol{v}_0| + L + m.$$

Let

$$\|P\|_1 = \sum_{\boldsymbol{a} \bmod m} P(\boldsymbol{a}), \qquad \|w\|_1 = \sum_{\boldsymbol{v} \in \mathcal{V}} w(\boldsymbol{v}).$$

Then

$$\sum_{\substack{d \sim D \\ \text{cube-free} \\ \gcd(d,m)=1}} \sum_{\boldsymbol{v} \in \mathcal{V}} w(\boldsymbol{v}) |S_d(\boldsymbol{v})| \ll_{F,\varepsilon} (DH)^\varepsilon \left(\mathcal{R} + D^{13/2} T^{20/3} \|P\|_1^{2/3} \|w\|_1^{1/3} \right), \tag{10.21}$$

where $\mathcal{R} = \min\left\{ (D^{19/2}T + D^{41/6}T^9)\|P\|_1,\ D^9 \|w\|_1 \right\}$.

Proof. To simplify a bit the notation, in what follows we will denote $x = T/D^{1/3}$. Let

$$\mathcal{U}_d := \{\boldsymbol{v} \in \mathcal{V} \cap P_0 \cap Q_0 : R_{2,2}(d, \boldsymbol{v}) \leqslant T\}.$$

Observe that

$$\begin{aligned} \sum_{\substack{d \sim D \\ d \text{ cube-free} \\ \gcd(d,m)=1}} \sum_{\boldsymbol{v} \in \mathcal{V}} w(\boldsymbol{v}) |S_d(\boldsymbol{v})| &= \sum_{\substack{d \sim D \\ d \text{ cube-free} \\ \gcd(d,m)=1}} \sum_{\boldsymbol{v} \in \mathcal{U}_d} w(\boldsymbol{v}) |S_d(\boldsymbol{v})| + \sum_{\substack{d \sim D \\ d \text{ cube-free} \\ \gcd(d,m)=1}} \sum_{\boldsymbol{v} \in \mathcal{V} \setminus \mathcal{U}_d} w(\boldsymbol{v}) |S_d(\boldsymbol{v})| \\ &= M + E, \end{aligned} \tag{10.22}$$

say. We will prove that $E \ll (DH)^\varepsilon \mathcal{R}$ and that $M \ll (DH)^\varepsilon D^{13/2} T^{20/3} \|P\|_1^{2/3} \|w\|_1^{1/3}$.

We provide two different bounds for E. First we sort the vector $\boldsymbol{v}$ according to its reduction modulo m. If $\boldsymbol{v} \equiv \boldsymbol{a} \bmod m$, then $w(\boldsymbol{v}) \leqslant P(\boldsymbol{a})$. Thus

$$E = \sum_{\boldsymbol{a} \bmod m} \sum_{\substack{d \sim D \\ d \text{ cube-free} \\ \gcd(d,m)=1}} \sum_{\substack{\boldsymbol{v} \in \mathcal{V} \setminus \mathcal{U}_d \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod m}} w(\boldsymbol{v}) |S_d(\boldsymbol{v})| \leqslant \|P\|_1 \max_{\boldsymbol{a} \bmod m} f(\boldsymbol{a}),$$

where

$$f(\boldsymbol{a}) = \sum_{\substack{d \sim D \\ d \text{ cube-free} \\ \gcd(d,m)=1}} \sum_{\substack{\boldsymbol{v} \in \mathcal{V} \setminus \mathcal{U}_d \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod m}} |S_d(\boldsymbol{v})|.$$

On the other hand, after an application of Lemma 10.6 for $(j,k) = (n-5, n-5)$, Lemma 10.7 for $(j,k) \neq (0,0), (n-5,n-5)$, and Lemma 10.8 for $(j,k) \neq (n-5,n-5)$, one gets

$$\begin{aligned} f(\boldsymbol{a}) &\ll (DH)^{\varepsilon} \left(TD^9 + T^9 D^{13/2} + T^8 D^7 + T^7 D^{15/2} + T^5 D^8 + T^4 D^{17/2} \right) \\ &\ll (DH)^{\varepsilon} \left(xD^{28/3} + x^9 D^{19/2} + x^8 D^{29/3} + x^7 D^{59/6} + x^5 D^{29/3} + x^4 D^{59/6} \right). \end{aligned}$$

Since $D^{59/6}(x^4 + x^7) \leqslant 2D^{59/6}(x + x^9)$, it follows that

$$\begin{aligned} f(\boldsymbol{a}) &\ll (DH)^{\varepsilon} \left(xD^{28/3} + x^9 D^{19/2} + x^8 D^{29/3} + x^5 D^{29/3} + (x + x^9) D^{59/6} \right) \\ &\ll (DH)^{\varepsilon} D^{59/6}(x + x^9) = (DH)^{\varepsilon}(D^{19/2}T + D^{41/6}T^9), \end{aligned}$$

since $D \gg 1$ and $x = T/D^{1/3}$. Second, thanks to the $h = \ell = n-3$ case of (10.15) we know that for every $\boldsymbol{v} \in P_j \cap Q_k$, for $j,k \in \{0, \dots, n-5\}$ we have

$$\sideset{}{^*}\sum_{d_1 \sim D_1} \sideset{}{^*}\sum_{\substack{d_2 \sim D_2 \\ \gcd(d_1,d_2)=1}} |S_{d_1 d_2^2}(\boldsymbol{v})| \ll_{F,\varepsilon} (DH)^{\varepsilon} D^{n-3/2} D_1^{\max\{j-n+6,0\}/2} \ll (DH)^{\varepsilon} D^{n-1},$$

since $j \leqslant n-5$. Indeed, for $j = k = n-5$ this was already observed in the proof of Lemma 10.6. Hence

$$E \leqslant \sum_{\boldsymbol{v} \in \mathcal{V}} w(\boldsymbol{v}) \sum_{\substack{d \sim D \\ d \text{ cube-free} \\ \gcd(d,m)=1}} |S_d(\boldsymbol{v})| \ll \|w\|_1 (DH)^{\varepsilon} D^9.$$

Combining the two bounds, we conclude that $E \ll (DH)^{\varepsilon} \mathcal{R}$.

It remains to bound

$$M = \sum_{\substack{d \sim D \\ d \text{ cube-free} \\ \gcd(d,m)=1}} \sum_{\substack{\boldsymbol{v} \in \mathcal{V} \cap P_0 \cap Q_0 \\ R_{2,2}(d,\boldsymbol{v}) \leqslant T}} w(\boldsymbol{v}) |S_d(\boldsymbol{v})|.$$

Let $M(K_0)$ be the contribution to M from $K(d,\boldsymbol{v}) \sim K_0$. We bound $M(K_0)$ using the same two methods as for E. First, the $(j,k) = (0,0)$ case of Lemma 10.7 implies that

$$M(K_0) \ll \|P\|_1 (DH)^{\varepsilon} T^n D^{\frac{n+3}{2}} K_0^{-1/2}.$$

Second, by (10.14) for $(j,k) = (0,0)$ we have

$$M(K_0) \ll \|w\|_1 (DH)^{\varepsilon} D^{\frac{n+3}{2}} K_0.$$

Thus

$$M(K_0) = M(K_0)^{2/3} M(K_0)^{1/3} \ll (DH)^{\varepsilon} D^{\frac{n+3}{2}} \left(\|P\|_1 T^n K_0^{-1/2} \right)^{2/3} (\|w\|_1 K_0)^{1/3}.$$

The powers of K_0 cancel out, leading to $M(K_0) \ll (DH)^{\varepsilon} D^{(n+3)/2} T^{2n/3} \|P\|_1^{2/3} \|w\|_1^{1/3}$. Dyadic summation over the interval $1/2 \leqslant K_0 \ll D^{O(1)}$ completes the proof. □

11. Mixed modulus averages

Throughout this section $n = 10$ and $X(\mathbb{Q}) = \emptyset$. We will combine Lemmas 6.8, 7.7, 8.4 and Proposition 10.9 to prove the following result concerning arbitrary integer moduli $q \geqslant 1$.

Proposition 11.1. *Fix $0 < \gamma < 1/96$. There exists a modulus W as in (8.6) such that for every $\varepsilon > 0$, $\boldsymbol{v}_0 \in \mathbb{R}^n$ and $Q, L \geqslant 1/2$, we have*

$$\sum_{q \sim Q} \sum_{\substack{\mathbf{0} \neq \boldsymbol{v} \in \mathbb{Z}^{10} \\ |\boldsymbol{v} - \boldsymbol{v}_0| \leqslant L}} |S_{q,W}(\boldsymbol{v})| \ll_{F,\gamma,\varepsilon} (LQ + |\boldsymbol{v}_0|)^{\varepsilon} (L + Q^{1/3})^{10} Q^{20/3 - \gamma}. \tag{11.1}$$

Here W may depend on F and γ, but not on $\boldsymbol{v}_0, L, Q, \varepsilon$.

We start by recalling some notation. For a fixed $\eta > 0$, let $\mathcal{P}_\eta$ be a finite, sufficiently large set of primes depending only on F and η. Construct $W_\eta = \prod_{p \in \mathcal{P}_\eta} p^{M_p}$ as in (8.6).

Every positive integer q has a unique decomposition $q = gdr$, where g is supported on the primes in $\mathcal{P}_\eta$, d is cube-free, r is cubefull, both d and r are supported on primes outside $\mathcal{P}_\eta$, and g, d, r are pairwise coprime. By Lemma 8.2,

$$S_{q,W_\eta}(\boldsymbol{v}) = S_{g,W_\eta}(\boldsymbol{v}) S_d(\boldsymbol{v}) S_r(\boldsymbol{v}).$$

Let us first focus on $S_{g,W_\eta}(\boldsymbol{v})$. The estimate Lemma 8.4 gives, for every $p \in \mathcal{P}_\eta$ and every $k \geqslant 0$,

$$\sum_{\boldsymbol{a} \bmod p^a} \max_{\substack{\boldsymbol{v} \bmod p^{\max(k, M_p)} \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod p^a}} |S_{p^k, p^{M_p}}(\boldsymbol{v})| \ll_{F,\eta} p^{k(10 + (4 - r(F))/6)},$$

where $a = \lceil k/3 \rceil$. On the other hand, Lemma 8.1 implies that $r(F) \geqslant 8$ when $n = 10$. Hence,

$$\sum_{\boldsymbol{a} \bmod p^a} \max_{\substack{\boldsymbol{v} \bmod p^{\max(k, M_p)} \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod p^a}} |S_{p^k, p^{M_p}}(\boldsymbol{v})| \ll_{F,\eta} p^{28k/3}.$$

By Lemma 8.2 and the Chinese remainder theorem over all primes $p \in \mathcal{P}_\eta$, we obtain

$$\sum_{\boldsymbol{a} \bmod A_g} \max_{\substack{\boldsymbol{v} \bmod \operatorname{lcm}(g, W_\eta) \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod A_g}} |S_{g,W_\eta}(\boldsymbol{v})| \ll_{F,\eta} g^{28/3}, \tag{11.2}$$

where

$$A_g = \prod_{p \mid g} p^{\lceil v_p(g)/3 \rceil} \asymp_{F,\eta} g^{1/3}.$$

Let us now focus on the cubefull part r and its interaction with g. Following Lemma 7.7, put

$$A = bz_2, \qquad T = bz_1$$

where

$$b = \prod_{v_p(r) > 0} p^{\lfloor v_p(r)/3 \rfloor}, \qquad z_i = \prod_{\substack{v_p(r) > 0 \\ v_p(r) \equiv i \bmod 3}} p, \tag{11.3}$$

for $0 \leqslant i \leqslant 2$. By (11.2), Lemma 7.7, and the Chinese remainder theorem,

$$\sum_{\boldsymbol{a} \bmod A_g A} \max_{\substack{\boldsymbol{v} \bmod \operatorname{lcm}(g, W_\eta) r \\ \boldsymbol{v} \equiv \boldsymbol{a} \bmod A_g A}} |S_{g,W_\eta}(\boldsymbol{v}) S_r(\boldsymbol{v})| \ll_{F,\eta} g^{28/3} r^{10+\eta} \omega(r), \tag{11.4}$$

where $\omega(r) = z_0^{-2} z_1^{-4} z_2^{-1}$. The following lemma will help us estimate sums over r.

Lemma 11.2. *For $0 \leqslant \alpha \leqslant 10$ and every $\varepsilon > 0$,*

$$\sum_{\substack{r \sim C \\ r \text{ cube-full}}} \omega(r) (z_1/z_2)^{\alpha/3} \ll_{\alpha,\varepsilon} C^{\max(0, (\alpha - 9)/12) + \varepsilon}.$$

Proof. By Rankin's trick, the left-hand side divided by the right-hand side is

$$\ll \sum_{r \text{ cube-full}} \omega(r)\,(z_1/z_2)^{\alpha/3} r^{-s} \leqslant \prod_p E_p,$$

where $s = \max(0, (\alpha-9)/12) + \varepsilon$ and

$$E_p = 1 + \sum_{e\geqslant 1} p^{-3es-2} + \sum_{e\geqslant 1} p^{-(3e+1)s-4+\alpha/3} + \sum_{e\geqslant 1} p^{-(3e+2)s-1-\alpha/3}.$$

The three sums are respectively $\ll p^{-3s-2}$, $\ll p^{-4s-4+\alpha/3}$, and $\ll p^{-5s-1-\alpha/3}$, whose exponents are all strictly less than -1. Indeed, this is clear for the first and third exponents, since $s > 0$ and $\alpha \geqslant 0$. For the second exponent, we have $-4s - 4 + \alpha/3 < -1$ because $s > (\alpha-9)/12$. Thus the Euler product converges. □

Proof of Proposition 11.1. Fix $0 < \gamma < 1/96$, and choose $\eta > 0$ sufficiently small in terms of $1/96 - \gamma$, to be fixed at the end. Let $\mathcal{P}_\eta$ and W_η be chosen in terms of η as above.

We first assume that $L \leqslant Q^{1/3}$ and put

$$\mathcal{V} = \{\boldsymbol{v} \in \mathbb{Z}^{10} \setminus \{\mathbf{0}\} : |\boldsymbol{v} - \boldsymbol{v}_0| \leqslant L\}.$$

For $q \sim Q$, write $q = gdr$ as above and fix dyadic ranges

$$g \sim G, \qquad d \sim D, \qquad r \sim C,$$

so that $GDC \asymp Q$. For each fixed g, r, define

$$w_{g,r}(\boldsymbol{v}) := |S_{g,W_\eta}(\boldsymbol{v}) S_r(\boldsymbol{v})|, \qquad m_{g,r} := A_g A_r, \qquad T_{g,r} := 1 + \frac{L}{m_{g,r}},$$

where A_g is as above and $A_r = bz_2$, with b, z_2 as in (11.3). Put

$$H_{g,r} := 2 + |\boldsymbol{v}_0| + L + m_{g,r},$$

and, for every $\boldsymbol{a} \bmod m_{g,r}$, define

$$P_{g,r}(\boldsymbol{a}) := \sup_{\substack{\boldsymbol{v}\in\mathcal{V} \\ \boldsymbol{v}\equiv\boldsymbol{a} \bmod m_{g,r}}} |S_{g,W_\eta}(\boldsymbol{v}) S_r(\boldsymbol{v})|,$$

with the convention that the supremum is zero if the set is empty. An application of Proposition 10.9 gives

$$\sum_{\substack{d\sim D \\ \text{cube-free} \\ \gcd(d,m_{g,r})=1}} \sum_{\boldsymbol{v}\in\mathcal{V}} w_{g,r}(\boldsymbol{v}) |S_d(\boldsymbol{v})| \ll_{F,\varepsilon} (DH_{g,r})^{\varepsilon} \left(\min\{\mathcal{R}_1, \mathcal{R}_2\} + \mathcal{R}_3\right) \tag{11.5}$$

where

$$\begin{aligned}
\mathcal{R}_1 &= (D^{19/2} T_{g,r} + D^{41/6} T_{g,r}^9) \|P_{g,r}\|_1, \\
\mathcal{R}_2 &= D^9 \|w_{g,r}\|_1, \\
\mathcal{R}_3 &= D^{13/2} T_{g,r}^{20/3} \|P_{g,r}\|_1^{2/3} \|w_{g,r}\|_1^{1/3}.
\end{aligned}$$

Here $\operatorname{rad}(m_{g,r}) = \operatorname{rad}(gr)$, so that $\gcd(d, m_{g,r}) = 1$ is equivalent to $\gcd(d, gr) = 1$.

In order to further average the estimate (11.5) over g and r, our strategy will be to bound

$$\sum_{g\sim G} \sum_{r\sim C} \|w_{g,r}\|_1 \quad \text{and} \quad \sum_{g\sim G} \sum_{r\sim C} T_{g,r}^{\alpha} \|P_{g,r}\|_1 \qquad (\alpha = 1, 9, 10) \tag{11.6}$$

first, before attacking $\min\{\mathcal{R}_1, \mathcal{R}_2\}$ and $\mathcal{R}_3$ by interpolation. We start with the average of $\|w_{g,r}\|_1$. It will be convenient to handle the g-aspect using a pointwise bound. By (8.5), together

with Lemma 8.1 and Lemma 8.2, we have $|S_{g,W_\eta}(\boldsymbol{v})| \ll_{F,\eta} g^7$ uniformly in $\boldsymbol{v}$. Moreover, since $\mathcal{P}_\eta$ is fixed and finite, there are $O_{F,\eta,\varepsilon}(G^\varepsilon)$ possible integers $g \sim G$. Hence, for every $\varepsilon > 0$,

$$\begin{aligned}\sum_{g\sim G}\sum_{r\sim C}\|w_{g,r}\|_1 &= \sum_{g\sim G}\sum_{r\sim C}\sum_{\boldsymbol{v}\in\mathcal{V}}|S_{g,W_\eta}(\boldsymbol{v})S_r(\boldsymbol{v})|\\ &\ll_{F,\eta,\varepsilon} G^{7+\varepsilon}\sum_{r\sim C}\sum_{\boldsymbol{v}\in\mathcal{V}}|S_r(\boldsymbol{v})|\\ &\ll_{F,\eta,\varepsilon} G^{7+\varepsilon}C^{32/5+\varepsilon}(L+C^{1/3})^{10}\\ &\ll_{F,\eta,\varepsilon} G^{7+\varepsilon}C^{32/5+\varepsilon}Q^{10/3}.\end{aligned}$$

Here the penultimate step follows from the second bound, (6.19), in Lemma 6.8, and the last step uses $L \leqslant Q^{1/3}$ and $C \ll Q$.

Let us focus on the other term in (11.6): Let us now bound the moments of $\|P_{g,r}\|_1$. All implied constants below may depend on F, η, ε. Observe that (11.4) gives

$$\|P_{g,r}\|_1 \ll G^{28/3}r^{10+\eta}\omega(r).$$

Moreover, $r = b^3z_1z_2^2$ by (11.3), so we have

$$A_g \asymp_{F,\eta} g^{1/3}, \qquad A_r^3 = b^3z_2^3 = r\frac{z_2}{z_1} = \frac{r}{u(r)^3}, \qquad u(r) := \left(\frac{z_1}{z_2}\right)^{1/3}.$$

Since $L \leqslant Q^{1/3}$ and $GDC \asymp Q$, we obtain

$$T_{g,r} \ll 1 + \left(\frac{Q}{gr}\right)^{1/3}u(r) \ll 1 + D^{1/3}u(r).$$

Lemma 11.2, applied with exponents 0 and α, gives, for $\alpha \in \{1, 9, 10\}$,

$$\begin{aligned}\sum_{g\sim G}\sum_{r\sim C}T_{g,r}^\alpha\|P_{g,r}\|_1 &\ll G^{28/3+\varepsilon}C^{10+\eta}\left(\sum_{r\sim C}\omega(r) + D^{\alpha/3}\sum_{r\sim C}\omega(r)u(r)^\alpha\right)\\ &\ll D^{\alpha/3}G^{28/3+\varepsilon}C^{10+\max\{0,(\alpha-9)/12\}+\eta+\varepsilon}\\ &\ll Q^\varepsilon D^{\alpha/3}G^{28/3}C^{10+\max\{0,(\alpha-9)/12\}+\eta}.\end{aligned}$$

Here we used $D, C \geqslant 1/2$ to absorb the contribution of the exponent 0, and $GC \ll Q$ in the last step. Likewise, the preceding estimate for the weight norms gives

$$\sum_{g\sim G}\sum_{r\sim C}\|w_{g,r}\|_1 \ll Q^\varepsilon G^7C^{32/5}Q^{10/3}.$$

We are now ready to sum the bound in (11.5). First, the moment estimates with $\alpha = 1, 9$ give

$$\begin{aligned}\sum_{g\sim G}\sum_{r\sim C}\mathcal{R}_1 &= \sum_{g\sim G}\sum_{r\sim C}\big(D^{19/2}T_{g,r} + D^{41/6}T_{g,r}^9\big)\|P_{g,r}\|_1\\ &\ll Q^\varepsilon\big(D^{19/2+1/3} + D^{41/6+3}\big)G^{28/3}C^{10+\eta}\\ &\ll Q^\varepsilon D^{59/6}G^{28/3}C^{10+\eta}\\ &\asymp Q^{10+\varepsilon}C^\eta D^{-1/6}G^{-2/3},\end{aligned}$$

since $GDC \asymp Q$. On the other hand, the estimate for $\|w_{g,r}\|_1$ gives

$$\begin{aligned}\sum_{g\sim G}\sum_{r\sim C}\mathcal{R}_2 &\ll Q^\varepsilon D^9G^7C^{32/5}Q^{10/3}\\ &\ll Q^\varepsilon D^9G^7C^{13/2}Q^{10/3}\\ &\asymp Q^{10-1/6+\varepsilon}G^{1/2}D^{5/2}.\end{aligned}$$

Since the sum of the minima $\min\{\mathcal{R}_1, \mathcal{R}_2\}$ is at most the minimum of the sums, we obtain

$$\sum_{g\sim G}\sum_{r\sim C}\min\{\mathcal{R}_1,\mathcal{R}_2\} \ll \big(Q^{10+\varepsilon}C^{\eta}D^{-1/6}G^{-2/3}\big)^{15/16}\big(Q^{10-1/6+\varepsilon}G^{1/2}D^{5/2}\big)^{1/16}$$
$$= Q^{10-1/96+\varepsilon}G^{-19/32}C^{15\eta/16}.$$

It remains to bound the last term in (11.5). By Hölder's inequality,

$$\begin{aligned}\sum_{g\sim G}\sum_{r\sim C}\mathcal{R}_3 &= D^{13/2}\sum_{g\sim G}\sum_{r\sim C}T_{g,r}^{20/3}\|P_{g,r}\|_1^{2/3}\|w_{g,r}\|_1^{1/3}\\ &\leqslant D^{13/2}\left(\sum_{g\sim G}\sum_{r\sim C}T_{g,r}^{10}\|P_{g,r}\|_1\right)^{2/3}\left(\sum_{g\sim G}\sum_{r\sim C}\|w_{g,r}\|_1\right)^{1/3}\\ &\ll Q^{\varepsilon}D^{13/2}\big(D^{10/3}G^{28/3}C^{10+1/12+\eta}\big)^{2/3}\big(G^7C^{32/5}Q^{10/3}\big)^{1/3}\\ &\asymp Q^{10-1/30+\varepsilon}G^{-3/10}D^{-2/15}C^{2\eta/3}.\end{aligned}$$

Here the last step uses $GDC \asymp Q$.

Since $G, D \geqslant 1/2$, $C \ll Q$, and $1/30 > 1/96$, both contributions are

$$\ll Q^{10-1/96+\eta+\varepsilon}.$$

Moreover, $m_{g,r} \mid gr$, so $m_{g,r} \ll Q$ and $H_{g,r} \ll Q + |\boldsymbol{v}_0|$. Restoring the factors $(DH_{g,r})^{\varepsilon}$ from (11.5), summing over the $O((1+\log Q)^3)$ dyadic blocks, and rescaling ε, we therefore obtain

$$\sum_{q\sim Q}\sum_{\substack{\mathbf{0}\neq\boldsymbol{v}\in\mathbb{Z}^{10}\\ |\boldsymbol{v}-\boldsymbol{v}_0|\leqslant L}}|S_{q,W_\eta}(\boldsymbol{v})| \ll_{F,\eta,\varepsilon} (LQ+|\boldsymbol{v}_0|)^{\varepsilon}Q^{10-1/96+\eta} \qquad (L\leqslant Q^{1/3}).$$

We now fix $0 < \eta < 1/96 - \gamma$, depending only on γ, and set $W = W_\eta$. We conclude that

$$\sum_{q\sim Q}\sum_{\substack{\mathbf{0}\neq\boldsymbol{v}\in\mathbb{Z}^{10}\\ |\boldsymbol{v}-\boldsymbol{v}_0|\leqslant L}}|S_{q,W}(\boldsymbol{v})| \ll_{F,\gamma,\varepsilon} (LQ+|\boldsymbol{v}_0|)^{\varepsilon}Q^{10-\gamma} \qquad (L\leqslant Q^{1/3}).$$

If instead $L > Q^{1/3}$, cover the original box by

$$O\left(\left(1+\frac{L}{Q^{1/3}}\right)^{10}\right)$$

boxes of radius $Q^{1/3}$, with centres of size $O(|\boldsymbol{v}_0| + L)$. Applying the preceding bound to each box, and rescaling ε if necessary, gives

$$\sum_{q\sim Q}\sum_{\substack{\mathbf{0}\neq\boldsymbol{v}\in\mathbb{Z}^{10}\\ |\boldsymbol{v}-\boldsymbol{v}_0|\leqslant L}}|S_{q,W}(\boldsymbol{v})| \ll_{F,\gamma,\varepsilon} (LQ+|\boldsymbol{v}_0|)^{\varepsilon}\left(1+\frac{L}{Q^{1/3}}\right)^{10}Q^{10-\gamma}$$
$$= (LQ+|\boldsymbol{v}_0|)^{\varepsilon}(L+Q^{1/3})^{10}Q^{20/3-\gamma}.$$

Together with the case $L \leqslant Q^{1/3}$, this proves Proposition 11.1. □

12. The circle method

Throughout this section we shall assume that $n \geqslant 10$. Our basic approach is to assume for a contradiction that $X(\mathbb{Q}) = \emptyset$. Under this assumption our goal will be to show that

$$\#\{\boldsymbol{x}\in\mathbb{Z}^n : |\boldsymbol{x}|\leqslant P,\ F(\boldsymbol{x})=0\} \gg P^{n-3},$$

for all $P \gg 1$, where the implied constants are allowed to depend on F. This is a contradiction. We shall attempt to establish this lower bound for the counting function for $n = 10$.

Since F is a square-free cubic form, there exists a point $\boldsymbol{x}_0 \in \mathbb{R}^n$ such that $F(\boldsymbol{x}_0) = 0$ and $\nabla F(\boldsymbol{x}_0) \neq \mathbf{0}$. Furthermore, by Lemma 3.4 we can assume that $\operatorname{rank}(H(\boldsymbol{x}_0)) = r(F) \geqslant 2n/3$.

After a fixed $\mathbb{Q}$-linear change of variables, we may assume that $\frac{\partial F}{\partial x_1}(\boldsymbol{x}_0) \neq 0$ and that the leading principal $r(F) \times r(F)$ minor of $H(\boldsymbol{x}_0)$ is nonzero. Let

$$\Psi : \mathbb{R}^n \to \mathbb{R}^{r(F)}, \qquad \boldsymbol{y} \mapsto \left(\frac{\partial F(\boldsymbol{y})}{\partial y_1}, \dots, \frac{\partial F(\boldsymbol{y})}{\partial y_{r(F)}} \right). \tag{12.1}$$

By the inverse function theorem, there exists an open ball U centred at $\boldsymbol{x}_0$ for which the map $\boldsymbol{y} \mapsto \big(\Psi(\boldsymbol{y}), y_{r(F)+1}, \dots, y_n\big)$ defines a diffeomorphism from U onto its image, with Jacobian

$$\left| \det \left(\frac{\partial^2 F}{\partial y_i \partial y_j}(\boldsymbol{y}) \right)_{1 \leqslant i,j \leqslant r(F)} \right| \gg 1 \quad \text{for } \boldsymbol{y} \in U. \tag{12.2}$$

Let $\omega \in C_c^\infty(\mathbb{R}^n)$ be a fixed non-negative smooth weight function supported on $[-1,1]^n$, with $\omega(\mathbf{0}) > 0$. Let

$$w(\boldsymbol{y}) = \omega((\boldsymbol{x}_0 - \boldsymbol{y})/\ell),$$

where $\ell > 0$ is a sufficiently small parameter. In particular, $w(\boldsymbol{y}) \neq 0$ implies that $|\boldsymbol{y} - \boldsymbol{x}_0| \leqslant \ell$. In view of Lemma 2.10, the slightly larger box $|\boldsymbol{y} - \boldsymbol{x}_0| \leqslant 2\ell$ will be relevant when estimating certain oscillatory integrals. By taking ℓ to be sufficiently small, we can assume that

$$\left| \frac{\partial F}{\partial y_1}(\boldsymbol{y}) \right| \gg 1 \quad \text{for } |\boldsymbol{y} - \boldsymbol{x}_0| \leqslant 2\ell, \tag{12.3}$$

and that

$$\boldsymbol{y} \in U \quad \text{for } |\boldsymbol{y} - \boldsymbol{x}_0| \leqslant 2\ell. \tag{12.4}$$

All of the estimates in our work are allowed to depend on the coefficients of F and on the small parameter ℓ appearing in the definition of w. Fix W and Ω_W as in (8.6). For $n = 10$ choose W using Proposition 11.1, with a fixed $0 < \gamma < 1/96$, before choosing η, ε. All implicit constants below may also depend on W. With these conventions in place, our line of attack is to use the circle method to study the counting function

$$N(P) = \sum_{\substack{\boldsymbol{x} \in \mathbb{Z}^n \\ F(\boldsymbol{x}) = 0 \\ \boldsymbol{x} \bmod W \in \Omega_W}} w(\boldsymbol{x}/P) \tag{12.5}$$

introduced in (8.7), as $P \to \infty$.

The precise form of the circle method we use is the following re-interpretation due to Marmon and Vishe [MV19, Proposition 1.2 and Lemma 2.2] of Heath-Brown's smooth δ-method [HB96], which in turn is based on pioneering work of Duke, Friedlander and Iwaniec [DFI93].

Proposition 12.1. *Let $Q \geqslant 1$ and $\eta > 0$. Then for all $x \in \mathbb{Z}$,*

$$\delta_0(x) = \sum_{q \leqslant Q} \sum_{\substack{1 \leqslant a \leqslant q \\ \gcd(a,q) = 1}} \int_{|\theta| < (qQ)^{-1+\eta}} p_q(\theta) e((a/q + \theta)x) \mathrm{d}\theta + O_{N,\eta}\left(Q^{-N\eta}\right),$$

where p_q is a smooth function satisfying $|p_q(\theta)| \ll 1$ and

$$p_q(\theta) = 1 + O_N\big((1 + Q^2|\theta|)^{2N+2}(q/Q)^N\big) \qquad (\theta \in \mathbb{R}). \tag{12.6}$$

It follows from Proposition 12.1 that

$$N(P) = \sum_{q \leqslant Q} \sum_{\substack{1 \leqslant a \leqslant q \\ \gcd(a,q) = 1}} \int_{|\theta| < (qQ)^{-1+\eta}} p_q(\theta) S(a/q + \theta) \mathrm{d}\theta + O_{N,\eta}(P^n Q^{-N\eta}), \tag{12.7}$$

where

$$S(\alpha) = \sum_{\substack{\boldsymbol{x} \in \mathbb{Z}^n \\ \boldsymbol{x} \bmod W \in \Omega_W}} w(\boldsymbol{x}/P) e(\alpha F(\boldsymbol{x}))$$

for any $\alpha \in \mathbb{R}$. We shall henceforth take $Q = P^{3/2}$. In particular, the error term in (12.7) is $O(1)$ on taking N sufficiently large in terms of η. It will be convenient to examine the contribution

from q and $|\theta|$ restricted to dyadic intervals. Let $R \geqslant 1/2$ and let $\phi > 0$. We let $N(P; R, \phi)$ be the overall contribution to $N(P)$ from q and θ such that

$$R < q \leqslant 2R \quad \text{and} \quad \phi < |\theta| \leqslant 2\phi.$$

That is, $q \sim R$ and $|\theta| \sim \phi$. Clearly we have $N(P; R, \phi) = 0$ unless $R \ll Q$ and $\phi \ll (RQ)^{-1+\eta}$.

We shall adopt an approach based on Poisson summation to extract an asymptotic formula for $N(P)$. The Poisson summation formula in residue classes $\boldsymbol{x}$ mod $\operatorname{lcm}(q, W)$ yields

$$\sum_{\substack{1 \leqslant a \leqslant q \\ \gcd(a,q)=1}} S(a/q + \theta) = \frac{1}{\operatorname{lcm}(q, W)^n} \sum_{\boldsymbol{v} \in \mathbb{Z}^n} S_{q,W}(\boldsymbol{v}) I(\theta, \operatorname{lcm}(q, W)^{-1}\boldsymbol{v}),$$

in the integral notation of (2.4). The complete exponential sum $S_{q,W}(\boldsymbol{v})$ was defined in (8.9).

As usual, the main term will come from the vector $\boldsymbol{v} = \mathbf{0}$, and the main task is to show that the vectors $\boldsymbol{v} \neq \mathbf{0}$ make a negligible overall contribution to $N(P; R, \phi)$. Returning to the expression (12.7), and recalling that $|p_q(\theta)| \ll 1$, we clearly have

$$N(P; R, \phi) = M(P; R, \phi) + O\big(E(P; R, \phi)\big),$$

where

$$M(P; R, \phi) = \sum_{q \sim R} \frac{S_{q,W}(\mathbf{0})}{\operatorname{lcm}(q, W)^n} \int_{|\theta| \sim \phi} p_q(\theta) I(\theta, \mathbf{0}) \mathrm{d}\theta \tag{12.8}$$

and

$$E(P; R, \phi) = \sum_{q \sim R} \frac{1}{\operatorname{lcm}(q, W)^n} \sum_{\mathbf{0} \neq \boldsymbol{v} \in \mathbb{Z}^n} |S_{q,W}(\boldsymbol{v})| \int_{|\theta| \sim \phi} |I(\theta, \operatorname{lcm}(q, W)^{-1}\boldsymbol{v})| \mathrm{d}\theta. \tag{12.9}$$

In (12.8) and (12.9), the restrictions $q \leqslant Q$ and $|\theta| < (qQ)^{-1+\eta}$ are understood; this is a very minor point, relevant only at values of R and ϕ near the boundary. Before proceeding, let us quickly dispose of the main term

$$M(P) = \sum_{R,\phi} M(P; R, \phi) = \sum_{q \leqslant Q} \frac{S_{q,W}(\mathbf{0})}{\operatorname{lcm}(q, W)^n} \int_{|\theta| < (qQ)^{-1+\eta}} p_q(\theta) I(\theta, \mathbf{0}) \mathrm{d}\theta.$$

Lemma 12.2. *We have $M(P) \gg P^{n-3}$ for all sufficiently large P.*

Proof. Let $F_1 = \partial F / \partial x_1$. Writing $\boldsymbol{x} = P\boldsymbol{y}$ in (2.4) gives $I(\theta; \mathbf{0}) = P^n J_w(P^3\theta)$, where

$$J_w(t) = \int_{\mathbb{R}^n} w(\boldsymbol{y}) e(tF(\boldsymbol{y})) \, \mathrm{d}\boldsymbol{y} = \int_{\mathbb{R}^n} w(\boldsymbol{y}) e(tf) \frac{\mathrm{d}f}{|F_1(\boldsymbol{y})|} \mathrm{d}\boldsymbol{y}'$$

by the change of coordinates $\boldsymbol{y} \mapsto (f, \boldsymbol{y}')$ where $f = F(\boldsymbol{y})$ and $\boldsymbol{y}' = (y_2, \dots, y_n)$, which is valid by (12.3). In particular, J_w is Schwartz in t, by integration by parts over f. By Fourier inversion,

$$\int_{\mathbb{R}} J_w(t) \, \mathrm{d}t = \int_{F(\boldsymbol{y})=0} w(\boldsymbol{y}) \frac{\mathrm{d}\boldsymbol{y}'}{|F_1(\boldsymbol{y})|} > 0,$$

since $w \geqslant 0$, $w(\boldsymbol{x}_0) > 0$ and $F(\boldsymbol{x}_0) = 0$.

Now write $\theta = t/P^3$ in the definition of $M(P)$ to get

$$\frac{M(P)}{P^{n-3}} = \sum_{q \leqslant Q} \frac{S_{q,W}(\mathbf{0})}{\operatorname{lcm}(q, W)^n} \int_{|t/P^3| < (qQ)^{-1+\eta}} p_q(t/P^3) J_w(t) \mathrm{d}t.$$

Let $B_q = P^3(qQ)^{-1+\eta} \geqslant P^3(qQ)^{-1} = Q/q$. By the triangle inequality,

$$\begin{aligned} \left| \int_{|t| < B_q} p_q(t/P^3) J_w(t) \, \mathrm{d}t - \int_{\mathbb{R}} J_w(t) \, \mathrm{d}t \right| &\leqslant \int_{|t| < B_q} |p_q(t/P^3) - 1| \, |J_w(t)| \, \mathrm{d}t + \int_{|t| \geqslant B_q} |J_w(t)| \, \mathrm{d}t \\ &\ll_w \frac{q}{Q} \int_{\mathbb{R}} (1 + |t|)^4 |J_w(t)| \, \mathrm{d}t + B_q^{-1} \ll_{F,w} \frac{q}{Q}, \end{aligned}$$

where we have applied the $N = 1$ case of (12.6), together with the rapid decay of J_w. Consequently, by the triangle inequality and the absolute convergence of $\mathfrak{S}_\Omega$ from Lemma 8.3,

$$\begin{aligned}\frac{M(P)}{P^{n-3}} - \mathfrak{S}_\Omega \int_{\mathbb{R}} J_w(t)\,\mathrm{d}t &\ll_{F,w} \sum_{q\leqslant Q} \frac{|S_{q,W}(\mathbf{0})|}{\mathrm{lcm}(q,W)^n}\frac{q}{Q} + \sum_{q>Q}\frac{|S_{q,W}(\mathbf{0})|}{\mathrm{lcm}(q,W)^n}\int_{\mathbb{R}} J_w(t)\,\mathrm{d}t \\ &\ll Q^{-1/2}\sum_{q\geqslant 1}\frac{|S_{q,W}(\mathbf{0})|}{\mathrm{lcm}(q,W)^n} + \sum_{q>Q^{1/2}}\frac{|S_{q,W}(\mathbf{0})|}{\mathrm{lcm}(q,W)^n} = o(1)\end{aligned}$$

as $P\to\infty$. Thus $M(P)/P^{n-3}\to \mathfrak{S}_\Omega\int_{\mathbb{R}} J_w(t)\,\mathrm{d}t > 0$ as $P\to\infty$. □

We now turn to the error terms $E(P;R,\phi)$. Let

$$V = RP^{-1}\max\left\{1,\phi P^3\right\} \quad \text{and} \quad V_0 = RP^{-1}\max\left\{1,\sqrt{\phi P^3}\right\}. \tag{12.10}$$

Take $0<\eta<1/12$ and P sufficiently large; then for all $R\ll Q$ and $\phi\ll (RQ)^{-1+\eta}$, we certainly have $\mathrm{lcm}(q,W)\leqslant P^2$ and $|\theta|\leqslant \mathrm{lcm}(q,W)^{-1}P^{-1}$. It follows directly from Lemma 2.10, with $\varepsilon/2$ in place of ε, that $I(\theta,\mathrm{lcm}(q,W)^{-1}\boldsymbol{v})\ll_N P^{-N}$ unless $|\boldsymbol{v}|\leqslant P^\varepsilon V$. Moreover, observe that

$$q\sim R \quad\Longrightarrow\quad \mathrm{lcm}(q,W)\asymp_W R.$$

We shall employ the following upper bound for the volume appearing in Lemma 2.10, with a general modulus $q\asymp_W R$ in place of $\mathrm{lcm}(q,W)$.

Lemma 12.3. *Let $q\asymp_W R$, $|\theta|\sim\phi$, and let V_0 be given by (12.10). Assume that (12.4) holds. Then, for every $\varepsilon>0$ and every $\boldsymbol{v}\in\mathbb{R}^n$, we have*

$$\mathrm{Vol}\left\{\boldsymbol{x}\in\mathbb{R}^n : |\boldsymbol{x}/P-\boldsymbol{x}_0|\leqslant 2\ell,\ |q\theta\nabla F(\boldsymbol{x})-\boldsymbol{v}|\leqslant P^\varepsilon V_0\right\}\ll P^{n+O_n(\varepsilon)}\left(\frac{R}{PV_0}\right)^\rho,$$

where $\rho = r(F)$.

Proof. We may assume that $\theta\neq 0$, since $\phi>0$. Write $\boldsymbol{x}=P\boldsymbol{y}$. As F is cubic, we have $\nabla F(P\boldsymbol{y}) = P^2\nabla F(\boldsymbol{y})$ and the volume in question is

$$P^n\,\mathrm{Vol}\left\{\boldsymbol{y}\in\mathbb{R}^n : |\boldsymbol{y}-\boldsymbol{x}_0|\leqslant 2\ell,\ |q\theta P^2\nabla F(\boldsymbol{y})-\boldsymbol{v}|\leqslant P^\varepsilon V_0\right\}.$$

By (12.4), the box $|\boldsymbol{y}-\boldsymbol{x}_0|\leqslant 2\ell$ is contained in U. Using the diffeomorphism

$$\boldsymbol{y}\mapsto(\Psi(\boldsymbol{y}),y_{\rho+1},\dots,y_n)$$

on U and the Jacobian bound (12.2), where Ψ was defined in (12.1), we find by Fubini's theorem that

$$\mathrm{Vol}\{\boldsymbol{y}\in U : |\Psi(\boldsymbol{y})-\boldsymbol{a}|\leqslant\Delta\}\ll\Delta^\rho,$$

uniformly for $\boldsymbol{a}\in\mathbb{R}^\rho$ and $0<\Delta\leqslant 1$. For $\Delta>1$ we take the trivial bound $O(1)$ for the volume. It follows that

$$\mathrm{Vol}\left\{\boldsymbol{y}\in\mathbb{R}^n : |\boldsymbol{y}-\boldsymbol{x}_0|\leqslant 2\ell,\ |\nabla F(\boldsymbol{y})-\boldsymbol{b}|\leqslant\Delta\right\}\ll\min\{1,\Delta^\rho\}, \tag{12.11}$$

uniformly for $\boldsymbol{b}\in\mathbb{R}^n$.

Applying this with $\boldsymbol{b}=(q\theta P^2)^{-1}\boldsymbol{v}$ and $\Delta=(q|\theta|P^2)^{-1}P^\varepsilon V_0$, we deduce that

$$\mathrm{Vol}\left\{\boldsymbol{x}\in\mathbb{R}^n : |\boldsymbol{x}/P-\boldsymbol{x}_0|\leqslant 2\ell,\ |q\theta\nabla F(\boldsymbol{x})-\boldsymbol{v}|\leqslant P^\varepsilon V_0\right\}\ll P^{n+O_n(\varepsilon)}\min\left\{1,\frac{V_0}{R\phi P^2}\right\}^\rho,$$

since $q\asymp_W R$ and $|\theta|\sim\phi$. The lemma will follow if we are able to show that

$$\min\left\{1,\frac{V_0}{R\phi P^2}\right\}\ll\frac{R}{PV_0},$$

where we recall that V_0 is given by (12.10). But this is easily checked on a case distinction between the case $\phi P^3\leqslant 1$ (so that $V_0 = RP^{-1}$) and the case $\phi P^3>1$ (so that $V_0 = RP^{-1}\sqrt{\phi P^3}$). □

Alternatively, we can exploit averaging over $\boldsymbol{v} \in \mathbb{Z}^n$ to get alternative estimates. We begin with the following simple estimate.

Lemma 12.4. *Let $q \asymp_W R$, $|\theta| \sim \phi$, and let V, V_0 be given by (12.10). Then, for every $\varepsilon > 0$, we have*

$$\sum_{\substack{\boldsymbol{v} \in \mathbb{Z}^n \\ |\boldsymbol{v}| \leqslant P^\varepsilon V}} \mathrm{Vol}\left\{\boldsymbol{x} \in \mathbb{R}^n : |\boldsymbol{x}/P - \boldsymbol{x}_0| \leqslant 2\ell,\ |q\theta \nabla F(\boldsymbol{x}) - \boldsymbol{v}| \leqslant P^\varepsilon V_0\right\} \ll P^{n+O_n(\varepsilon)}(1+V_0)^n.$$

Proof. Interchanging the integral with the sum, we are left with estimating the number of integer vectors $\boldsymbol{v}$, each of whose entries lie in an interval of length $O(P^\varepsilon V_0)$. The statement of the lemma is now clear, on integrating over $\boldsymbol{x}$ such that $|\boldsymbol{x}/P - \boldsymbol{x}_0| \leqslant 2\ell$. □

The following result interpolates between aspects of both of the previous estimates. Taking $L = 2P^\varepsilon V$ morally recovers Lemma 12.3, whereas taking $L = 0$ refines Lemma 12.4. Lemma 12.5 will eventually let us use ℓ^∞-ℓ^1 Hölder's inequality on windows of an optimised length $L \geqslant 0$, in order to separate the integral from other factors in (12.9).

Lemma 12.5. *Fix $\varepsilon > 0$. Let $P, R \geqslant 1/2$ and $\phi > 0$. Let V, V_0 be given by (12.10). Assume that (12.4) holds. Let $\rho = r(F)$. Let $L \geqslant 0$ be a real number. If $|\alpha| \asymp_W R\phi$, then*

$$\sum_{\boldsymbol{v} \in \mathbb{Z}^n} M_\alpha(\boldsymbol{v}, L) \ll \begin{cases} 0 & \text{if } P^\varepsilon V < 1, \\ P^{n+O_n(\varepsilon)}(1+\frac{L}{V})^\rho (1+V_0+L)^{n-\rho} V_0^\rho & \text{if } \phi P^3 > 1, \\ P^{n+O_n(\varepsilon)}(V+L)^n & \text{if } \phi P^3 \leqslant 1, \end{cases}$$

where

$$M_\alpha(\boldsymbol{v}, L) := \sup_{\substack{\boldsymbol{a} \in \mathbb{Z}^n \\ 0 < |\boldsymbol{a}| \leqslant P^\varepsilon V \\ |\boldsymbol{a} - \boldsymbol{v}| \leqslant L}} \mathrm{Vol}\left\{\boldsymbol{x} \in \mathbb{R}^n : |\boldsymbol{x}/P - \boldsymbol{x}_0| \leqslant 2\ell,\ |\alpha \nabla F(\boldsymbol{x}) - \boldsymbol{a}| \leqslant P^\varepsilon V_0\right\},$$

with the convention that $\sup_\emptyset = 0$ if the set in the supremum is empty.

Proof. For each $\boldsymbol{v}$ for which the set in the supremum is nonempty, choose $\boldsymbol{a} = \boldsymbol{a}(\boldsymbol{v})$ with $0 < |\boldsymbol{a}| \leqslant P^\varepsilon V$ and $|\boldsymbol{a} - \boldsymbol{v}| \leqslant L$ such that

$$M_\alpha(\boldsymbol{v}, L) = \mathrm{Vol}\left\{\boldsymbol{x} \in \mathbb{R}^n : |\boldsymbol{x}/P - \boldsymbol{x}_0| \leqslant 2\ell,\ |\alpha \nabla F(\boldsymbol{x}) - \boldsymbol{a}| \leqslant P^\varepsilon V_0\right\}. \tag{12.12}$$

In particular, the outer sum is zero unless $P^\varepsilon V \geqslant 1$, which we now assume. We begin as in the proof of Lemma 12.3 by writing $\boldsymbol{x} = P\boldsymbol{y}$, where $\boldsymbol{y} \in U$ by (12.4). By the discussion after (12.1), we have a diffeomorphism

$$U \xrightarrow{\cong} S \subset \mathbb{R}^n, \quad \boldsymbol{y} \mapsto (\Psi(\boldsymbol{y}), y_{\rho+1}, \dots, y_n), \tag{12.13}$$

for which the Jacobian and the Jacobian of the inverse map are uniformly bounded on U. Then, for all vectors $\boldsymbol{s}, \boldsymbol{t} \in \mathbb{R}^\rho$ and $\boldsymbol{y}'' \in \mathbb{R}^{n-\rho}$ with $(\boldsymbol{s}, \boldsymbol{y}''), (\boldsymbol{t}, \boldsymbol{y}'') \in U$, we have

$$|\nabla F(\boldsymbol{s}, \boldsymbol{y}'') - \nabla F(\boldsymbol{t}, \boldsymbol{y}'')| \ll_U |\boldsymbol{s} - \boldsymbol{t}|, \tag{12.14}$$

$$|\Psi(\boldsymbol{s}, \boldsymbol{y}'') - \Psi(\boldsymbol{t}, \boldsymbol{y}'')| \gg_U |\boldsymbol{s} - \boldsymbol{t}|. \tag{12.15}$$

The inequality (12.14) is a general consequence of the smoothness of ∇F; the inequality (12.15) similarly follows from the smoothness of the inverse of the diffeomorphism (12.13).

Let us write $\boldsymbol{v} = (\boldsymbol{v}', \boldsymbol{v}'')$ and $\boldsymbol{a} = (\boldsymbol{a}', \boldsymbol{a}'')$ for $\boldsymbol{v}', \boldsymbol{a}' \in \mathbb{Z}^\rho$ and $\boldsymbol{v}'', \boldsymbol{a}'' \in \mathbb{Z}^{n-\rho}$. Similarly write $\boldsymbol{y} = (\boldsymbol{y}', \boldsymbol{y}'')$, where $\boldsymbol{y}' \in \mathbb{R}^\rho$ and $\boldsymbol{y}'' \in \mathbb{R}^{n-\rho}$. By the triangle inequality

$$|\alpha \nabla F(\boldsymbol{x}) - \boldsymbol{v}| \leqslant |\alpha \nabla F(\boldsymbol{x}) - \boldsymbol{a}| + |\boldsymbol{a} - \boldsymbol{v}| \leqslant P^\varepsilon V_0 + L$$

and the change of coordinates $\boldsymbol{x} = P\boldsymbol{y}$,

$$\mathrm{Vol}\left\{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{x}/P \in U,\ |\alpha \nabla F(\boldsymbol{x}) - \boldsymbol{a}| \leqslant P^\varepsilon V_0\right\} \leqslant P^n \int_{\substack{\boldsymbol{y} \in U \\ |\alpha P^2 \Psi(\boldsymbol{y}) - \boldsymbol{a}'| \leqslant P^\varepsilon V_0}} \mathbf{1}_{A(\boldsymbol{y}, \boldsymbol{v})}\, \mathrm{d}\boldsymbol{y}, \tag{12.16}$$

where $A(\boldsymbol{y}, \boldsymbol{v})$ denotes the condition that $|\alpha P^2 \frac{\partial F(\boldsymbol{y})}{\partial y_j} - v_j| \leqslant P^\varepsilon V_0 + L$ for $1 \leqslant j \leqslant n$. The condition $A(\boldsymbol{y}, \boldsymbol{v})$ constrains $\boldsymbol{v}$ directly, rather than $\boldsymbol{a}$, at the cost of introducing a summand L. If $\boldsymbol{v}'$ and $\boldsymbol{y}''$ are fixed, then any two possible choices $\boldsymbol{s}, \boldsymbol{t} \in \mathbb{R}^\rho$ for the vector $\boldsymbol{y}'$ satisfy

$$|\boldsymbol{s} - \boldsymbol{t}| \ll_U \frac{P^\varepsilon V_0 + L}{|\alpha P^2|},$$

by $A(\boldsymbol{y}, \boldsymbol{v})$ applied with $1 \leqslant j \leqslant \rho$, together with the inequality (12.15). Then, (12.14) and the case $\rho + 1 \leqslant j \leqslant n$ of $A(\boldsymbol{y}, \boldsymbol{v})$ constrain the vector $\boldsymbol{v}''$ to lie in a box $B(\boldsymbol{v}', \boldsymbol{y}'')$ of diameter

$$\ll_U \sup_{\boldsymbol{s},\boldsymbol{t}} |\alpha P^2||\boldsymbol{s} - \boldsymbol{t}| + P^\varepsilon V_0 + L \ll_U |\alpha P^2| \frac{P^\varepsilon V_0 + L}{|\alpha P^2|} + P^\varepsilon V_0 + L \asymp P^\varepsilon V_0 + L.$$

The strategy is now to choose $\boldsymbol{v}'$, $\boldsymbol{y}''$, $\boldsymbol{v}''$ and $\boldsymbol{y}'$ in that order, where

$$|\boldsymbol{v}'| \leqslant |\boldsymbol{v}| \leqslant |\boldsymbol{a}| + |\boldsymbol{v} - \boldsymbol{a}| \leqslant P^\varepsilon V + L.$$

Plugging (12.16) into (12.12) and summing over $\boldsymbol{v}$, it follows from our analysis of $\boldsymbol{v}'$ and $\boldsymbol{v}''$ above that

$$\begin{aligned}\sum_{\boldsymbol{v} \in \mathbb{Z}^n} M_\alpha(\boldsymbol{v}, L) &\leqslant P^n \sum_{\boldsymbol{v} \in \mathbb{Z}^n} \int_{\substack{\boldsymbol{y} \in U \\ |\alpha P^2 \Psi(\boldsymbol{y}) - \boldsymbol{a}'| \leqslant P^\varepsilon V_0}} \mathbf{1}_{|\boldsymbol{v}'| \leqslant P^\varepsilon V + L} \mathbf{1}_{\boldsymbol{v}'' \in B(\boldsymbol{v}', \boldsymbol{y}'')} \, \mathrm{d}\boldsymbol{y} \\ &= P^n \sum_{\substack{\boldsymbol{v}' \in \mathbb{Z}^\rho \\ |\boldsymbol{v}'| \leqslant P^\varepsilon V + L}} \int_{\substack{\boldsymbol{y}'' \in \mathbb{R}^{n-\rho} \\ |\boldsymbol{y}''| \ll_U 1}} \sum_{\boldsymbol{v}'' \in B(\boldsymbol{v}', \boldsymbol{y}'')} \int_{\substack{\boldsymbol{y}' \in \mathbb{R}^\rho \\ \boldsymbol{y} \in U \\ |\alpha P^2 \Psi(\boldsymbol{y}) - \boldsymbol{a}'| \leqslant P^\varepsilon V_0}} \mathrm{d}\boldsymbol{y}' \, \mathrm{d}\boldsymbol{y}'',\end{aligned}$$

by Fubini's theorem. For any fixed $\boldsymbol{v}'$, $\boldsymbol{y}''$ and $\boldsymbol{v}''$, the inner volume is

$$\ll_U P^{O_n(\varepsilon)} \min\left\{1, \frac{V_0}{R\phi P^2}\right\}^\rho,$$

arguing via the diffeomorphism (12.13) as in the proof of (12.11). The number of available vectors $\boldsymbol{v}'$ and $\boldsymbol{v}''$ being $O((1 + P^\varepsilon V + L)^\rho)$ and $O((1 + P^\varepsilon V_0 + L)^{n-\rho})$, respectively, we get

$$\sum_{\boldsymbol{v} \in \mathbb{Z}^n} M_\alpha(\boldsymbol{v}, L) \ll P^{n + O_n(\varepsilon)} (V + L)^\rho (1 + V_0 + L)^{n-\rho} \min\left\{1, \frac{V_0}{R\phi P^2}\right\}^\rho,$$

since $P^\varepsilon V \geqslant 1$.

If $\phi P^3 \leqslant 1$ then $V = V_0 = RP^{-1}$ and the overall bound is clearly $O(P^{n+O_n(\varepsilon)}(V + L)^n)$, since $P^\varepsilon V \geqslant 1$. On the other hand, if $\phi P^3 > 1$ then $V = R\phi P^2$ and $V_0 = R\phi^{1/2} P^{1/2}$. But then $V_0/(R\phi P^2) = (\phi P^3)^{-1/2} < 1$, leading to the overall contribution

$$\ll P^{n+O_n(\varepsilon)} (V + L)^\rho (1 + V_0 + L)^{n-\rho} (V_0/V)^\rho = P^{n+O_n(\varepsilon)} (1 + \tfrac{L}{V})^\rho (1 + V_0 + L)^{n-\rho} V_0^\rho,$$

as required. □

Let V, V_0 be given by (12.10). Plugging Lemma 2.10 into (12.9) and writing $\tau = \mathrm{lcm}(q, W)\theta/R$, we get

$$E(P; R, \phi) \ll \phi + \frac{1}{L^n} \int_{|\tau| \asymp_W \phi} \sum_{\boldsymbol{v} \in \mathbb{Z}^n} M_{R\tau}(\boldsymbol{v}, L) \sum_{q \sim R} \frac{1}{R^n} \sum_{\substack{\boldsymbol{a} \in \mathbb{Z}^n \\ 0 < |\boldsymbol{a}| \leqslant P^\varepsilon V \\ |\boldsymbol{a} - \boldsymbol{v}| \leqslant L}} |S_{q,W}(\boldsymbol{a})| \, \mathrm{d}\tau \tag{12.17}$$

for any integer $1 \leqslant L = L(P, R, \phi) \ll P^{100}$, since by averaging over windows of length L and applying ℓ^∞-ℓ^1 Hölder's inequality, if $f, g \geqslant 0$ then

$$(2L + 1)^n \sum_{\boldsymbol{v} \in \mathbb{Z}^n} f(\boldsymbol{v}) g(\boldsymbol{v}) = \sum_{\boldsymbol{v} \in \mathbb{Z}^n} \sum_{\substack{\boldsymbol{a} \in \mathbb{Z}^n \\ |\boldsymbol{a} - \boldsymbol{v}| \leqslant L}} f(\boldsymbol{a}) g(\boldsymbol{a}) \leqslant \sum_{\boldsymbol{v} \in \mathbb{Z}^n} \sup_{\substack{\boldsymbol{a} \in \mathbb{Z}^n \\ |\boldsymbol{a} - \boldsymbol{v}| \leqslant L}} g(\boldsymbol{a}) \sum_{\substack{\boldsymbol{a} \in \mathbb{Z}^n \\ |\boldsymbol{a} - \boldsymbol{v}| \leqslant L}} f(\boldsymbol{a}).$$

Conjecture 12.6. *Fix $n \geqslant 10$. There exists a constant $b < 2n/3$ such that for all real $\varepsilon > 0$, $B, L, R \geqslant 1/2$, and vectors $\boldsymbol{v} \in \mathbb{Z}^n$ with $|\boldsymbol{v}| \leqslant B$, we have*

$$\sum_{q \sim R} \sum_{\substack{\mathbf{0} \neq \boldsymbol{a} \in \mathbb{Z}^n \\ |\boldsymbol{a} - \boldsymbol{v}| \leqslant L}} |S_{q,W}(\boldsymbol{a})| \ll_\varepsilon (BLR)^\varepsilon (L + R^{1/3})^n R^b.$$

For $n = 10$ this assertion is Proposition 11.1, with $b = 20/3 - \gamma$ and the fixed W chosen above.

Proposition 12.7. *Assume Conjecture 12.6 for some fixed $n \geqslant 10$. Let $\rho = r(F)$. Then*

$$E(P; R, \phi) \ll \min\{1, \phi P^3\} \left(1 + P^{n-3+O(\eta+\varepsilon)-\Delta}\right),$$

where $\Delta = \min(\rho/4 - 3/2, \rho/2 - 1, 2n/3 - b) > 0$ is admissible.

Proof. Plugging Lemma 12.5 with $\alpha = R\tau$ and Conjecture 12.6 with $B \ll P^{100}$ into (12.17), the desired bound is trivial unless $P^\varepsilon V \geqslant 1$, in which case

$$E(P; R, \phi) \ll \phi + (1 + \tfrac{R^{1/3}}{L})^n \phi R^{b-n} P^{n+O_n(\varepsilon)} \begin{cases} (1 + \frac{L}{V})^\rho (V_0 + L)^{n-\rho} V_0^\rho & \text{if } \phi P^3 > 1, \\ (V + L)^n & \text{if } \phi P^3 \leqslant 1, \end{cases}$$

for any real $1 \leqslant L \ll P^{100}$, since $L \asymp \lceil L \rceil \in \mathbb{Z}$. Here $1/2 \leqslant R \ll P^{3/2}$ and $\phi \ll (RP^{3/2})^{-1+\eta}$.

If $\phi P^3 \leqslant 1$ then $V = RP^{-1}$ and $P^\varepsilon V \gg 1$, so $R^{1/3} \gg R/P = V \gg P^{-\varepsilon}$, since $R \ll P^{3/2}$. Taking $L \asymp R^{1/3} \gg V$ gives

$$E(P; R, \phi) \ll \phi + \phi R^{b-2n/3} P^{n+O_n(\varepsilon)} \ll \phi + \phi P^{b-2n/3} P^{n+O(\varepsilon)},$$

since $R \gg P^{1-\varepsilon}$. Thus $\Delta = 2n/3 - b > 0$ is admissible in this case.

Now suppose $\phi P^3 > 1$. Then $V = R\phi P^2$ and $V_0 = R\phi^{1/2} P^{1/2}$, with $P^\varepsilon V \gg 1$, $V > V_0$ and $V_0 \ll (RP^{3/2})^{\eta/2} R^{1/2} P^{-1/4} \ll (RP^{3/2})^{\eta/2} R^{1/3}$. Taking $L \asymp R^{1/3}$ gives

$$E(P; R, \phi) \ll 1 + \phi R^{b-n} P^{n+O_n(\varepsilon)} (1 + \tfrac{R^{1/3}}{V})^\rho (V_0 + R^{1/3})^{n-\rho} V_0^\rho.$$

Writing $V = (\phi P^3)^{1/2} V_0$ and $V_0 + R^{1/3} \ll (RP^{3/2})^{\eta/2} R^{1/3}$, this is

$$\ll 1 + \phi R^{b-2n/3-\rho/3} P^{n+O(\eta+\varepsilon)} (V_0 + \tfrac{R^{1/3}}{(\phi P^3)^{1/2}})^\rho.$$

Since $\rho \geqslant 2$, optimising over the interval $\max(P^{-3}, R^{-1}P^{-2-\varepsilon}) \ll \phi \ll (RP^{3/2})^{-1+\eta}$ gives

$$\begin{aligned} E(P; R, \phi) &\ll 1 + R^{b-2n/3-\rho/3} P^{n+O(\eta+\varepsilon)} (R^{-1} P^{-3/2} R^{\rho/2} P^{-\rho/4} + P^{-3} \max(1, P/R)^{1-\rho/2} R^{\rho/3}) \\ &= 1 + R^{b-2n/3} P^{n-3+O(\eta+\varepsilon)} ((R/P^{3/2})^{\rho/6-1} + \max(1, P/R)^{1-\rho/2}). \end{aligned}$$

By writing each term on the right-hand side as P raised to a piecewise linear function of the quantity $\log_P R$, we find that this is optimised at either $R \asymp 1$, $R \asymp P$, or $R \asymp P^{3/2}$. Thus

$$E(P; R, \phi) \ll 1 + P^{n-3+O(\eta+\varepsilon)} ((1/P^{3/2})^{\rho/6-1} + P^{1-\rho/2} + P^{b-2n/3}).$$

Since $\rho = r(F) \geqslant \lceil 2n/3 \rceil \geqslant 7$, the saving $\Delta = \min(\rho/4 - 3/2, \rho/2 - 1, 2n/3 - b) > 0$ is admissible. □

Dyadic summation of Proposition 12.7 over R and ϕ gives $N(P) = M(P) + o(P^{n-3})$, unconditionally for $n = 10$. Thus $N(P) \gg P^{n-3}$ for all sufficiently large P, by Lemma 12.2. Theorem 1.1 follows first for $n = 10$, and then for $n \geqslant 11$ as well by setting the last $n - 10$ variables of F equal to zero.

Appendix A. Applying singular support

Let $n \geqslant 2$ and let $F \in \mathbb{Z}[x_1, \dots, x_n]$ be a nonzero homogeneous polynomial of degree $d \geqslant 1$. For a finite field $\mathbb{F}_q$ of characteristic p, we define a nontrivial additive character $\psi_q \colon \mathbb{F}_q \to \mathbb{C}^*$ by $\psi_q(t) = e_p(\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(t))$. For $\boldsymbol{v} \in \mathbb{F}_q^n$, we are interested in the exponential sum

$$T_q(\boldsymbol{v}) = \sum_{\substack{\boldsymbol{x} \in \mathbb{F}_q^n \\ F(\boldsymbol{x})=0}} \psi_q(\boldsymbol{x} \cdot \boldsymbol{v}) - q^{n-1}\mathbf{1}_{\boldsymbol{v}=\mathbf{0}}$$

that we defined in (9.1). Fouvry–Katz's general stratification theorem gives exceptional loci for such sums that measure the failure of square-root cancellation [FK01, Theorem 3.1 and Corollary 3.2]. We give an alternative construction for such a stratification using Beilinson's singular support for constructible étale sheaves introduced in [Bei16]. Let $\mathcal{X} \subset \mathbb{P}^{n-1}_{\mathbb{Z}}$ be the hypersurface defined by $F = 0$. As in the main body of the paper, let $\widehat{Y}$ denote the affine cone of a projective variety Y.

Theorem A.1. *There exist an integer $N \geqslant 1$ and closed subschemes*

$$\mathbb{A}^n_{\mathbb{Z}} = \mathcal{Z}_0 \supseteq \mathcal{Z}_1 \supseteq \cdots \supseteq \mathcal{Z}_n = \{\mathbf{0}\}$$

defined by homogeneous polynomials such that

(1) $\dim(\mathcal{Z}_i \otimes \mathbb{Q}) \leqslant n - i$ for every $0 \leqslant i \leqslant n$;

(2) for every finite field $\mathbb{F}_q$ of characteristic $p \nmid N$, every $0 \leqslant i \leqslant n-1$, and every $\boldsymbol{v} \notin \mathcal{Z}_{i+1}(\mathbb{F}_q)$, one has

$$|T_q(\boldsymbol{v})| \ll_{n,d} q^{(n-1+i)/2}, \tag{A.1}$$

where the implied constant may be given as an explicit function of n and d.

If $\mathcal{X} \otimes \mathbb{Q}$ is geometrically integral and $d \geqslant 2$, then

$$\widehat{W}_i \subseteq \mathcal{Z}_{i+1} \otimes \mathbb{Q} \subseteq \widehat{B}_i \qquad (0 \leqslant i \leqslant n-1), \tag{A.2}$$

in the notation of (4.1) and (4.6). If in addition $\mathcal{X}(\mathbb{Q}) = \emptyset$, then

$$\dim \overline{\mathcal{Z}_i(\mathbb{Q})}^{\mathrm{Zar}} \leqslant n - i - 1 \qquad (1 \leqslant i \leqslant n-1), \tag{A.3}$$

where $\overline{\mathcal{Z}_i(\mathbb{Q})}^{\mathrm{Zar}}$ denotes the Zariski closure of $\mathcal{Z}_i(\mathbb{Q})$ in the variety $\mathcal{Z}_i \otimes \mathbb{Q}$.

Remark A.2. The reader may notice that (A.1) holds trivially for all $\boldsymbol{v} \in \mathbb{F}_q^n$ if $i = n-1$, even if $\boldsymbol{v} \in \mathcal{Z}_n(\mathbb{F}_q)$. Nonetheless, (A.2) for $i = n-1$ requires us to include the zero section $\mathbf{0}$ in $\mathcal{Z}_n$. Including the zero section is also natural from the perspective of projective geometry, which is pervasive throughout our paper.

Notation and conventions. Over a field k, write $X = \mathcal{X}_k$. We write $P = \mathbb{P}^{n-1}$ and let $P^\vee$ denote its dual projective space. It will usually be clear from the context over which ring we are working. For $h = [\boldsymbol{v}] \in P^\vee$ we write H_h for the hyperplane $\boldsymbol{v} \cdot \boldsymbol{x} = 0$ and $X_h = X \cap H_h$. Given a scheme Y over S and a morphism $\mathrm{Spec}(R) \to S$, we write Y_R for the base change of Y to $\mathrm{Spec}(R)$. Unless indicated otherwise, cohomology over a field means geometric *étale* cohomology. We put $\dim \emptyset = -1$ and $\operatorname{codim} \emptyset = +\infty$.

A.1. The finite-field argument.

A.1.1. *Reduction to a Gysin isomorphism.* Throughout this section, we fix an auxiliary prime ℓ and work with a finite field $k = \mathbb{F}_q$ of characteristic $p \neq \ell$. We assume $F \in k[x_1, \dots, x_n]$ is nonzero and homogeneous of degree d, and continue to write $X \subset \mathbb{P}^{n-1}$ for the projective hypersurface it defines.

Lemma A.3. *For every $\boldsymbol{v} \neq \mathbf{0}$, with $h = [\boldsymbol{v}]$, one has*

$$T_q(\boldsymbol{v}) = 1 + q\#X_h(k) - \#X(k). \tag{A.4}$$

Proof. When $q = p$, this follows from (5.1) together with the identity $S_p(\boldsymbol{v}) = pT_p(\boldsymbol{v})$ observed in (9.2). For general prime powers q, the same underlying algebraic manipulations apply. □

Applying the Grothendieck–Lefschetz trace formula [Mil13, Theorem 29.8], the previous lemma gives

$$T_q(\boldsymbol{v}) = 1 + \sum_{i \geqslant 0} (-1)^i \left(\operatorname{Tr}(\mathrm{Fr}_q \mid H^{i-2}(X_{h,\overline{k}}, \mathbb{Q}_\ell)(-1)) - \operatorname{Tr}(\mathrm{Fr}_q \mid H^i(X_{\overline{k}}, \mathbb{Q}_\ell)) \right), \tag{A.5}$$

where Fr_q denotes the geometric Frobenius. By Deligne's general form of the Riemann hypothesis [Del80], an eigenvalue α of the action of Fr_q on $H^{i-2}(X_{h,\overline{k}}, \mathbb{Q}_\ell)(-1)$ or $H^i(X_{\overline{k}}, \mathbb{Q}_\ell)$ satisfies $|\alpha| \leqslant q^{i/2}$. The idea of the proof of Theorem A.1 is now to construct strata on $P^\vee$ measuring in which degrees the groups $H^{i-2}(X_{h,\overline{k}}, \mathbb{Q}_\ell)(-1)$ and $H^i(X_{\overline{k}}, \mathbb{Q}_\ell)$ are isomorphic.

For this we will use the *Gysin map* in a form that applies even when X or X_h is singular, following the construction in [RS22, §2.1]. Let $a_X : X \hookrightarrow P$, $i_h : H_h \hookrightarrow P$, and $K_R = (a_X)_* R_X$, where R is $\mathbb{Z}/\ell^r\mathbb{Z}$ for some integer $r \geqslant 1$. Purity for the smooth divisor $H_h \subset P$ identifies $i_h^! R_P$ with the complex $R_{H_h}(-1)[-2]$, in a derived sense. Since $R_{H_h}(-1)$ is locally free of rank one, the natural morphism $i_h^* K_R \otimes_R^L i_h^! R_P \to i_h^! K_R$, where $\otimes_R^L$ denotes the derived tensor product of R-modules, therefore gives a morphism

$$i_h^* K_R(-1)[-2] \longrightarrow i_h^! K_R.$$

We can compose its pushforward by i_h with the counit $(i_h)_* i_h^! K_R \to K_R$ and take global sections, so that proper base change identifies the resulting map with

$$G_h^a(R) : H^{a-2}(X_{h,\overline{k}}, R)(-1) \longrightarrow H^a(X_{\overline{k}}, R). \tag{A.6}$$

The construction is compatible with the Galois action and with coefficient maps, including all reductions $\mathbb{Z}/\ell^{r+1}\mathbb{Z} \to \mathbb{Z}/\ell^r\mathbb{Z}$. Thus the maps $G_h^a(\mathbb{Z}/\ell^r\mathbb{Z})$ lift to the ℓ-adic coefficients $\mathbb{Z}_\ell$ and $\mathbb{Q}_\ell$, which will eventually let us pass to $\mathbb{Q}_\ell$ coefficients in Lemma A.8.

The key connection between the Gysin map and the size of $T_q(\boldsymbol{v})$ is recorded in the following lemma.

Lemma A.4. *If $t \geqslant 0$ and $G_h^i(\mathbb{Q}_\ell)$ is an isomorphism for all $i > t$, then, for $\boldsymbol{v} \neq \mathbf{0}$ with $[\boldsymbol{v}] = h$,*

$$|T_q(\boldsymbol{v})| \ll_{n,d} q^{t/2},$$

where the implied constant may be given as an explicit function of n and d.

Proof. In (A.5), the terms with $i > t$ vanish, since by assumption $G_h^i(\mathbb{Q}_\ell)$ is a Galois-equivariant isomorphism for $i > t$. Deligne's general form of the Riemann hypothesis [Del80, Théorème 3.3.1] bounds the absolute values of the eigenvalues of the nonzero terms by $q^{i/2} \leqslant q^{t/2}$. Katz's bounds [Kat01, Theorem 3 and its corollary] bound the total Betti numbers of X and X_h in terms of n, d: they are cut out by at most two equations of degree at most d. The triangle inequality in (A.5), including its constant term, therefore gives the stated bound. □

Consequently, to obtain (A.1) away from the cone over $Y_{j+1} \subset P^\vee$, it suffices to prove that $G_h^a(\mathbb{Q}_\ell)$ is an isomorphism for $a > n - 1 + j$ when $h \notin Y_{j+1}$. The next subsection constructs these loci with $\operatorname{codim} Y_j \geqslant j$.

A.1.2. *Singular support and the exceptional loci.* We first work over a perfect field k with coefficients $\Lambda = \mathbb{F}_\ell$, where $\ell \neq \operatorname{char} k$. For a smooth equidimensional variety M and $K \in D_c^b(M, \Lambda)$, Beilinson defines the singular support $\mathrm{SS}(K) \subset T^*M$ as the smallest closed conical set on which K is micro-supported [Bei16, Sections 1.2–1.3]. The precise definition of $\mathrm{SS}(K)$ will not be important for us; all that matters is that the same definition is consistently used in the key references [Bei16, HY18, RS22]. In any case, the rough idea is that the singular support records the cotangent directions in which local acyclicity can fail.

The properties needed here are the following. Every irreducible component of $\mathrm{SS}(K)$ has dimension $\dim M$; its image in M is the closed support $\operatorname{Supp} K$; and formation of singular

support commutes with extensions of the ground field. All these properties are recorded in [Bei16, Theorems 1.3, 1.4(iii), and Lemma 2.3(iii)].

We apply this construction with $M = P = \mathbb{P}_k^{n-1}$ and $K = (a_X)_*\Lambda_X$, so that $\operatorname{Supp} K = X$. Let

$$I = \{(x, h) \in P \times P^\vee : x \in H_h\} \simeq \mathbb{P}(T^*P)$$

be the incidence variety. The identification assigns to (x, h) the conormal line to H_h at x. Put

$$\Sigma = \mathbb{P}(\mathrm{SS}(K)) \subset I, \qquad \pi : \Sigma \longrightarrow P^\vee. \tag{A.7}$$

The map π is proper and $\dim \Sigma \leqslant n-2$. In fact, every irreducible component of Σ has dimension exactly equal to $\dim M - 1 = n-2$. Indeed, a component of $\mathrm{SS}(K)$ contained in the zero section of T^*M (and thus disappearing after projectivisation) would have base of dimension $\dim M = n-1$, whereas $\operatorname{Supp} K = X$ has dimension $n-2$.

Definition A.5. For $1 \leqslant i \leqslant n$, define

$$Y_i = \{h \in P^\vee : \dim \Sigma_{\overline{h}} \geqslant i-1\}, \tag{A.8}$$

where $\overline{h}$ is a geometric point over h.

Lemma A.6. *The loci Y_i are closed, satisfy $Y_{i+1} \subset Y_i$, and have $\operatorname{codim}_{P^\vee} Y_i \geqslant i$. In particular $Y_n = \emptyset$.*

Proof. Closedness follows from upper semicontinuity of fibre dimension for proper morphisms [TSta, Tag 0D4I]. For an irreducible component E of Y_i over an algebraic closure, a component of $\pi^{-1}(E)$ dominating E has dimension at least $\dim E + i - 1$. Hence $\dim E + i - 1 \leqslant \dim \Sigma \leqslant n-2$, giving $\dim E \leqslant n-1-i$. The other assertions follow at once. □

The key property of the loci Y_i is that they measure in which degree the Gysin map is an isomorphism.

Proposition A.7 (Raskin–Smith)**.** *For $h \notin Y_i$, the map $G_h^a(\Lambda)$ is an isomorphism if $a > n-2+i$, and is surjective if $a = n-2+i$.*

Proof. This follows from [RS22, Theorem 1.2] applied to $a_X : X \hookrightarrow P$. Its parameter is $r = \dim(X \times_P X) = \dim X = n-2$. The construction in [RS22, Theorem 2.2 and Corollary 2.3] chooses precisely the loci (A.8): the cone of $i_h^* K(-1)[-2] \to i_h^! K$ is supported on the image of Σ_h in H_h. Outside Y_i this support has dimension at most $i-2$. The cited theorem therefore gives the stated degree range. □

A.1.3. *Passage from finite to rational coefficients.* Next, we pass from finite to ℓ-adic coefficients using an elementary argument involving coefficient long exact sequences and inverse limits.

Lemma A.8. *Fix a hyperplane h and an integer t. Suppose $G_h^a(\mathbb{F}_\ell)$ is an isomorphism for $a > t$ and is surjective for $a = t$. Then the same statements hold with coefficients $\mathbb{Z}/\ell^r\mathbb{Z}$ for every $r \geqslant 1$, and with coefficients $\mathbb{Z}_\ell$ and $\mathbb{Q}_\ell$.*

Proof. Write $\Lambda_r = \mathbb{Z}/\ell^r\mathbb{Z}$ and abbreviate

$$A_r^a = H^{a-2}(X_{h,\overline{k}}, \Lambda_r)(-1), \qquad B_r^a = H^a(X_{\overline{k}}, \Lambda_r).$$

For $r \geqslant 2$, the coefficient sequence

$$0 \longrightarrow \Lambda_{r-1} \xrightarrow{\ell} \Lambda_r \longrightarrow \Lambda_1 \longrightarrow 0$$

gives long exact sequences compatible with the Gysin maps, by standard properties of derived tensor products with respect to distinguished triangles in the derived category of Λ_r-modules. For $\mathbb{Z}/\ell^r\mathbb{Z}$, we use an induction on r. The case $r = 1$ is precisely our assumption, so assume that

$r > 1$ and the statement already holds for $r-1$. For $a > t$, we obtain a commutative diagram that is exact on its horizontal arrows:

$$\begin{array}{ccccccccc} A_1^{a-1} & \longrightarrow & A_{r-1}^{a} & \longrightarrow & A_r^a & \longrightarrow & A_1^a & \longrightarrow & A_{r-1}^{a+1} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ B_1^{a-1} & \longrightarrow & B_{r-1}^{a} & \longrightarrow & B_r^a & \longrightarrow & B_1^a & \longrightarrow & B_{r-1}^{a+1}. \end{array}$$

The first vertical map is surjective, and the second, fourth, and fifth are isomorphisms. By the five lemma the middle map is thus an isomorphism. For $a = t$, the last four terms instead give surjectivity of the middle map: the maps on A_{r-1}^t and A_1^t are surjective, and the map on A_{r-1}^{t+1} is injective.

Since the finite-coefficient cohomology groups are finite, the inverse systems of kernels of the surjective maps $G_h^t(\Lambda_r)$ satisfy the Mittag–Leffler condition. Taking inverse limits therefore preserves surjectivity at $a = t$, as well as the isomorphisms for $a > t$. Now $H^b(Z, \mathbb{Z}_\ell) = \varprojlim_r H^b(Z, \Lambda_r)$ (see [Mil13, Section 19]), giving the assertions for $\mathbb{Z}_\ell$. Finally, we can use $H^b(Z, \mathbb{Q}_\ell) = H^b(Z, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ and the flatness of $\mathbb{Q}_\ell$ over $\mathbb{Z}_\ell$ to deduce the result for $\mathbb{Q}_\ell$. □

Corollary A.9. *For $h \notin Y_i$, the map $G_h^a(\mathbb{Q}_\ell)$ is an isomorphism for $a > n-2+i$ and is surjective at equality. If $k = \mathbb{F}_q$ and $\boldsymbol{v} \neq \mathbf{0}$ satisfies $[\boldsymbol{v}] \notin Y_{j+1}(k)$, then $|T_q(\boldsymbol{v})| \ll_{n,d} q^{(n-1+j)/2}$.*

Proof. This follows from Lemma A.8 applied to Proposition A.7 with $t = n-2+i$. For $i = j+1$, we can use Lemma A.4 with $t = n-1+j$. □

A.2. **A construction over the integers.** We use Hu–Yang's relative singular support [HY18] to construct the exceptional loci over an open subscheme of $\operatorname{Spec}\mathbb{Z}$.

A.2.1. *Relative singular support.* Fix a prime ℓ. Invert ℓ and all primes dividing the content of F, and write $S = \operatorname{Spec}\mathbb{Z}[1/N]$ for the resulting base. We enlarge N finitely many times without changing the notation. Let

$$\mathcal{P} = \mathbb{P}_S^{n-1}, \qquad \mathcal{X}_S = V_+(F) \subset \mathcal{P}, \qquad \mathcal{K} = (a_{\mathcal{X}_S})_* \mathbb{F}_\ell.$$

In particular every fibre of $\mathcal{X}_S$ is a hypersurface of dimension $n-2$.

We use the following consequence of Hu–Yang's relative theory. Their Theorems 5.4 and 5.8 in [HY18] imply, after shrinking S, the existence of a closed conical subset

$$\mathcal{C} = \mathrm{SS}(\mathcal{K}, \mathcal{P}/S) \subset T^*(\mathcal{P}/S) \tag{A.9}$$

such that, for every geometric point $\overline{s} \to S$,

$$(\mathcal{C}_{\overline{s}})_{\mathrm{red}} = \mathrm{SS}(\mathcal{K}|_{\mathcal{P}_{\overline{s}}}). \tag{A.10}$$

In other words, the reduction of the singular support is the singular support of the reduction away from finitely many primes. The numbering here follows arXiv:1702.06752v1.

Under the relative incidence identification set

$$\Sigma_S = \mathbb{P}(\mathcal{C}) \subset \mathcal{P} \times_S \mathcal{P}^\vee, \qquad \pi_S : \Sigma_S \longrightarrow \mathcal{P}^\vee.$$

For $1 \leqslant i \leqslant n$, define

$$\mathcal{Y}_i = \{h \in \mathcal{P}^\vee : \dim(\pi_S^{-1}(\overline{h})) \geqslant i-1\}. \tag{A.11}$$

Properness makes these closed. By (A.10), their geometric fibres are exactly the loci of Definition A.5. Thus

$$\operatorname{codim}_{\mathcal{P}_{\overline{s}}^\vee} \mathcal{Y}_{i,\overline{s}} \geqslant i \quad \text{for all } \overline{s} \to S, \qquad \mathcal{Y}_n = \emptyset. \tag{A.12}$$

A.2.2. *Affine cones and specialization.* For $1 \leqslant i \leqslant n-1$, let $\mathcal{Z}_{i,S} \subset \mathbb{A}^n_S$ be the affine cone over $\mathcal{Y}_i$, with the origin included even when $\mathcal{Y}_i = \emptyset$. Set $\mathcal{Z}_{0,S} = \mathbb{A}^n_S$ and $\mathcal{Z}_{n,S} = \{\mathbf{0}\}$. Formation of these cones on underlying sets commutes with field extension, and (A.12) gives $\dim(\mathcal{Z}_{i,S})_{\overline{s}} \leqslant n - i$.

For $1 \leqslant i \leqslant n-1$, define $\mathcal{Z}_i \subset \mathbb{A}^n_{\mathbb{Z}}$ to be the reduced closure of $\mathcal{Z}_{i,S}$, and set $\mathcal{Z}_0 = \mathbb{A}^n_{\mathbb{Z}}$ and $\mathcal{Z}_n = \{\mathbf{0}\}$. These are nested closed conical subschemes defined over $\mathbb{Z}$. Each irreducible component is the closure of a component over S. A component dominating $\operatorname{Spec}\mathbb{Z}$ has dimension at most $n+1-i$ by the dimension formula [TSta, Tag 02JU]; a vertical component lies over a prime in S and has dimension at most $n-i$. Hence $\operatorname{codim}_{\mathbb{A}^n_{\mathbb{Z}}} \mathcal{Z}_i \geqslant i$, and their restrictions to S are unchanged.

Proposition A.10. *The schemes $\mathcal{Z}_i$ satisfy (1) and (2) of Theorem A.1.*

Proof. The codimension assertions have just been proved. Suppose $p \nmid N$, $0 \leqslant j \leqslant n-2$, and $\boldsymbol{v} \notin \mathcal{Z}_{j+1}(\mathbb{F}_q)$. Then $\boldsymbol{v} \neq \mathbf{0}$ and $[\boldsymbol{v}] \notin \mathcal{Y}_{j+1}(\mathbb{F}_q)$, so Corollary A.9 gives the required estimate. For $j = n-1$, we use instead the elementary bound

$$|T_q(\boldsymbol{v})| \leqslant \#\{\boldsymbol{x} \in \mathbb{F}_q^n : F(\boldsymbol{x}) = 0\} \leqslant dq^{n-1} \qquad (\boldsymbol{v} \in \mathbb{F}_q^n).$$

The same N works over every finite extension of $\mathbb{F}_p$ by (A.10). □

A.2.3. *Comparison with the geometric exceptional loci.* Suppose that $X_{\overline{\mathbb{Q}}}$ is integral and $d \geqslant 2$. We work over $\mathbb{Q}$. Recall the correspondences Γ and S from (4.5) and (4.2). We claim that

$$\Gamma \subseteq \Sigma_{\mathbb{Q}} \subseteq S. \tag{A.13}$$

At a smooth point of X, the singular support of $(a_X)_*\mathbb{F}_\ell$ is the ordinary conormal bundle of X in P; see [Bei16, Lemmas 2.3(i) and 2.5(ii)]. Taking closures gives the first inclusion. For the second inclusion, let $(x,h) \in \Sigma_{\mathbb{Q}}$ be a geometric point. The incidence identification gives $x \in H_h$, and the support property gives $x \in X$. If x is smooth on X, then H_h contains the tangent space to X at x, so the defining equations of S hold. If x is singular on X, all the first partial derivatives of F vanish at x, so these equations again hold.

Since $X_{\overline{\mathbb{Q}}}$ is integral of degree $d \geqslant 2$, the restriction of F to every hyperplane is nonzero. Thus $X \cap H_h$ is a hypersurface in H_h, and the Jacobian criterion identifies S_h with its scheme-theoretic singular locus. Taking geometric fibre dimensions in (A.13) gives

$$W_i \subseteq \mathcal{Y}_{i+1,\mathbb{Q}} \subseteq B_i \qquad (0 \leqslant i \leqslant n-1).$$

Taking affine cones, with the convention $\widehat{\emptyset} = \{\mathbf{0}\}$, proves assertion (A.2) of Theorem A.1.

A.3. **Rational points on the exceptional loci.** To complete the proof of Theorem A.1, it remains to prove the additional codimension bound for the closure of $\mathcal{Z}_i(\mathbb{Q})$, assuming $X(\mathbb{Q}) = \emptyset$. We shall work projectively, from which the statement for affine cones follows easily.

A.3.1. *Fibre jumps and dual varieties.* We begin with the following elementary, but useful observation.

Lemma A.11. *Let $f : A \to B$ be a proper dominant morphism of integral varieties, and put $c = \dim A - \dim B$. For $j \geqslant 1$, the closed locus*

$$E_j = \{b \in B : \dim A_{\overline{b}} \geqslant c + j\}$$

has codimension at least $j+1$ in B.

Proof. The generic fibre dimension is c, so E_j is a proper closed subset by upper semicontinuity of fibre dimension for proper morphisms [TSta, Tag 0D4I]. For an irreducible component E of E_j, some component of $f^{-1}(E)$ dominating E has dimension at least $\dim E + c + j$. Since $f^{-1}(E)$ is proper closed in the integral variety A, its dimension is at most $\dim A - 1 = \dim B + c - 1$. Comparing these two inequalities proves the claim. This is also a consequence of [TSta, Tag 0B2K]. □

For an integral proper subvariety $V \subsetneq \mathbb{P}^m$ in characteristic zero, its projective conormal variety is

$$\mathcal{N}_V = \overline{\{(x,h) : x \in V_{\mathrm{sm}},\ \mathbb{T}_x V \subset H_h\}} \subset \mathbb{P}^m \times (\mathbb{P}^m)^\vee,$$

where $\mathbb{T}_x V$ denotes the embedded projective tangent space. Its second projection is the dual variety $V^\vee$. Biduality identifies $\mathcal{N}_V$ with $\mathcal{N}_{V^\vee}$ after interchanging the two factors, and gives $(V^\vee)^\vee = V$; see [Tev01, Theorem 1.7].

Lemma A.12. *Let $V \subsetneq \mathbb{P}^m_{\mathbb{Q}}$ be integral, but not necessarily geometrically integral. If $V(\mathbb{Q}) = \emptyset$, then*

$$V^\vee(\mathbb{Q}) \subset \mathrm{Sing}(V^\vee)(\mathbb{Q}).$$

Proof. Suppose that $h \in (V^\vee)_{\mathrm{sm}}(\mathbb{Q})$. The annihilator

$$(\mathbb{T}_h V^\vee)^\perp \subset \mathbb{P}^m_{\mathbb{Q}}$$

of the embedded tangent space is a projective linear space over $\mathbb{Q}$ of dimension $m - \dim V^\vee - 1 \geqslant 0$. By biduality it is the fibre over h of $\mathcal{N}_V \to V^\vee$, viewed in the first factor, and is therefore contained in V. It has a rational point, contrary to the hypothesis.

If V is not geometrically integral, its geometric components and their distinct duals are conjugate. A smooth point of $V^\vee$ belongs to a unique geometric component, so the same fibre identification applies at h and descends to $\mathbb{Q}$. □

A.3.2. *Conormal components and the codimension gain.* We continue to use the notation $K = (a_X)_* \mathbb{F}_\ell$ and $\Sigma = \mathbb{P}(\mathrm{SS}(K))$ on $P_{\mathbb{Q}}$ from the previous sections. In characteristic zero, the components of singular support are conical Lagrangian subvarieties [Bei16, Remark after Theorem 1.3], hence closures of conormal bundles to their images [Tev01, Theorem 1.9]. Those images lie in X, since the base of singular support is $\mathrm{Supp}\, K = X$. For each irreducible component Σ_α of Σ over $\mathbb{Q}$, there is an integral subvariety $V_\alpha \subset X$ with $\Sigma_\alpha = \mathcal{N}_{V_\alpha}$.

Proposition A.13. *Assume $X(\mathbb{Q}) = \emptyset$. For the generic-fibre loci $Y_i = \mathcal{Y}_{i,\mathbb{Q}}$ and $1 \leqslant i \leqslant n-1$,*

$$\mathrm{codim}_{P^\vee_{\mathbb{Q}}} \overline{Y_i(\mathbb{Q})} \geqslant i + 1. \tag{A.14}$$

Proof. The projection $\Sigma_\alpha \to V_\alpha^\vee$ has generic relative dimension $c_\alpha = n - 2 - \dim V_\alpha^\vee$. Put

$$E_{\alpha,i} = \{h \in V_\alpha^\vee : \dim(\Sigma_\alpha)_{\overline{h}} \geqslant i - 1\}.$$

Then $Y_i = \bigcup_\alpha E_{\alpha,i}$. We distinguish three cases.

If $c_\alpha > i - 1$, then $\dim V_\alpha^\vee = n - 2 - c_\alpha \leqslant n - i - 2$, and the same bound holds for $\overline{E_{\alpha,i}(\mathbb{Q})}$.

If $c_\alpha < i - 1$, Lemma A.11 with $j = i - 1 - c_\alpha \geqslant 1$ gives

$$\dim E_{\alpha,i} \leqslant \dim V_\alpha^\vee - (i - 1 - c_\alpha) - 1 = n - i - 2.$$

If $c_\alpha = i - 1$, then $\dim V_\alpha^\vee = n - i - 1$. Since $V_\alpha \subset X$ has no rational point, Lemma A.12 implies that $V_\alpha^\vee(\mathbb{Q})$ lies in the proper closed subset $\mathrm{Sing}(V_\alpha^\vee)$, of dimension at most $n - i - 2$. The closure of $E_{\alpha,i}(\mathbb{Q})$ has the same bound. □

Completion of the proof of Theorem A.1. It remains to prove the assertion (A.3) about the closure of rational points assuming $X(\mathbb{Q}) = \emptyset$. For $1 \leqslant i \leqslant n-1$, every nonzero rational point of $\mathcal{Z}_i$ projects to a rational point of Y_i. Thus $\overline{\mathcal{Z}_i(\mathbb{Q})}$ lies in the affine cone over $\overline{Y_i(\mathbb{Q})}$, together with the origin. If $Y_i(\mathbb{Q}) \neq \emptyset$, this cone has dimension at most $n - i - 1$ by Proposition A.13. Otherwise only the origin remains, and again its dimension is at most $n - i - 1$. This proves (A.3) for $1 \leqslant i \leqslant n - 1$. □

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